# Development of a Device for Non-Invasive Treatment of Alzheimer's Disease using Electric Field

**HT599: Master's Thesis Project**

A report submitted in partial fulfilment of the requirements for the degree of
Master of Technology in Biomedical Science and Engineering


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# Certificate

*This is to certify that the work presented in the report entitled "**Development of a Device for treatment of Alzheimer's Disease using an Electric Field**" by **K S RANJITH KUMAR** (M.Tech. Student, Roll No.: 244159016) represents the original work under the guidance of **Prof. Harshal B Nemade**. This study has not been submitted elsewhere for a degree.*

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# Acknowledgement

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I am grateful to the **Jyoti and Bhupat Mehta School of Health Sciences and Technology** for providing the academic environment, laboratory facilities and resources necessary for carrying out this research. The opportunity to work in a multidisciplinary research setting greatly enhanced my understanding of biomedical engineering and medical device development.

I would also like to acknowledge the contributions of previous researchers and members of the research group whose simulation studies and published work provided the foundation for this project. Their efforts established important design considerations and research directions that guided the development of the present work.

Building upon these prior investigations, this work focused on translating simulation derived concepts into a practical hardware platform through the design and development of simulation electrodes, stimulation circuitry, protection architectures, PCB layouts, proof-of-concept prototypes and experimental evaluations. This work involved multiple electrode and circuit design iterations, breadboard implementations, waveform characterization using oscilloscope measurements, prototype realization, fabrication of electrodes and performance assessment under practical operating conditions. These activities contributed significantly to my technical knowledge and growth as a researcher and engineer.

Finally, I would like to thank everyone who contributed directly or indirectly to creating a productive and supportive academic environment during the completion of this work.

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# Declaration

I hereby declare that the thesis entitled "Development of Device for Non-Invasive treatment of Alzheimer's Disease using Electric Field" submitted in partial fulfilment of the requirements for the award of the degree of Master of Technology (MTech) is a record of the research work carried out by me under the guidance of Prof Harshal B Nemade at the Jyoti and Bhupat Mehta School of Health Sciences and Technology.

I further declare that this work has not been submitted, either in whole or in part to any other university or institution for the award of any degree, diploma or academic qualification.

The work presented in this thesis is based on original investigations carried out during the course of the study. Where previous simulation studies, published research and existing literature have been used as the basis for design considerations and development, appropriate acknowledgement and citation have been provided.

The electrode design refinement, circuits design and development, comparative evaluation of multiple stimulation architectures, PCB design, breadboard implementation, proof-of-concept fabrication, prototype realization, experimental testing, waveform characterization, data analysis and interpretation of results presented in this thesis were carried out by me under the guidance and supervision of my supervisor.

I certify that, to best of my knowledge and belief this thesis contains no material previously published or written by any other person except where due references have been made in the text.


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# Abstract

Alzheimer's disease is a progressive neurodegenerative disorder characterized by cognitive decline, memory impairment and loss of functional independence. Despite significant advances in medical research, currently available treatment strategies provide limited therapeutic benefit and primarily focus on symptomatic management. Consequently, alternative approaches such as neuromodulation have attracted increasing attention for their potential to influence neural activity through externally applied stimuli. Among these approaches non-invasive electrical stimulation has emerged as a promising technique due to its ability to generate controlled electric fields within biological tissues without the need for surgical intervention.

Previous stimulation studies conducted within the research group investigated the generation of electric fields for non-invasive stimulation applications and established key design constraints including a ten-electrode configuration, an angular coverage of approximately 256°, a width-to-gap ratio of 1:1 and an operating voltage of 25 V. Although these studies demonstrated the theoretical feasibility of the proposed stimulation approach practical realization of the system remained largely unexplored.

The present work focuses on translating these simulation constraints into a practical and experimentally realizable hardware platform. Multiple electrode geometries were developed and evaluated with consideration for anatomical constraints, manufacturability and practical implementation requirements. In parallel, several stimulation circuit architectures, including transistor based astable multivibrators, operational amplifiers-based oscillators, H-bridge configurations, self-oscillating MOSFET architectures and protection-oriented circuits incorporating current monitoring mechanisms were designed, implemented and experimentally investigated. Breadboard based implementations and oscilloscope measurements were used to compare the performance of the various architectures and guide the selection of a suitable stimulation platform based on simplicity, reliability, component count, portability and implementation feasibility.

Based on the design evaluation process, a compact stimulation architecture consisting of a rechargeable power subsystem, voltage boosting stage and MOSFET based signal generation circuitry was selected and implemented. PCB layouts for both the stimulation circuitry and electrode structures were developed considering a single layer manufacturability constraint. In addition, a proof of concept electrode was fabricated using a ferric chloride etching process and utilized for preliminary experimental evaluation. Hardware testing demonstrated stable waveform generation at the required operating voltage, while electrode-loading experiments indicated minor variations in waveform characteristics under practical contact conditions.

The work presented in this thesis demonstrated the successful translation of simulation derived design concepts into a functional proof-of-concept stimulation platform. Furthermore, the study establishes a foundation for future development involving custom PCB fabrication, system miniaturization, advanced safety mechanisms, impedance monitoring, current monitoring, charge balancing verification and comprehensive validation for future preclinical investigations. The proposed platform contributes toward the development of compact and experimentally realizable non-invasive electrical stimulation systems for biomedical research applications.

Keywords: Alzheimer's disease, neuromodulation, non-invasive electrical stimulation, electrode design, electric field stimulation, astable multivibrator, PCB design, biomedical instrumentation, proof-of-concept prototype.

# Chapter 1
# Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by gradual deterioration of memory, cognitive function, reasoning ability, and behavioural responses. It is one of the most common causes of dementia worldwide and represents a major healthcare challenge due to its increasing prevalence among the ageing population. The disease is primarily associated with the accumulation of amyloid-beta plaques and neurofibrillary tau tangles within the brain, which disrupt normal neuronal communication and ultimately lead to neuronal death as shown in *Figure 1*. As the disease progresses, patients experience severe cognitive impairment, loss of independence, and a significant reduction in quality of life.

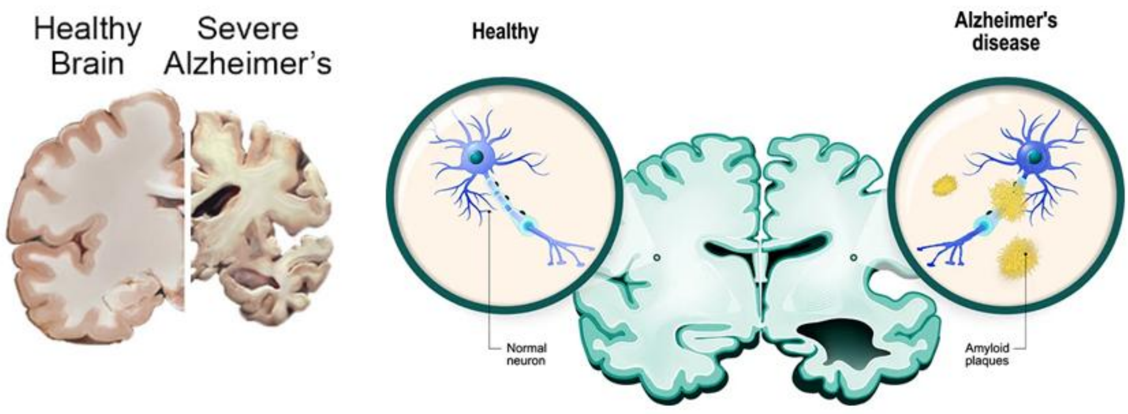



*Figure 1. Comparison between normal brain and patient suffering from Alzheimer's Disease*

*(Adapted from National Institute on Aging, 2024 & Research outreach, 2023)*

## 1.1. Background

Despite decades of research, currently available therapeutic approaches provide only limited symptomatic relief and have shown limited success in preventing or reversing disease progression. Most pharmacological interventions aim to improve neurotransmitter activity or reduce pathological symptoms; however, their effectiveness remains restricted and often varies among patients (Thair et al., 2017). Consequently, there is an increasing need to investigate alternative therapeutic strategies capable of influencing neural function through non-pharmacological means.

Neuromodulation has emerged as one such promising approach. Neuromodulation involves the alteration of neural activity through the controlled application of electrical, magnetic, optical, acoustic, or chemical stimuli. By directly interacting with neural circuits, these techniques have demonstrated potential in the treatment and management of several neurological disorders including Parkinson's disease, epilepsy, depression, chronic pain, and Alzheimer's disease.

Among the various neuromodulation approaches as shown in *Figure 2*, electrical stimulation is particularly attractive because it directly interacts with the natural electrical signalling mechanisms of the nervous system. Electrical stimulation techniques can be implemented using relatively simple hardware, offer good controllability, and can be adapted for both invasive and non-invasive applications. These advantages have motivated significant research efforts toward developing compact and efficient electrical stimulation systems for biomedical applications.

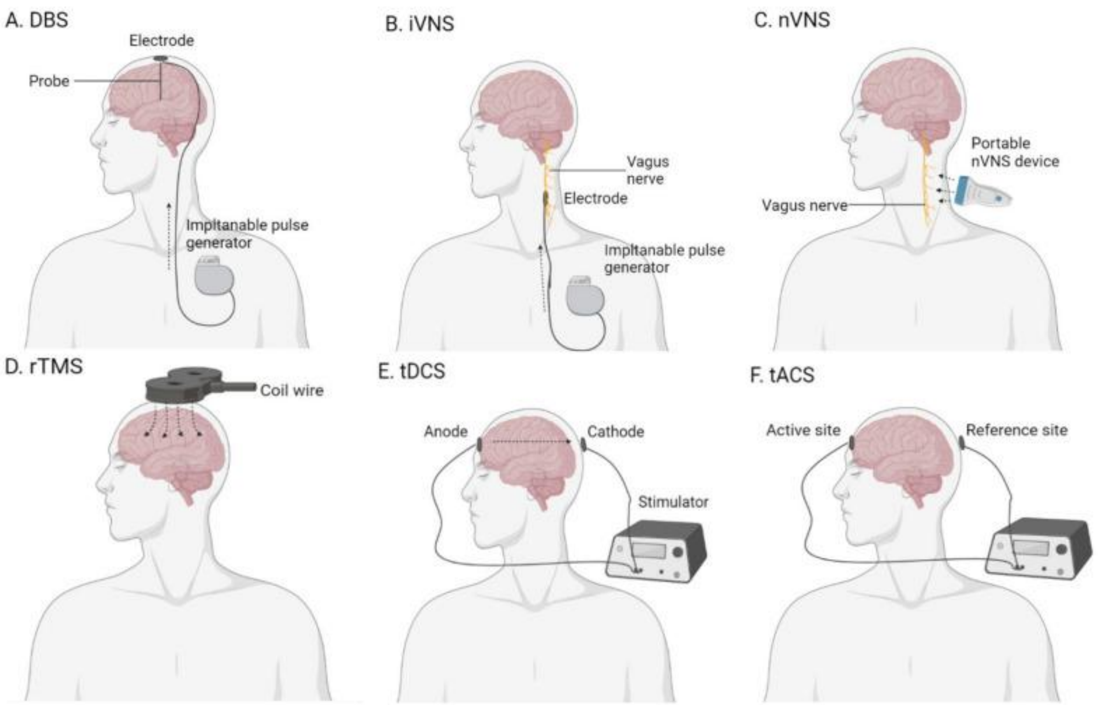



*Figure 2. Different types of existing methods of brain stimulation*

*(A) deep-brain stimulation (DBS); (B) invasive vagus-nerve stimulation (iVNS); (C) non-invasive vagus-nerve stimulation (nVNS); (D) repetitive transcranial magnetic stimulation (rTMS); (E) transcranial direct-current stimulation (tDCS); (F) transcranial alternating-current stimulation (tACS). (Adapted from Yu et al., 2021).*

## 1.2. Electrical Stimulation and Electrode Design

The effectiveness of an electrical stimulation system depends strongly on the characteristics of the electric field generated within the target tissue. Parameters such as electrode geometry, electrode placement, stimulation voltage, waveform characteristics, tissue properties, and electrode-tissue interface behaviour influence the resulting electric field distribution and stimulation effectiveness.

Among these parameters, electrode design plays a particularly important role. The size, shape, spacing, and arrangement of electrodes determine how current is distributed through biological tissue and directly affect the resulting field intensity and uniformity. Consequently, careful electrode optimization is necessary to achieve the desired stimulation characteristics while maintaining practical manufacturability and biological compatibility.

Recent studies have shown that externally applied electric fields can influence cellular behaviour, protein aggregation mechanisms, and neural activity (Saikia et al., 2019). In particular, previous investigations have demonstrated that electric field stimulation may affect amyloid-beta aggregation pathways, suggesting its potential relevance in Alzheimer's disease research. These findings have encouraged further investigation into stimulation systems capable of generating controlled electric field distributions within biological tissues.

## 1.3. Previous Work and Motivation

Previous work carried out within the research group focused on simulation-based investigation of electric field stimulation using COMSOL Multiphysics. A multilayer head model consisting of skin, skull, cerebrospinal fluid, cortex, and white matter was developed to study electric field propagation under various electrode configurations.

The simulation studies established several important design constraints, including the use of a ten-electrode configuration, an angular coverage of approximately 256°, and a width-to-gap ratio of 1:1. Furthermore, the simulations demonstrated that an excitation voltage of approximately 25 V was capable of generating the desired electric field distribution within the target region. These findings provided a theoretical foundation for the development of a practical stimulation system.

Although the simulation studies demonstrated the feasibility of the proposed approach, the translation of simulation-derived concepts into a physical hardware platform remained largely unexplored. Practical implementation introduces several additional challenges that are not fully captured within simulation environments. These include electrode manufacturability, geometric constraints imposed by animal models, power delivery requirements, circuit miniaturization, stimulation waveform generation, portability, and system integration.

For small-animal studies, these challenges become even more significant due to strict limitations on device size, weight, and mechanical compatibility. Therefore, a systematic approach was required to convert the simulation-derived design into a compact, lightweight, and experimentally realizable stimulation platform.

## 1.4. Research Gap

A review of literature and previous investigations reveals a significant gap between stimulation-based optimization studies and practical hardware realization of electrical stimulation systems. While numerous studies have focused on electric field modelling and electrode optimization, relatively few investigations have demonstrated the complete process of translating simulation-derived stimulation concepts into compact and experimentally validated hardware platforms.

Furthermore, many reported stimulation systems rely on complex architectures involving dedicated stimulation integrated circuits, specialized control hardware and laboratory-scale instrumentation. Such systems are often unsuitable for compact, lightweight applications where portability, simplicity and ease of implementation are important considerations.

Additional challenges arise from the need to develop manufacturable electrode structures while preserving simulation derived design requirements. Practical implementation also requires consideration of power management, stimulation waveform generation, hardware reliability and future investigation of safety mechanisms such as current monitoring, impedance monitoring, fault detection and charge balancing techniques.

Therefore there exists a need for a practically and experimentally realizable stimulation platform capable of bridging the gap between simulation based design optimization and hardware implementation.

## 1.5. Aim of the Study

The aim of this work is to develop a lightweight proof-of-concept non-invasive electrical stimulation platform based on simulation-derived design constraints and to experimentally evaluate its practical implementation through electrode development, circuit design, prototype realization and experimental testing.

## 1.6. Objectives

The specific objectives of this work are:

- To analyse the electrode design constraints established through previous simulation studies.
- To develop manufacturable electrode geometries suitable for practical implementation while preserving key simulation-derived design parameters.
- To investigate and compare multiple stimulation circuit architectures suitable for compact and lightweight implementation.
- To design and develop stimulation circuitry capable of generating the required stimulation waveform.
- To fabricate proof-of-concept electrode structures and evaluate their practical feasibility.
- To design PCB layouts for the stimulation circuitry and electrodes structures considering manufacturability requirements.
- To realize a hardware prototype using commercially available electronic components.
- To experimentally evaluate the developed system through breadboard implementations, oscilloscope measurements and prototype testing.
- To identify potential improvements related to safety, monitoring, miniaturization and future system development.

## 1.7. Scope of the work

The scope of this work is limited to the development and evaluation of a proof-of-concept non-invasive electrical stimulation platform. The study focuses on the translation of simulation derived design constraints into practical hardware through electrode development, circuit design, PCB layout generation, prototype fabrication and experimental testing.

Multiple stimulation architectures were investigated and experimentally evaluated to determine a suitable implementation approach. Preliminary electrode fabrication and testing were performed to assess practical feasibility and waveform behaviour under loading conditions. Although advanced protection and monitoring mechanisms, including current monitoring, impedance monitoring, watchdog protection, fault detection and charge balancing strategies were investigated during the design process, these features were not fully integrated into the final prototype and are considered potential directions for future development.

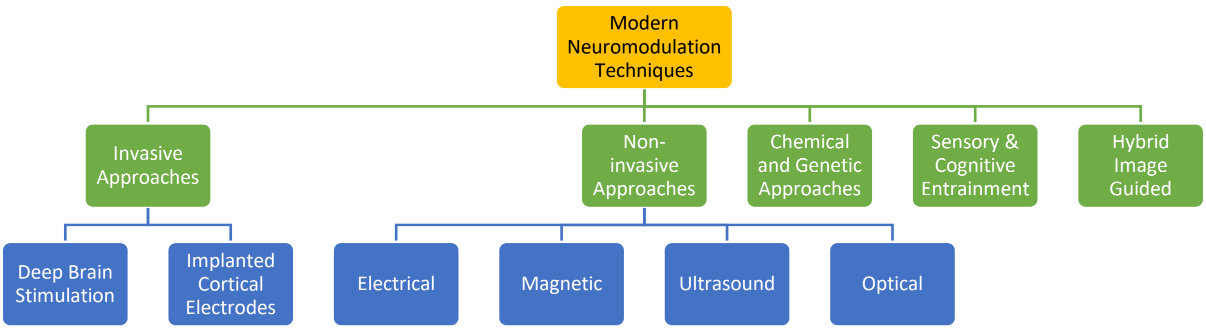
The present work focuses primarily on the engineering realization of a compact stimulation platform. While the previous studies established stimulation requirements through computational modelling, the practical implementation of an electrode and stimulation circuit remained to be demonstrated. Accordingly, the present work emphasizes design, simulation, optimization, fabrication, implementation and experimental validation a proof-of-concept transcutaneous electrical stimulation platform incorporating custom designed electrodes and discrete electronic stimulation circuitry.

# Chapter 2
# Literature Survey

Neuromodulation is the process of altering the process of neurons communicate with each other using physical or chemical stimuli through targeted approach. This energy delivered as electrical, magnetic, optical, or acoustic signals as shown in *Figure 3* an either increase or decrease the activity of neurons depending on how it is applied (Elyamany et al., 2021).

## 2.1. Introduction in Neuromodulation

The main aim of neuromodulation is to restore normal communication pattern of neurons in the diseased ones. By stimulating with either physical or chemical stimuli can treat disorders such as epilepsy, depression, Parkinson's disease, chronic pain, and Alzheimer's disease.




This literature review focuses both on biological and engineering principles of the above-mentioned methods identifying their strengths, limitations and relevance to the development of electric field stimulation (EFS) systems.

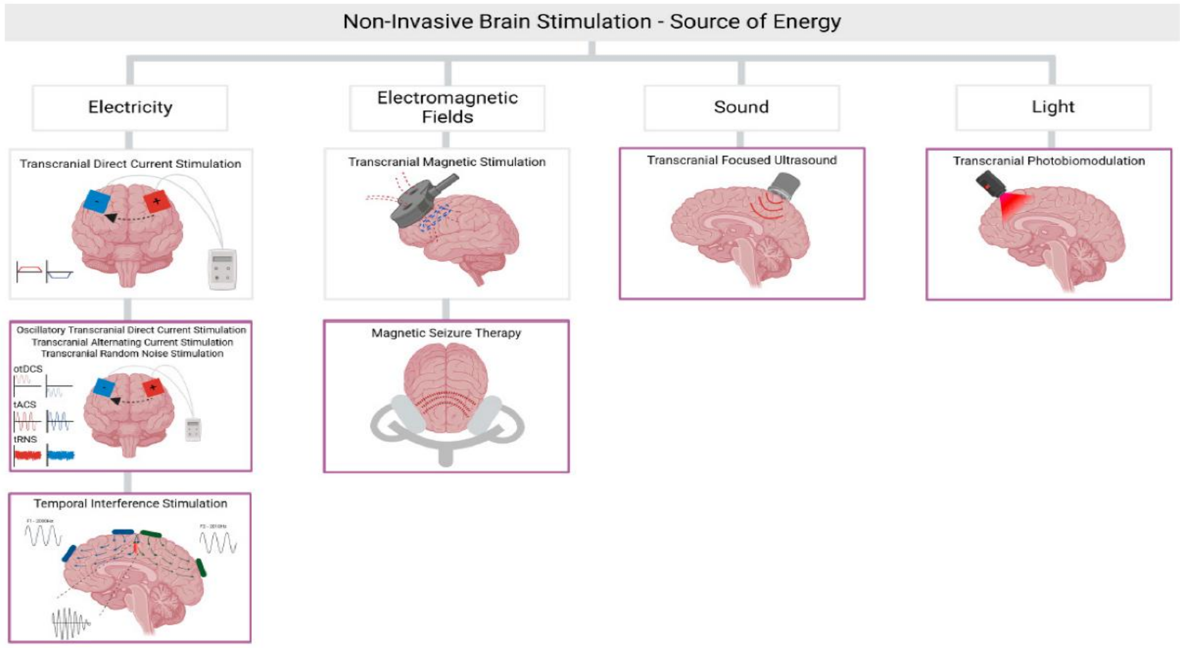
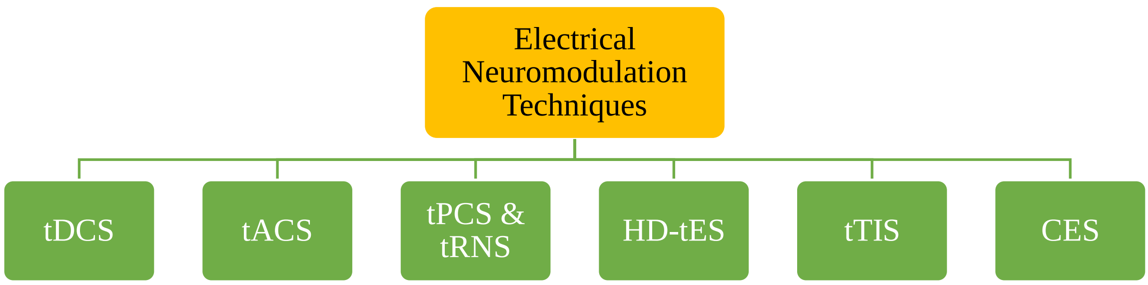



*Figure 3. Types of non-invasive brain stimulation*

*(Adapted from Brunoni et al., 2025).*

## 2.2. Electrical Neuromodulation Techniques

Electrical stimulation remains the most direct and well-understood form of neuromodulation because it interfaces with the inherent bioelectric nature of neurons. The below mentioned modalities summarize the major electrical modalities used in clinical and research applications.




### 2.2.1 Transcranial Direct Current Stimulation (tDCS)

Transcranial Direct Current Stimulation delivers weak constant current (1–2 mA) through the electrodes placed on the scalp at specific cortical sites (Thair et al., 2017). The anode is used to depolarizes neurons (i.e. make neurons more active), while the cathode tries to hyperpolarize (i.e. make them less active).

*Table 1. tDCS Summary*

| | |
|---|---|
| Applications | Improve motor learning, to treat depression, to manage chronic pain |
| Advantages | Minimal discomfort, Low cost, easy implementation. |
| Limitations | • Poor spatial resolution.<br>• Shallow depth penetration (~1 cm).<br>• Electrode polarization leading to skin irritation.<br>• Difficulty in achieving uniform cortical fields. |

tDCS in spite of all these drawbacks demonstrates that weak electric fields can modulate behaviour and brain activity.

### 2.2.2 Transcranial Alternating Current Stimulation (tACS)

tACS uses sinusoidal currents at frequencies from 0.1 Hz up to 2 kHz. The oscillatory nature of tACS allows it to "sync" brain waves with the applied frequency (Elyamany et al., 2021).

*Table 2. tACS Summary*

| | |
|---|---|
| Applications | Cognitive enhancement, sleep improvement and mood regulation |
| Advantages | Can target desired brain frequencies allowing to influence larger networks |
| Limitations | Less effective in deeper regions, uneven current flow due to tissue differences |

tACS has shown that how external circuit frequencies can be tuned to get in sync with the brain waves and treat disorders.

### 2.2.3 Transcranial Pulsed Current Stimulation (tPCS) and (tRNS)

tPCS delivers short pulses of current, while tRNS uses randomly fluctuating electrical noise. Concept of stochastic resonance is used where it states that randomness can enhance neural responsiveness. Studies suggests that tRNS can promote neuroplasticity more effectively compared to static or sinusoidal stimulation (Malekahmad et al., 2020).

*Table 3. tPCS & tRNS Summary*

| | |
|---|---|
| Applications | provides flexibility of waveform design |
| Limitations | there are lack of standardized protocols and limited implications of long-term impact. |

### 2.2.4 High-Definition Transcranial Electrical Stimulation (HD-tES)

HD-tES use multiple small electrodes (typically 4x1 configurations) to improve current focality and spatial precision (Ghafor et al., 2019). This setup increases current precision and minimizes the skin irritation. Finite-element modelling is used to concentrate electric fields at targeted cortical regions.

*Table 4. HD-tES Summary*

| | |
|---|---|
| Applications | Better localization and reduced skin irritation |
| Limitations | Requires more complex hardware and modelling; risk of current "leakage" between the electrodes |

### 2.2.5 Transcranial Temporal Interference Stimulation (tTIS)

tTIS was introduced by Grossman et al. (2017), tTIS uses superimposition of two high-frequency currents (e.g., 2 kHz and 2.01 kHz) that interfere within tissue, producing a low-frequency envelope (10 Hz) capable of modulating deep neurons (Violante et al., 2023).

*Table 5. tTIS Summary*

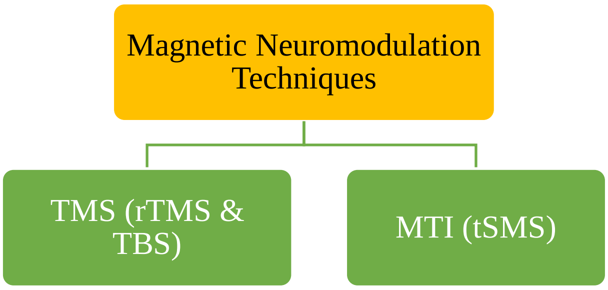
| | |
|---|---|
| Advantages | Deep non-invasive stimulation without stimulating other unwanted regions |
| Limitations | High precision required in electrode placement; no standardization of safety parameters. |

### 2.2.6 Cranial Electrotherapy Stimulation (CES)

It is an FDA approved procedure and also one of the earliest forms of safe and low-level electrical therapy which makes use of a very small alternating currents (< 1mA) where electrodes are placed on the earlobes or scalp. These waves mainly affect deeper brain structures involved in emotion and stress control. It is used in treating anxiety and insomnia.

## 2.3. Magnetic Neuromodulation Techniques

Magnetic stimulation influences brain activity by making use of applied magnetic field which induces tiny electric currents inside neural tissues.



### 2.3.1 Transcranial Magnetic Stimulation (TMS)

TMS works by making use of rapidly changing magnetic field (about 1-2Tesla) applied through the scalp, which induces eddy currents that depolarizes (i.e. activates) neurons (Chail et al., 2018).

*Table 6. tTMS Summary*

| | |
|---|---|
| Advantages | Non-invasive, can reach 2–3 cm deep |
| Limitations | Large, immobile and expensive equipment and limited focality |
| Clinical Use: | To treat depression, motor rehabilitation and OCD |

Repetitive TMS (rTMS) uses multiple pulses at specific frequencies, while Theta-Burst Stimulation (TBS) uses short bursts at 5 Hz to mimic natural brain rhythms (Noda et al., 2025).

Although effective, TMS systems are expensive, bulky and unsuitable for portable use or small-animal applications—highlighting a gap the gap for small electric field-based system.

### 2.3.2 Static and Magnetic Temporal Stimulation (MTI)

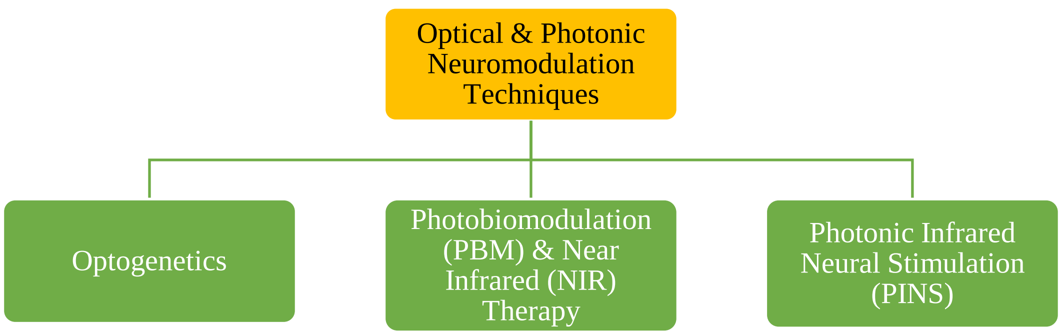
Transcranial Static Magnetic Stimulation (tSMS) uses steady magnetic fields to influence ion-channels and cellular activity, while Magnetic Temporal Interference (MTI) extends the concept of tTIS but uses magnetic fields rather than electrical field (Khalifa et al., 2023).

*Table 7. MTI Summary*

| | |
|---|---|
| Advantages | Has a potential for deep and non-invasive stimulation |
| Limitations | Challenges in miniaturization, complex device design and limited focality |

## 2.4. Optical and Photonic Neuromodulation Techniques

Optical and Photonic stimulation influences brain activity by making use of applied light and photons respectively.



### 2.4.1 Optogenetics

Optogenetics modifies specific neurons which to respond to light by inserting light-sensitive ion channels such as channel rhodopsin (Deisseroth, 2011). Light pulses of specific wavelengths are then used to control the neuron firing with milli second accuracy.

*Table 8. Optogenetics Summary*

| | |
|---|---|
| Advantages | High specificity and precision |
| Limitations | gene delivery and fiber optic implants require invasive procedure, not yet proved to be suitable for human use |

In spite of all these challenges, optogenetics has revolutionized neuroscience by allowing researchers to map and control brain circuits at the cellular level.

### 2.4.2 Photo biomodulation (PBM) and Near-Infrared (NIR) Therapy

In PBM red to near-infrared light (600–1100 nm) is used to stimulate mitochondrial activity and increase energy production in brain cells (Arany et al., 2023). Clinical studies have shown to improve in memory and blood flow in AD patients (Giménez et al., 2023).

*Table 9. PBM and NIR Summary*

| | |
|---|---|
| Advantages | Non-invasive, low side effects and supports cell repair |
| Limitations | Heat accumulation in dense tissue, and limited depth of penetration i.e. less than 2cm. |

### 2.4.3 Photonic Infrared Neural Stimulation (PINS) Therapy

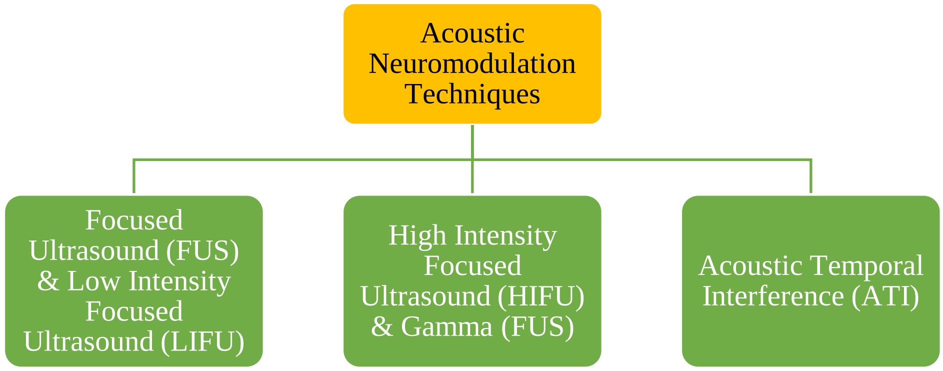
In PINS pulsed infrared light is used to gently heat the nerve membranes that depolarizes neurons (Xu et al., 2021).

*Table 10. PINS Summary*

| | |
|---|---|
| Advantages | High precision is achieved without electrical contact |
| Limitations | High equipment costs and need for cooling to avoid tissue damage |

## 2.5. Acoustic Neuromodulation Techniques

Uses specific sound waves to alter or modulate neural activity in the brain. The sound waves are used to create gentle mechanical pressure that interacts with the cell membrane and ion channel which affects the neurons firing.




### 2.5.1 Focused Ultrasound (FUS) and Low-Intensity Focused ultrasound (LIFU)

FUS uses sound energy to target the specific regions of the brain with very high precision. When used at low intensity (LIFU), it can focus within few millimetres which allows it to reach deep brain regions without the need of surgery (Zhang et al., 2024).

**Mechanism:** Acoustic pressure slightly changes how the ions move across the membranes altering the synaptic activity and neuron firing.

*Table 11. FUS and LIFU Summary*

| | |
|---|---|
| Applications | Tremor suppression, depression therapy, and temporary opening of Blood Brain Barrier to help in drug delivery |
| Limitations | Attenuation and reflection due to skull, temperature monitoring is required to prevent unwanted heating of tissue |

### 2.5.2 High-Intensity Focused ultrasound (HIFU) and Gamma Focused Ultrasound (gFUS)

In HIFU sound waves are used to remove or destroy small areas of the tissue without the need of surgery (Jolesz, 2009). In contrast gFUS uses much low-intensity periodic waves for safe modulation without damaging the tissue.

*Table 12. HIFU and gFUS Summary*

| | |
|---|---|
| Advantages | Both reach deep structures safely |
| Limitations | High intensity can cause heating or minor damage |

### 2.5.3 Acoustic Temporal Interference (ATI)

Similar to tTIS, ATI uses low-frequency interference within deep tissue through the intersection of two high-frequency ultrasound beams with slightly different frequencies (Jin et al., 2025). This ensures focal stimulation of deep brain regions without affecting any superficial tissue present in contact with.

*Table 13. ATI Summary*

| | |
|---|---|
| Advantages | Deep, precise and non-invasive stimulation |
| Limitations | Limited Focality |

## 2.6. Chemical and Genetic Neuromodulation Techniques

Chemical and genetic neuromodulation is done at molecular level.

Chemo genetic methods, such as Designer Receptors Exclusively Activated by Designer Drugs (DREADDs), involves turning neurons "on" or "off" by inserting special receptors into neurons that are controlled by specific drug. This method offers a cell-type specificity but lacks quick control and also requires gene delivery, which makes it invasive.

Thermogenesis and Magnetothermal methods make use of nanoparticles or magnetic fields to activate temperature-sensitive TRPV1 ion channels (Arany et al., 2023). These methods show potential for remote and selective neuron control but are still in early experimental stages and not yet well suited for clinical usage.

## 2.7. Sensory and Cognitive Entrainment

Another recent method of neuromodulation is Gamma Entrainment Using Sensory Stimulation (GENUS) it makes use of 40Hz auditory or visual signals to synchronize brain activity and improve memory in AD models (Black et al., 2024).

*Table 14. Sensory and Cognitive Entrainment Summary*

| | |
|---|---|
| Advantages | Non-invasive, low cost |
| Limitations | Limited Focality and indirect modulation compared to field-based methods |

## 2.8. Hybrid and Imaging-Guided Neuromodulation

Hybrid methods combine different types of stimulation or integrate them with imaging tools to increase precision and safety.

tES-fMRI integrates transcranial electrical stimulation with functional MRI to monitor how stimulation affects brain in real-time enabling closed-loop control where stimulation parameters automatically adjust based on brain responses (Violante et al., 2023).

Electromagnetic-Acoustic Imaging (EMAI) uses combination of EM excitation and ultrasound-based detection to map tissue conductivity which allows real-time monitoring and guiding of stimulation (Emerson et al., 2013).

Tumor-Treating Fields (TTFields) makes use of alternating electric fields to stop cancer cells from dividing. It shows that field-based stimulation has wider spread applications beyond neurology.

## 2.9. Hybrid and Imaging-Guided Neuromodulation

*Table 15. Summary of Comparison of Various Neuromodulation Methods*

| Modality | Depth (cm) | Focality | Safety | Cost | Clinical Status | Key Limitations |
|---|---|---|---|---|---|---|
| tDCS / tACS | 1–2 | Low | High | Low | Widely used | Poor depth, polarization |
| HD-tES | 2–3 | Moderate | High | Moderate | Emerging | Complex modeling |
| tTIS | 3–5 | High | Moderate | Moderate | Experimental | Calibration sensitive |
| TMS / rTMS | 2–3 | Moderate | Moderate | High | Established | Bulky, costly |
| Optogenetics | <1 | Very High | Variable | High | Research only | Invasive |
| PBM / NIR | 1–2 | Low | High | Low | Clinical | Limited penetration |
| LIFU / FUS | 5–8 | High | Moderate | High | Clinical trials | Skull attenuation |
| Chemogenetic | 1–5 | High | Moderate | High | Experimental | Invasive gene delivery |
| GENUS | N/A | Low | High | Low | Research | Indirect modulation |

## 2.10. Biological Relevance of Electric Field Stimulation

Application of electric fields show impact on cellular membranes, protein structures, and synaptic transmission. Key mechanisms include:

- Electro polarization: Which alters the membrane potential and influences ion-channel dynamics.
- Dielectrophoretic: Which moves and aligns the polar molecules inside the cells.
- Protein Alignment: Reorients dipole-rich structures like β-amyloid fibrils.

- Synaptic Plasticity Enhancement: Enhances neuron connectivity by modulating calcium channels.

At moderate intensities (~150 V/cm) fields modulate molecular aggregation without causing electroporation or heating (Saikia et al., 2019). Hence, controlled EFS represents a promising, mechanistically sound, and scalable neuromodulation method.

## 2.11. Summary of Literature Review

The literature clearly shows that while many neuromodulation techniques exist, electric-field-based approaches uniquely balance biological effectiveness, engineering simplicity, and translational potential.

Compared to magnetic, optical, or acoustic methods, EFS offers:

- Lower energy requirements.
- Direct action on natural brain electrophysiology.
- Compact, low-cost affordable device design.
- Compatibility with small-animal and human applications.

Despite these benefits, most current research remains limited to simulation and invitro testing.

A significant gap exists between simulation-based design optimization and practical hardware realization. Relatively few studies have addressed the engineering challenges associated with converting optimized stimulation concepts into lightweight, experimentally validated hardware systems suitable for small-animal applications. Challenges related to electrode fabrication, miniaturization, power management, stimulation waveform generation, and system integration remain insufficiently explored.

Furthermore, advanced safety mechanisms such as current monitoring, impedance monitoring, fault detection, and charge-balancing strategies are often discussed independently from hardware implementation, resulting in limited understanding of how such features may be incorporated into compact stimulation platforms.

Therefore, there is a need for a practical proof-of-concept stimulation system capable of translating simulation-derived design constraints into a functional hardware prototype while maintaining simplicity, portability, and experimental feasibility. The present work addresses this need through the development of a lightweight electrical stimulation platform involving electrode design refinement, circuit architecture evaluation, prototype realization, and experimental validation.

# Chapter 3
# Overview of Prior Work

Previous research conducted by Pooja (2024) in the laboratory focused on investigating the feasibility of non-invasive electrical stimulation for applications related to Alzheimer's disease using computational modelling techniques. The primary objective of these studies was to understand the electric field distribution generated within biological tissue under different electrode configurations and stimulation conditions.

A multilayer rat head model as shown in *Figure 4* consisting of skin, skull, cerebrospinal fluid (CSF), cortex and white matter was developed using COMSOL Multiphysics. The model was used to evaluate the propagation of electric fields through the various tissue layers and to investigate the influence of electrode geometry and placement on the resulting field distribution.

The geometric model included the following.

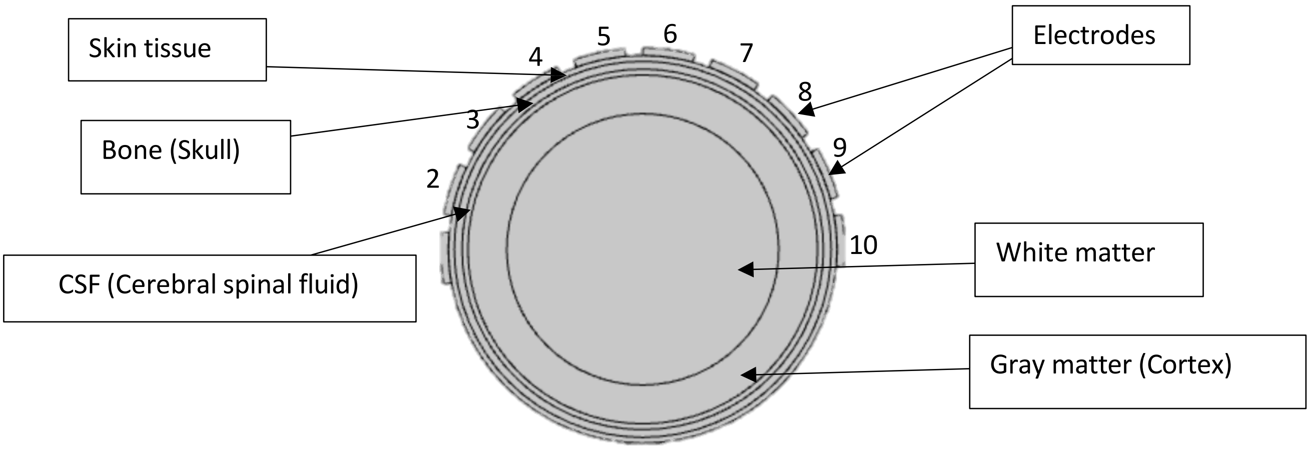



*Figure 4. Parts of Human Head considered for COMSOL Simulations*

***Note.*** *Adapted from* Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) Using Electric Field (EF)*, by Pooja, 2024, Indian Institute of Technology Guwahati.*

Each tissue layer was assigned realistic electrical properties:

- Skin: 0.465 S/m conductivity
- Skull: 0.01 S/m
- CSF: 1.79 S/m
- Cortex: 0.275 S/m
- White matter: 0.126 S/m

The model was energized by surface electrodes placed symmetrically on the skull, and the resulting field distribution was analysed. The study achieved a uniform cortical electric field of approximately 150 V/cm without excessive penetration into deeper white-matter regions.

This field strength was identified from in-vitro studies showing its effectiveness in delaying β-amyloid aggregation (Saikia et al., 2019).

## 3.1. Parametric Optimization

Subsequent analyses performed by Pooja (2024) focused on optimizing electrode geometry as shown in *Figure 5*. Based on several optimizations of several width-to-gap (W/G) ratio it revealed that the best uniformity was obtained when W ≈ G, meaning the electrode width and space between them was almost equal.

This configuration even with less voltage requirement was able to produce strong cortical field. It was determined that a 25 V potential difference is sufficient to generate the desired field safely.

A 2D advanced model was created, adding layers such as:

- Hair layer (to simulate impedance of the fur barrier),
- Polyimide layer representing the flexible PCB substrate,
- Realistic curvature derived from 3-D-printed skull models.

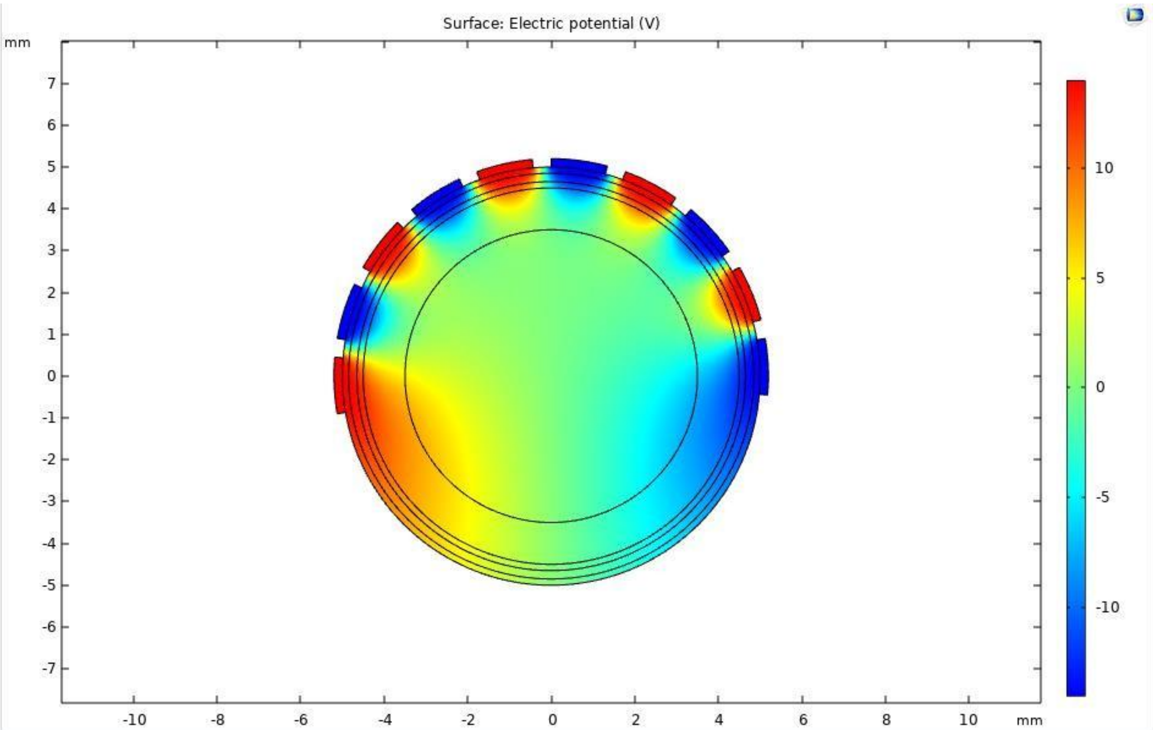



*Figure 5. Electric Field Distribution for 10 Electrodes*

***Note.*** *Adapted from* Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) Using Electric Field (EF)*, by Pooja, 2024, Indian Institute of Technology Guwahati.*

Suggested:

- Total electrodes: 10
- Each electrodes thickness: 0.2 mm
- Sector angle: 15°
- Spacing between electrodes: 5°
- Total covered periphery of head: 200°

The results showed that polyimide did not distort the electric field but in fact provided better flexibility and surface contact, confirming its suitability for electrode fabrication.

Several electrode configurations and geometric parameters were systematically explored through the stimulation studies. These investigations established a number of important design constraints that served as the foundation for the present work. The key findings included:

- A ten-electrode configuration was identified as a suitable arrangement for generating the desired electric field distribution.
- An angular coverage of approximately 256° was found to provide favourable field characteristics within the target region.
- A width-to-gap ratio of 1:1 was determined to be an important geometric parameter for electrode design.
- An operating voltage of 25 V was demonstrated to be capable of generating the required electric field distribution within the simulation environment.

## 3.2. Identified Research Gaps

These simulation studies provided valuable insight into the theoretical feasibility of the proposed stimulation approach and established a preliminary framework for electrode design and stimulation requirements. However, the investigations were primarily focused on computational modelling and electric field analysis.

Several practical challenges associated with practical hardware realization remained unaddressed. The simulation studies did not investigate the manufacturability of the proposed electrode geometries, the development of compact stimulation circuitry, power management requirements, prototype fabrication, experimental validation or the integration of monitoring and safety mechanisms. Furthermore, simulation environments cannot fully capture practical considerations such as fabrication tolerances, component limitations, electrode-tissue interactions and hardware performance under real operating conditions.

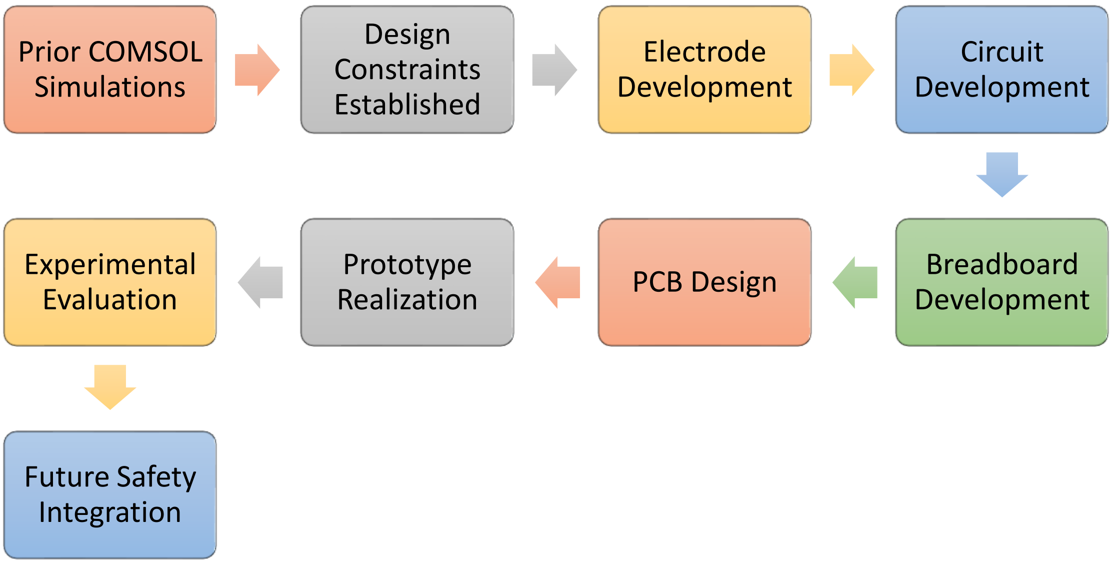
Consequently, a significant gap existed between the simulation derived design recommendations and the realization of a practical stimulation platform suitable for experimental evaluation. The present work addresses this gap by translating the simulation-based design constraints into manufacturable electrodes structures, practical stimulation circuitry, proof-of-concept prototypes and experimentally validated hardware implementations.

Prior Work done by Pooja (2024):

- COMSOL simulation
- Field distribution studies
- 10-electrode concept
- 150 V/cm target
- 256º angular coverage

## 3.3. Objectives of the Current Work

- Electrode design
- Electrode optimization
- Circuit development
- PCB design
- Prototype realization
- Experimental verification



# Chapter 4

# Electrode Design and Development

In transcutaneous electrical stimulation systems the applied stimulation waveform generated by the electronic circuitry is delivered to the target tissue through the electrode interface. Since the electric field distribution within biological tissue is strongly dependent on electrode geometry, careful optimization of electrode dimensions and layout is required to ensure uniform field penetration and adequate stimulation coverage. Parameters such as electrode width, spacing, active area and overall geometry directly affect the magnitude and distribution of the induced electric field.

## 4.1. Electrode Design Iterations

To investigate these effects, multiple electrode configurations were considered and evaluated during the design process. The overall idea was to implement an electrode on skull which more or less represents the *Figure 6.*

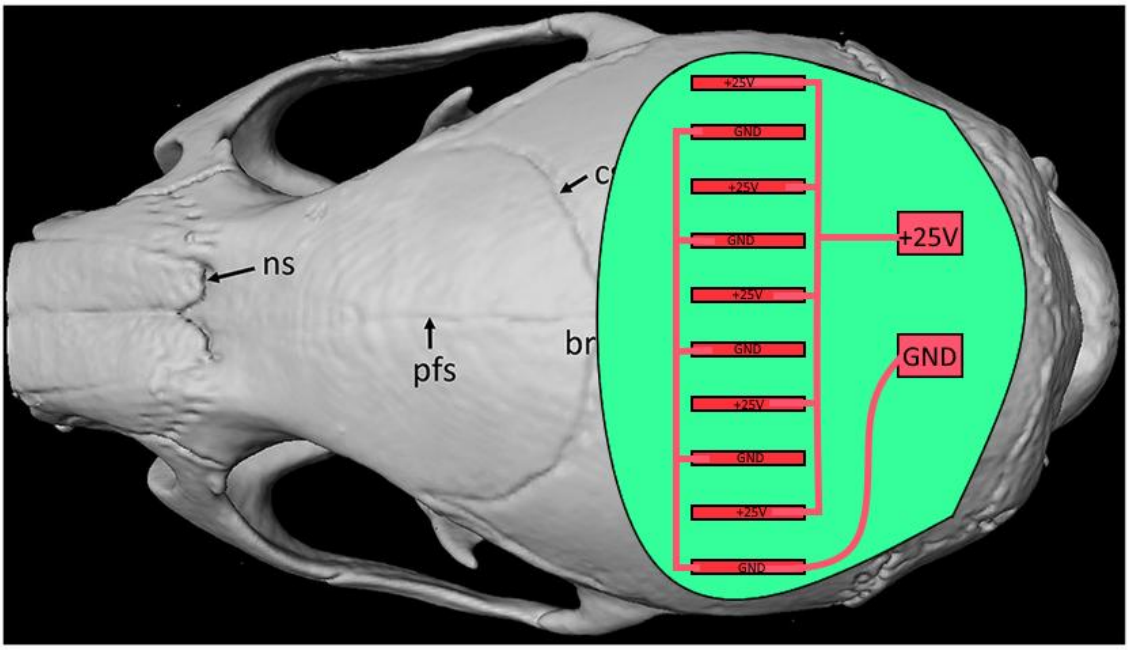



*Figure 6. Diagram depicting the design of electrode to be achieved*

Pooja (2024) COMOSL Model Design Specifications:

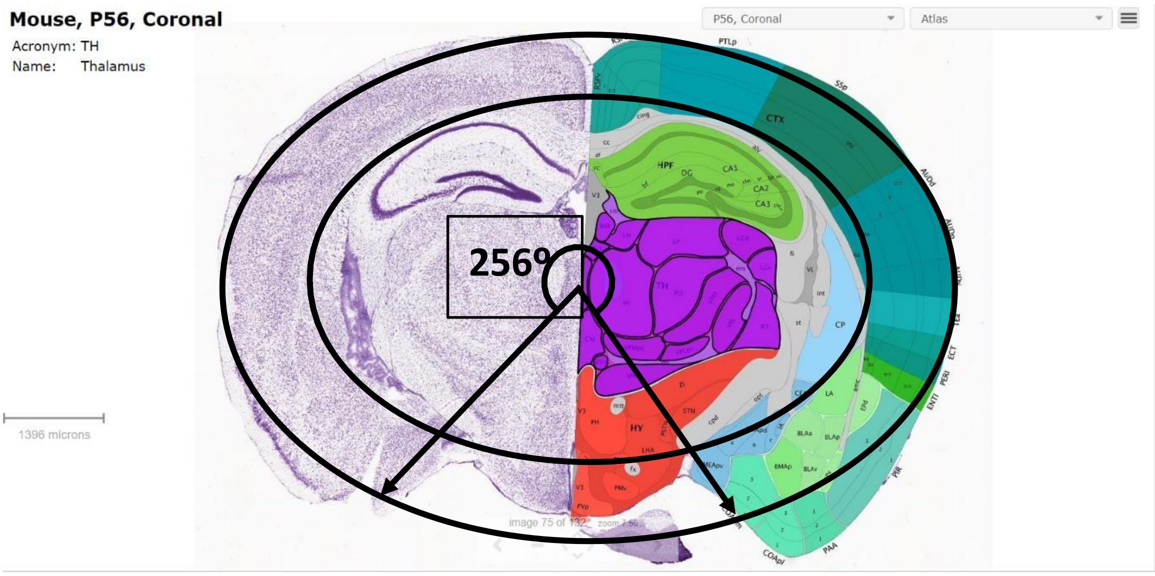



*Figure 7. Model of mouse brain depicting the location of thalamus and angle of coverage required for effective stimulation*

*Degrees of stimulation effective to stimulate the brain effectively considered in previous phase of work for COMSOL Simulations (Adapted from Allen Brain Institute)*

According to *Figure 7* we require an angle of 256º for effective stimulation of Thalamus, which is considered as responsible for Alzheimer's.

### Convert COMSOL Simulation → Fabrication-Ready Electrode Design

Design a flex-PCB electrode strip that wraps around ~200° of the mouse skull.

From previous phase work:

*Table 16. Thickness of various layers considered for COMSOL simulations*

| Layer | Thickness |
|---|---|
| Skin tissue | 0.15 mm |
| Skull bone | 0.20 mm |
| CSF | 0.15 mm |
| Gray matter (cortex) | 1.0 mm |
| White matter | Radius = 3.5 mm |
| Total radius | 5.0 mm |

This suggests a total head diameter = 10 mm in the simulated model.

That matches the dimensions of small rat head model.

1. Arc Length

$$Total\ arc\ (200°) = Arc\ length = \theta \times r = (200/360) \times 2\pi r$$

Use radius $r$ = 5 mm:

$$2\pi r = 2 \times 3.1416 \times 5 = 31.416$$
$$31.416 \times 0.5555 \approx 17.45\ \text{mm}$$

So, electrode strip must cover ~17.5 mm in length around the skull.

2. Per-Electrode Arc Length
Each electrode sector (15° of the circumference):

$$Arc\ per\ electrode = (15/360) \times 2\pi r$$
$$0.041666 \times 31.416 = 1.309\ \text{mm}$$

Each electrode width (arc-wise) ≈ 1.3 mm

3. Spacing between electrodes (5°):

$$Spacing\ arc = 5/360 \times 2\pi r = 0.436\ \text{mm} \approx 0.44\ \text{mm}$$

Final Electrode Dimensions as per COMSOL Simulations:

*Table 17. Summary of Electrodes Dimensions from above calculations*

| Parameter | Value |
|---|---|
| Electrode width (arc direction) | 1.3 mm |
| Gap between electrodes | 0.44 mm |
| Electrode height (toward crown) | 2 mm |
| Thickness | 0.2 mm (given) |
| Number of electrodes | 10 |
| Total arc length | ~17.5 mm |
| Substrate type | Flexible Polyimide PCB |

$$For\ 1°\ of\ arc = (2\pi r/360) = 31.416/360 = 0.08727$$

To Match COMSOL considered calculations exactly

$$Width : Gap = 3 : 1\ \ (because\ 15° : 5°).$$

Per electrode:

$$Width(arc) = 15° \times 0.08727 = 1.309\ \text{mm}$$
$$Gap(arc) = 5° \times 0.08727 = 0.436\ \text{mm}$$

For 10 electrodes

$$Total\ arc\ length(tip\text{-}to\text{-}tip) = 17.45\ \text{mm}$$

### 4.1.1 Version 02

The *Figure 8* represents electrode dimensions having length of 1.95 cm in length and 0.55 cm in breadth. The above considerations were done to arrive at these dimensions, it has effective stimulation of 1.66 cm in length and 0.2 cm in breadth.

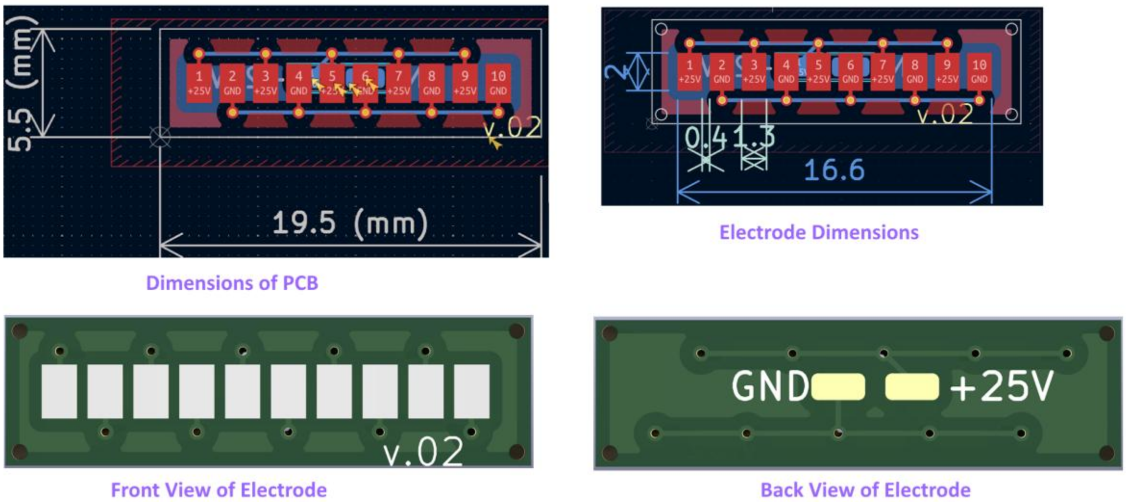



*Figure 8. Version 02 of electrode design*

### 4.1.2 Version 03

As per final conclusion drawn from previous work

$$Width : Gap = 1 : 1$$

Width to gap ratio needs to be unity.

Keeping 10 electrodes over 200°.

If we keep the period = 20° so the strip length and count don't change, but split it evenly per electrode:

$$Width(arc) = 10° \times 0.08727 = 0.873\ \text{mm}$$
$$Gap(arc) = 10° \times 0.08727 = 0.873\ \text{mm}$$

Array:

- Total arc length = 17.45 mm (unchanged)
- Pad height = 2.5–3.0 mm
- Stack thickness ≈ 0.2–0.3 mm

Considering worst case scenario available length of the electrode to effectively place on the rat head is 11 mm then:

To place 10 electrodes in 11 mm, we need

$$(Pad\ Width\ +\ Gap) = (11\ \mathrm{mm})/10\ electrodes = 1.1\ \mathrm{mm}\ per\ electrode\ slot$$

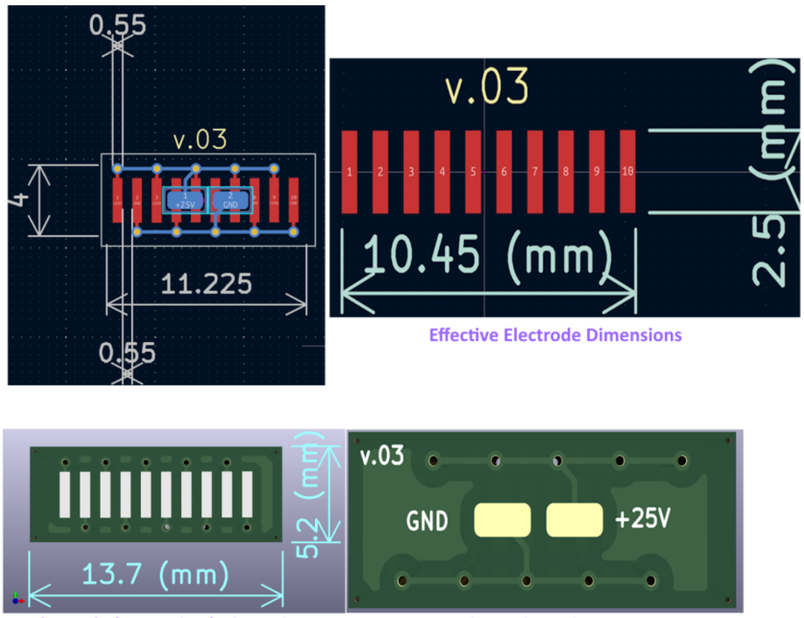
Since,

$$width\ =\ Gap\ \ ==>\ \frac{1.1}{2}\ \approx\ 0.55\ \mathrm{mm}$$

*Table 18. Summary of Electrode dimensions considered*

| Parameter | Final Value |
|---|---|
| Electrode count | 10 |
| Electrode width | 0.55 mm |
| Gap between pads | 0.55 mm |
| Total length used | 11 mm |
| Electrode height (radial) | 2.5–3 mm (you choose) |
| Substrate width | 6–8 mm |

The *Figure 9* represents electrode dimensions having length of 1.37 cm in length and 0.52 cm in breadth. The above considerations were done to arrive at these dimensions, it has effective stimulation of 1.45 cm in length and 0.25 cm in breadth.



Effective Electrode Dimensions

Electrode (Front View) Dimension

Electrode Back View

*Figure 9. Version 03 of electrode design*

### 4.1.3 Version 04

*Figure 10* adapted from Pooja's (2024) work show the dimensions of 3D printed brain about <1cm in sagittal plane and 3D printed skull ~1cm in coronal plane.

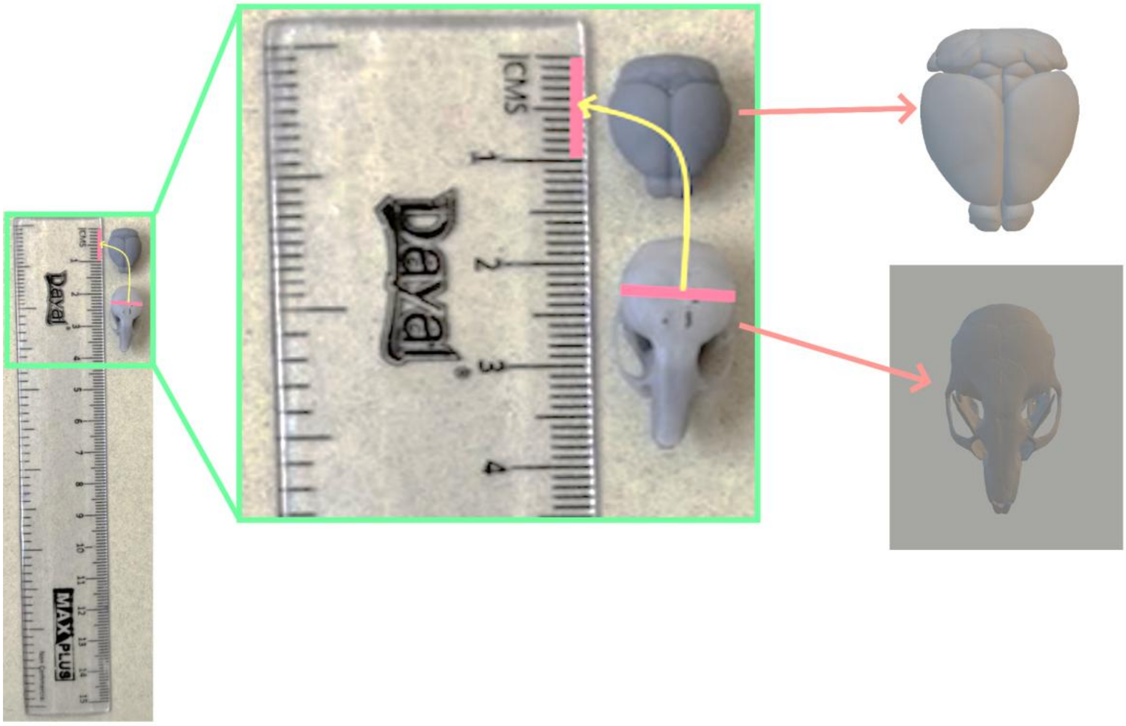


*Figure 10. Depicts the dimensions of 3D printed skull and brain*

***Note.*** *Adapted and modified from* Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) Using Electric Field (EF)*, by Pooja, 2024, Indian Institute of Technology Guwahati.*

Worst case possible scenario considered in case of small rat available for testing

Total Available Length = 6 mm

Number of Electrodes = 10

$$Width : Gap = 1 : 1$$

Each electrode slot:

$$\frac{Total\ Length}{Number\ of\ Electrodes} = \frac{6\text{ mm}}{10} \approx 0.6\text{ mm}$$

Since, width = gap, we split 0.6 mm evenly:

$$\frac{0.6}{2} \approx 0.3\text{ mm}$$

Substrate width with respect to final electrode design will be 8 mm.

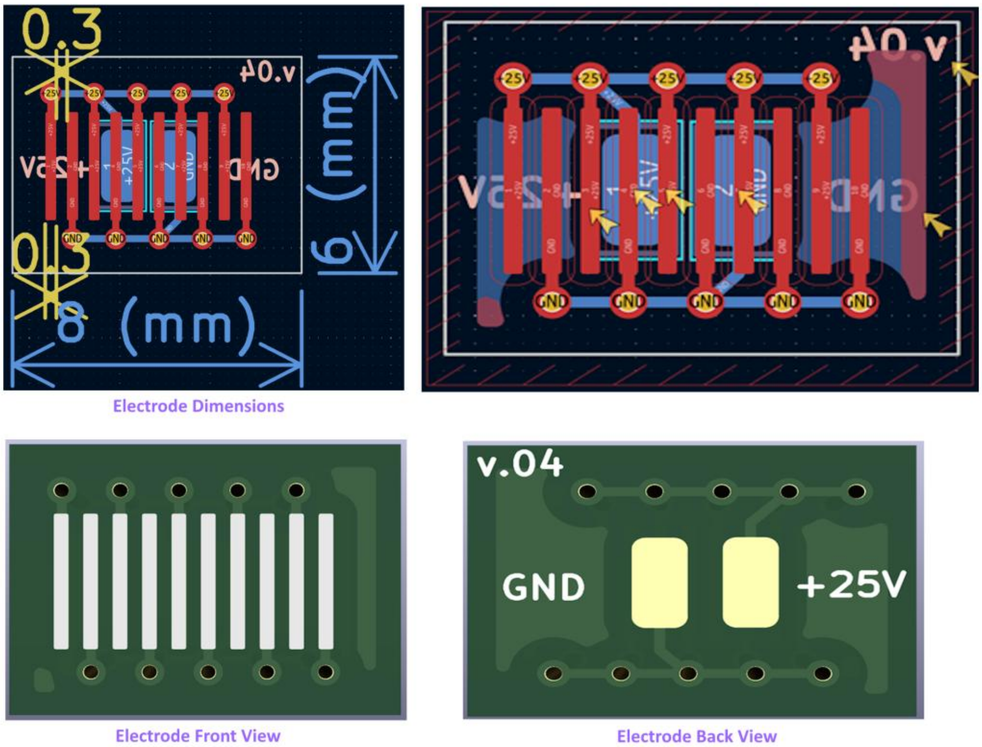



*Figure 11. Version 04 of electrode design*

The *Figure 11* represents electrode dimensions having length of 0.8 cm in length and 0.6 cm in breadth.

### 4.1.4 Version 05

This design was finalized considering the restrictions of availability to equipment's to manufacture 2-layer PCB design, here both electrodes and solder pad are on the same side, avoiding drilling of holes from one surface to the other surface. This design was done considering the fabricated skull dimensions.

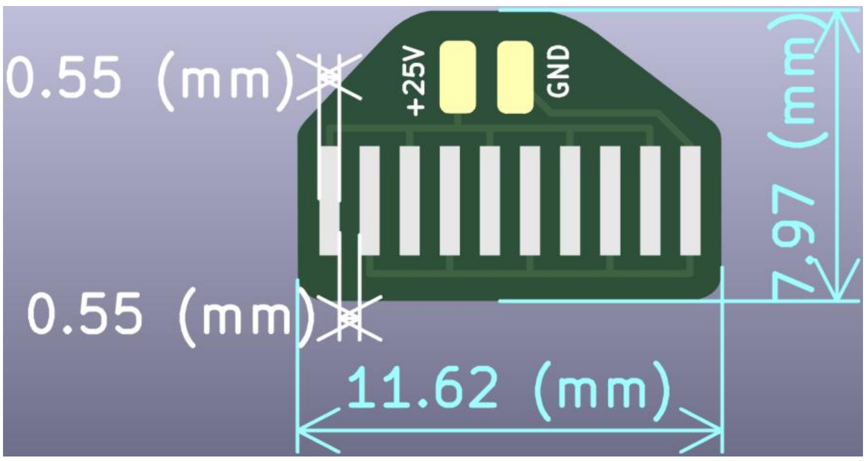
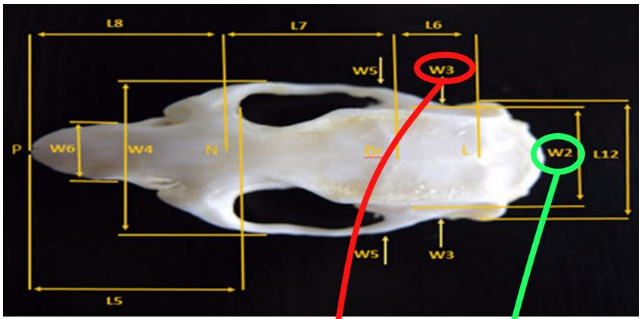



*Figure 12. Version 05 of electrode design*

The *Figure 12* represents electrode dimensions having length of 1.16 cm in length and 0.79 cm in breadth.

Practical Realization considering Adult Lab Rat (Practical approach)




| Measurements taken from the skull | Mean±SD (in cm) |
|---|---|
| W2 | 1,54 ± 0,05 |
| W3 | 1,74 ± 0,10 |

*Figure 13. Measurement points on Rat Skull Dorsal View*

*(Adapted and modified from Meryem et al., 2023)*

*Figure 13* represents the work done by Meryem et al. (2023), which investigated the cranial dimensions of lab rats. The findings were considered during electrode design process to ensure proper placement on the rat skull. In addition, the presence of the ears imposes further geometric constraints, influencing the selection of electrode dimensions and positioning to achieve effective electrical stimulation while maintaining proper contact with the rat head.

### 4.1.5 Final Version of Electrode Design

150 V/cm electric field needs to be generated inside the brain using only 25 V applied voltage with 10 electrodes placed over 200° around the head ➔ 25 V naturally produces 150 V/cm.

$$E = \frac{V}{d}$$

$$150\,\frac{\text{V}}{\text{cm}} = \frac{25\ \text{V}}{d} = 1.67\ \text{mm}$$

So, the gap between the opposite polarity electrodes must be of 1.67 mm in order to achieve 150 V/cm.

Since, we only need to cover 200° with 10 electrodes ➔ this creates the space limitation i.e. we cannot fit 10 electrodes with a 1.67 mm gap unless we shrink the size of electrode considering limitations of the dimensions of the rat skull. Hence, I chose to keep the ring compact and scale the electrodes.

A ring diameter of 10–12 mm naturally can be considered to ensure that without hanging off the skull or touching the unwanted areas.

If the ring diameter is considered somewhere near 10.6 mm, then arc length becomes:

$$L_{arc} = \frac{200}{360} \times \pi \times 10.6 \approx 18.5 \text{ mm}$$

Within this 18.5 mm we need to place 10 electrodes and 10 gaps.

$$\frac{18.5 \text{ mm}}{10} = 1.85 \text{ mm}$$

Since,

$$Width \approx Gap$$

$$\frac{1.85 \text{ mm}}{2} \approx 0.93 \text{ mm}$$

0.93 mm is electrode width and 0.93 mm is gap between adjacent electrodes.

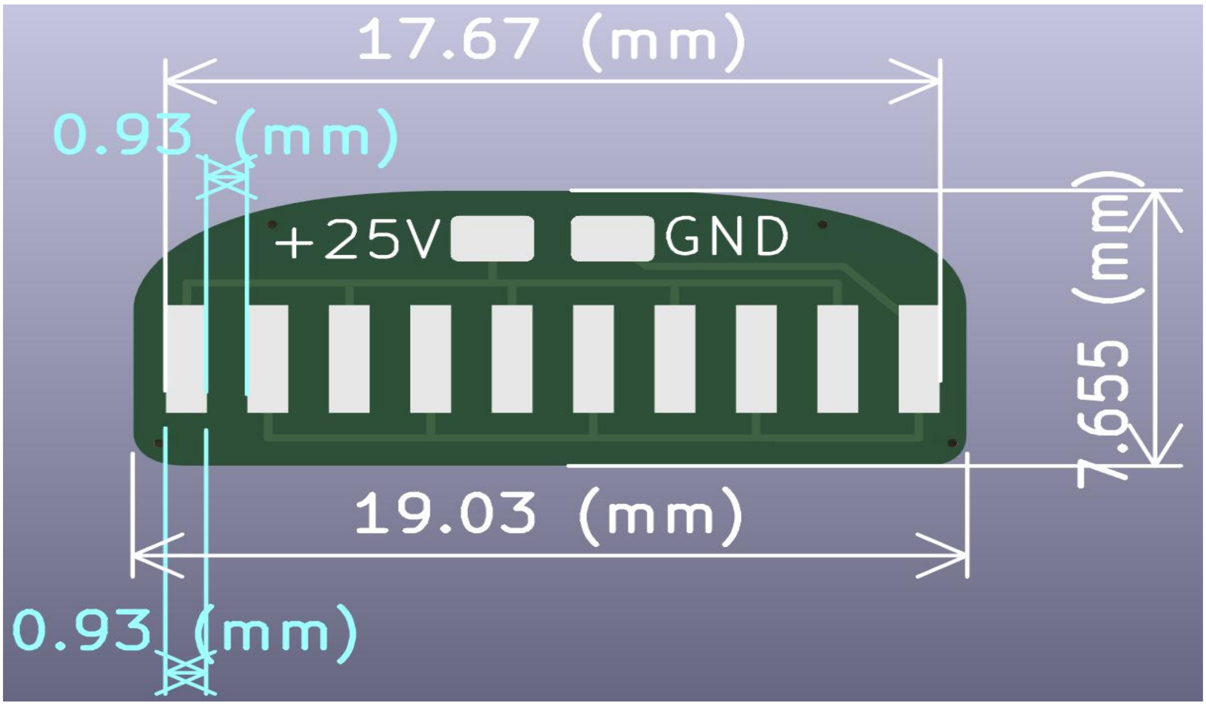



*Figure 14. Final Electrode design consideration based on practical approach.*

The *Figure 14* represents electrode dimensions having length of 1.9 cm in length and 0.765 cm in breadth. The above considerations were done to arrive at these dimensions, it has effective stimulation of 1.76 cm in length.

## 4.2. Electrode Design Considered for Prototyping

Although multiple electrode geometries were developed, the spiral configuration was selected for proof-of-concept implementation due to its ability to preserve the simulation derived width-to-gap relationship while offering a practical balance between available simulation area, mechanical robustness and manufacturability.

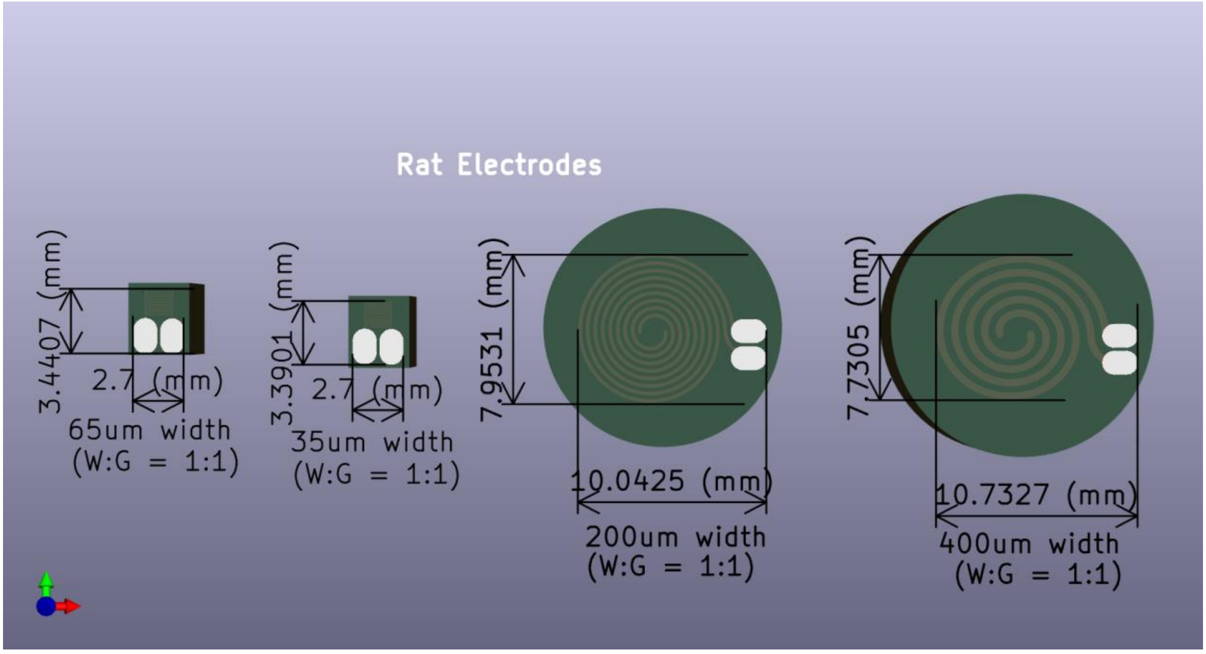



*Figure 15. Electrode design considered for prototyping.*

The *Figure 15* represents electrode dimensions that were designed and considered while prototyping, the first two electrodes are of Interdigitated Electrodes (IDE) and next two electrodes are of type spiral electrodes.

## 4.3. Electrode Fabrication for Prototyping

Due to the temporary unavailability of PCB fabrication facilities during the course of the study, an alternative fabrication approach was adopted to obtain a proof-of-concept electrode for preliminary experimental evaluation.

Alternative Method:

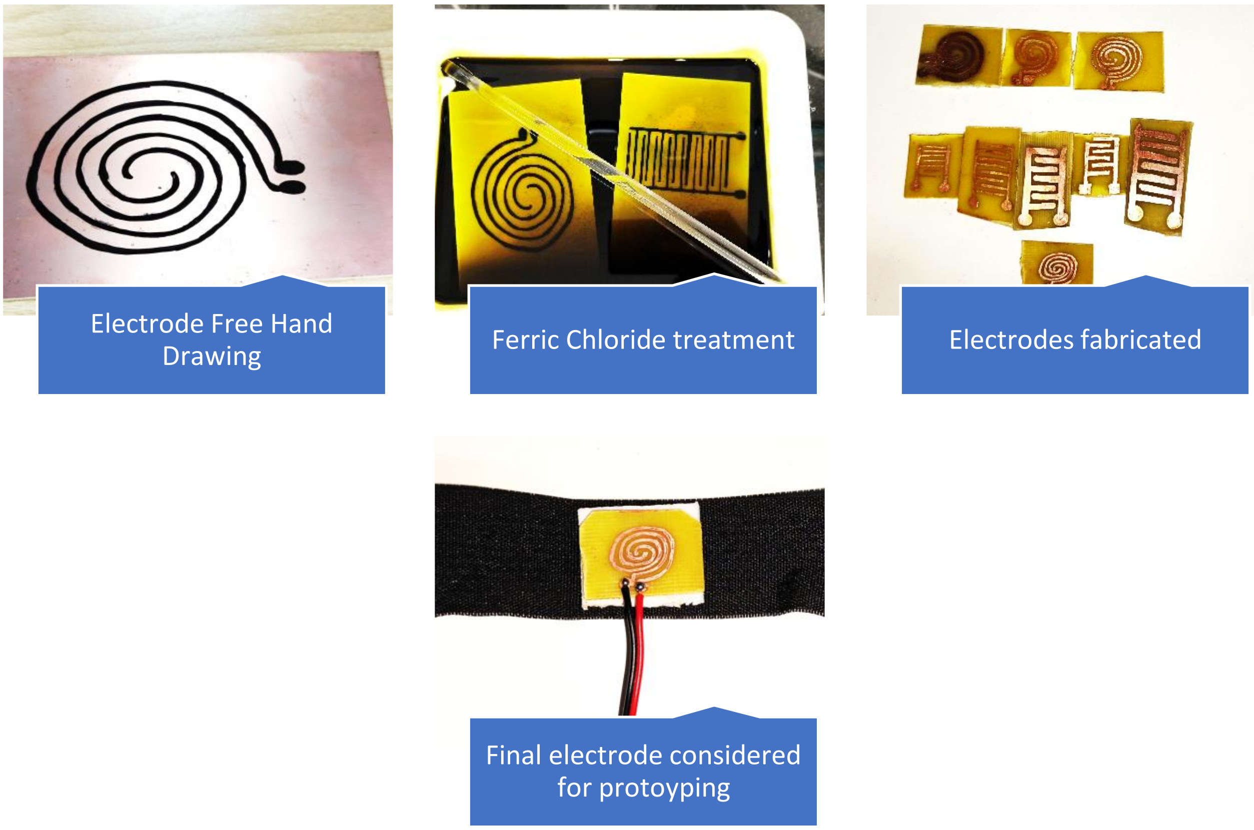



*Figure 16. Electrode design considered for prototyping.*

Since the electrode geometry was manually drawn prior to etching, slight deviations from the intended width-to-gap ratio and trace uniformity were unavoidable.

Consequently, the fabricated prototype should be considered a proof-of-concept realization rather than a geometrically exact representation of the PCB design.

The observed waveforms remained largely consistent with unloaded conditions, with only minor variations in frequency and amplitude.

# Chapter 5
# Circuit Design, Development and Comparative Evaluation

## 5.1. Introduction to Circuit Design

The successful realization of a non-invasive electrical stimulation platform requires not only an optimized electrode configuration but also reliable stimulation circuit capable of generating the desired excitation waveform. While simulation studies can establish the required operating voltage and electrode geometry, practical implementation requires the development of suitable hardware capable of delivering the stimulation signal under the real operating conditions.

Several factors were considered during the development of the stimulation circuitry. This included portability, component count, power consumption, ease of implementation, manufacturability, reliability and future integration of monitoring and protection mechanisms. Since the intended application involves a compact stimulation platform suitable for small animal studies, minimizing the circuit complexity and hardware footprint was considered an important design objective.

Rather than directly implementing the single architecture, multiple stimulation approaches were investigated and experimentally evaluated. Various oscillator topologies, switching circuits and protection-oriented architectures were designed, simulated and implemented using breadboard-based prototypes. Oscilloscope measurements were subsequently used to evaluate waveform quality, stability and practical feasibility.

The design process ultimately resulted in the selection of a compact MOSFET-based stimulation architecture that provided and acceptable balance between simplicity, performance and implementation feasibility. This chapter presents the evolution of the stimulation circuitry, the rationale behind each design iteration, experimental observations and the justification for selecting the final implementation.

## 5.2. Design Requirements

The stimulation circuit was developed based on the requirements establishes from previous simulation studies and practical implementation constraints.

The primary electrical requirement was the generation of a stimulation waveform with an operating voltage of approximately 25 V. Since the platform was intended to operate from a rechargeable battery source, a voltage conversion stage was required to generate the desired simulation voltage from a low voltage supply.

In addition to the electrical requirements, several practical constraints influenced the circuit design process:

- Compact and lightweight architecture suitable for future animal mounted implementation.
- Low component count to simply fabrication and assembly.
- Battery powered operation.
- Ease of troubleshooting and experimental validation.
- Compatibility with PCB based implementation.
- Potential for future integration of protection and monitoring circuitry.
- Reliable generation of alternating simulation signals.

These requirements guided the development and evaluation of the various stimulation architectures discussed in the following sections.

## 5.3. Power Conversion Architecture

### 5.3.1 Linear Voltage Regulation (LDO or Series Pass)

Linear regulators can only step down the voltage not step up. Hence to achieve 25 V output, the input voltage must be higher than 25 V. Since only one 3.7 V cell is used, series connection of many 3.7 V would be required which is not desirable in our case considering weight and space constraints.

*Table 19. Advantages v/s Disadvantages of Linear Regulators*

| Advantages | Disadvantages |
|---|---|
| Very low noise | Cannot step up voltage (only step down) |
| Very simple design | Would require multiple batteries in series |
| Minimal components | Not feasible with a single-cell Li-ion battery |

Conclusion: Hence this approach was rejected immediately.

### 5.3.2 Charge Pump Voltage Multiplier

A charge pump uses capacitors to multiply voltage in stages. A single stage can output 3.7 V, hence we require almost 7 stages to reach desired voltage level of 25 V.

*Table 20. Advantages v/s Disadvantages of Charge Pump voltage Multipliers*

| Advantages | Disadvantages |
|---|---|
| No inductors required | Very poor efficiency |

| | |
|---|---|
| Can be integrated in CMOS | Requires approx. 7 stages and large area |
| Works for low power circuits | High output ripple and poor voltage regulation |

Conclusion: Hence due to multiple stages are required to desirable voltage level which in turn requires more space, hence this is also ruled out.

### 5.3.3 Transformer based step-up converter

Transformer based topologies like flyback converters allow strong step-up ratios and electrical isolation.

*Table 21. Advantages v/s Disadvantages of Transformer based step-up converter*

| Advantages | Disadvantages |
|---|---|
| Can provide isolation | Transformer increases PCB height and weight |
| Can achieve higher boost ratios | Requires EMI optimized design |
| Efficient if well designed | Too bulky for rat mounted application |

Conclusion: Even though it looks fine with respect to electrically due to size and weight it is unsuitable for this application.

### 5.3.4 Final Choice of Circuit Design using DC-DC Boost Converter

A DC-DC boost converter IC such as the MT3608 provides the most practical solution. It makes use of an inductor, switching MOSFET, diode and a feedback network to step up voltage from 3.7 V to 25 V efficiently.

Why MT 3608 was chosen:

- Compact and widely available
- 80–95% efficiency
- Ideal for small PCBs

*Table 22. Advantages v/s Disadvantages of Boost Converter MT-3608*

| **Advantages** | **Disadvantages** |
|---|---|
| Can step up 3.7 V to 25 V reliably | Generates switching noise |

| | |
|---|---|
| High efficiency | Requires correct selection of inductor and capacitor, diode |
| Small inductor footprint | Schottky diode losses reduce efficiency |
| Readily available IC | Synchronous version more complex to layout |

**Power Consumption and Runtime Calculations (Approximations)**

The power required at the output of the boost converter is calculated using:

$$P_{out} = V_{out} \times I_{out} = 25\ V \times 10\ mA = 0.25\ W$$

To determine the input power with an assumed efficiency of 85%:

$$P_{in} = \frac{P_{out}}{\eta} = \frac{0.25}{0.85} \approx 0.294\ W$$

The input current drawn from the Li-ion battery is:

$$I_{in} = \frac{P_{in}}{V_{in}} = \frac{0.294}{3.7} \approx 0.0795\ A \approx 80\ mA$$

The total usable energy of a 1500 mAh cell is:

$$E_{battery} = 3.7\ V \times 1.5\ Ah = 5.55\ Wh$$

Finally, the estimated runtime of the device is:

$$\text{Runtime} = \frac{5.55\ Wh}{0.294\ W} \approx 18.8\ \text{hours}$$

This confirms that the designed boost converter can support approximately 18 hours of continuous operation. *Figure 17* shows the circuit diagram of Boost Converter design using MT3608 IC

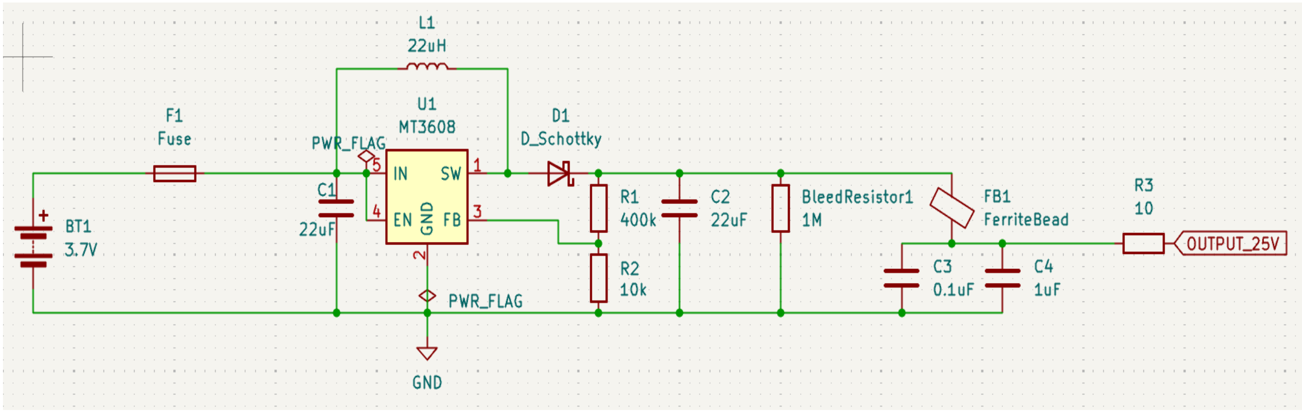



*Figure 17. Boost Converter Circuit Diagram*

## 5.4. Initial System Level Architecture

Before proceeding with discrete circuit implementation, a comprehensive stimulation system architecture was developed as shown in *Figure 18* to identify all subsystems required for a wearable electrical stimulation platform.

The proposed architecture integrated battery charging, battery protection, voltage regulation, pulse generation, output driving circuitry, current monitoring and safety shutdown mechanisms within a single design framework.

The objective was to create a complete stimulation platform capable of generating controlled biphasic stimulation pulses while simultaneously ensuring user safety and reliable operation.

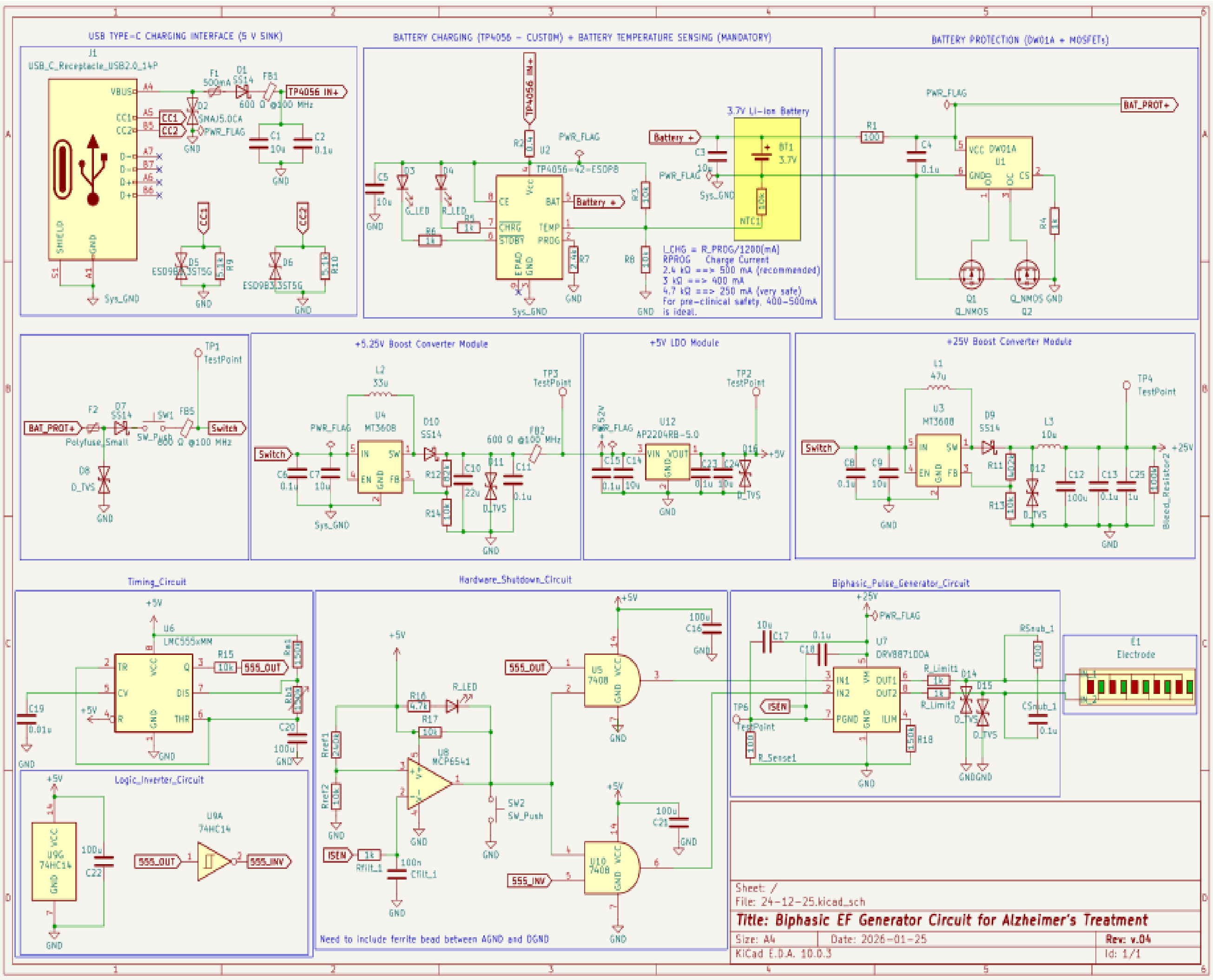



*Figure 18. initial System Level Architecture.*

While technically feasible implementation of the complete architectures would have been significantly increased development complexity and number of components during the proof-of-concept phase.

Consequently, a simplified design methodology based on discrete oscillator architectures was adopted to experimentally evaluate stimulation waveform generation before integrating advanced protection and monitoring functions and avoiding stages like Low Dropout Regulator for noise free operation of the ICs.

This decision enabled rapid hardware development, simplified troubleshooting and accelerated experimental validation.

The knowledge gained from the investigation of the complete architecture was subsequently utilized during the development of the final stimulation platform and current monitoring subsystem.

## 5.5. Preliminary Astable Multivibrator using BC547

As an initial step toward the development of the stimulation circuitry, a simple transistor based astable multivibrator was designed using BC547 NPN transistors. The primary objective of this implementation was to verify the generation of self-sustained oscillations and complementary output signals using a minimal component architecture.

*Figure 19* shows the breadboard implementation of the BC547 NPN transistors used to build astable multivibrator, the circuit consisted of two cross-coupled BC547 transistors, collector load resistors, base bias resistors and timing capacitors. Oscillation was achieved through the periodic charging and discharging of the coupling capacitors resulting in alternate switching of the two transistors. The circuit was initially operated using a 9 V supply and was experimentally verified through alternating LED illumination connected to the output nodes.

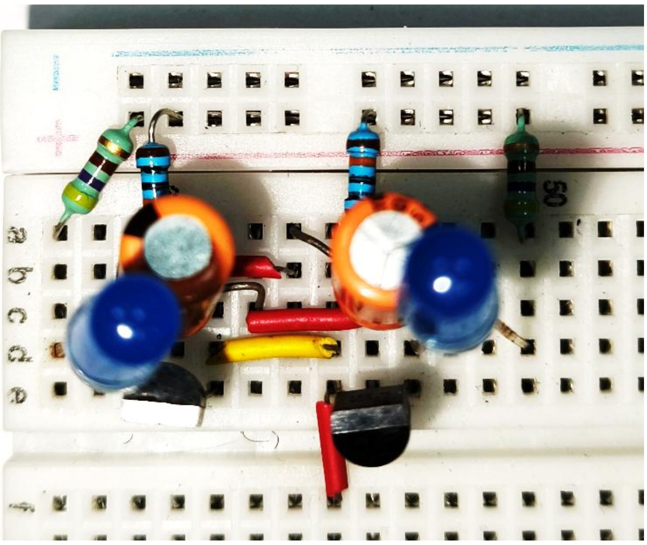


*Figure 19. BC547 breadboard implementation for Preliminary testing.*

The successful operation of the circuit confirmed the fundamental oscillation mechanism and provided a simple platform for understanding the behaviour of transistor based astable multivibrators. However, the BC547 transistor was not considered suitable for subsequent stimulation circuit development due to its limited current handling capability and reduced suitability for higher voltage operation. Consequently, further investigations were carried out using the 2N2222A transistor, which offered improved switching robustness and current capability. The implementation was verified through status of blinking of LED's used in the breadboard implementation, which worked as standalone system without requirement of oscilloscope to confirm circuits working.

## 5.6. Astable multivibrator using 2N2222A

### 5.6.1 Motivation

Following successful verification of oscillation using a BC547-based astable multivibrator, a more robust circuit was required for subsequent investigations. Although the BC547 implementation demonstrated the fundamental operating principle of an astable multivibrator, its current handling capability limited its suitability for higher voltage operation and future stimulation applications. Therefore, the BC547 transistors were replaced with 2N2222A NPN transistors, which offer improved current capability, better switching characteristics and greater robustness for practical hardware implementation.

The objective of this stage was to develop a stable oscillator capable of generating complementary output waveforms at a supply voltage of 25 V while maintaining a simple circuit architecture with a low component count.

### 5.6.2 Circuit Description

The circuit considered of two 2N2222A transistors connected is a cross-coupled astable multivibrator configuration. The collectors of the transistors were connected to the supply through the collector load resistors, while the bases were biased using the base resistors connected to the supply rail. Two timing capacitors were connected between the collector of one transistor and the base of the opposite transistor, thereby establishing positive feedback and enabling continuous oscillation.

*Table 23. Final Component values considered from implementation were:*

| Component | Value |
|---|---|
| Collector resistors | 5.6 kΩ |
| Base Resistors | 33 kΩ |
| Timing Capacitors | 0.1 µF |
| Supply Voltage | 25 V |
| Active Devices | 2N2222A |

The outputs were obtained from the collector of the two transistors producing complementary square wave signals.

### 5.6.3 Operating Principle

The operation of the circuit is based on regenerative switching and the charging and discharging behaviour of the timing capacitors. At any instant, one transistor remains in the conducting state while the other remains in the non-conducting state.

Assuming transistors Q1 is conducting, its collector voltage falls close to ground potential. Through the coupling capacitor connected to the base of Q2, a negative pulse is applied to Q2, keeping it in OFF state. As the capacitor gradually charges through the base resistor, the base voltage of Q2 increases. When the base emitter threshold voltage is reached, Q2 begins to conduct.

The collector voltage of Q2 then rapidly decreases, generating a negative pulse through the opposite coupling capacitor and turning Q1 OFF. This process repeats continuously, resulting in alternate conduction of the two transistors and generation of complementary output waveforms.

The oscillation frequency is primarily determined by the resistor capacitor time constant of the timing network.

### 5.6.4 Component Selection

Several combinations of collector and base resistor values were experimentally investigated to evaluate their influence on output amplitude, switching behaviour and waveform quality. The objective was to obtain output amplitude as close as possible to the supply voltage while maintaining stable oscillation and sharp switching transitions.

*Table 24. Summarizes the resistor combinations evaluated during the design process:*

| Collector resistors | Base Resistors | Observed Output Voltage |
|---|---|---|
| 1 kΩ | 10 kΩ | 5.6 V |
| 10 kΩ | 100 kΩ | 12.2 V |
| 47 kΩ | 470 kΩ | 16.2 V |
| 470 kΩ | 680 kΩ | 6.2 V |
| 51 kΩ | 470 kΩ | 16.8 V |
| 51 kΩ | 680 kΩ | 16.6 V |
| 5.6 kΩ | 33 kΩ | ≈ 25 V |

The experimental results indicated that larger resistor values produced reduced output amplitudes and slower switching transitions. Excessively large RC time constants resulted in slower capacitor charging and discharging leading to incomplete voltage swings and degraded waveform quality. Conversely, smaller resistor values improved transitions drive strength but did not always provide the desired output characteristics.

Among the combination tested, the configuration using 5.6 kΩ collector resistors and 33 kΩ base resistors provided the most favourable performance. This configuration generated output amplitude approaching the supply voltage while maintaining stable oscillation and sharp switching transitions. Consequently, it was selected for subsequent investigations.

### 5.6.5 Experimental Implementation

*Figure 20* shows the schematic of the circuit diagram considered for breadboard implementation as shown in *Figure 21*, the circuit was implemented on a solderless breadboard and powered using a regulated 25V supply. The outputs of the astable multivibrator were mentioned using a digital oscilloscope to evaluate waveform quality, frequency stability and output amplitude.

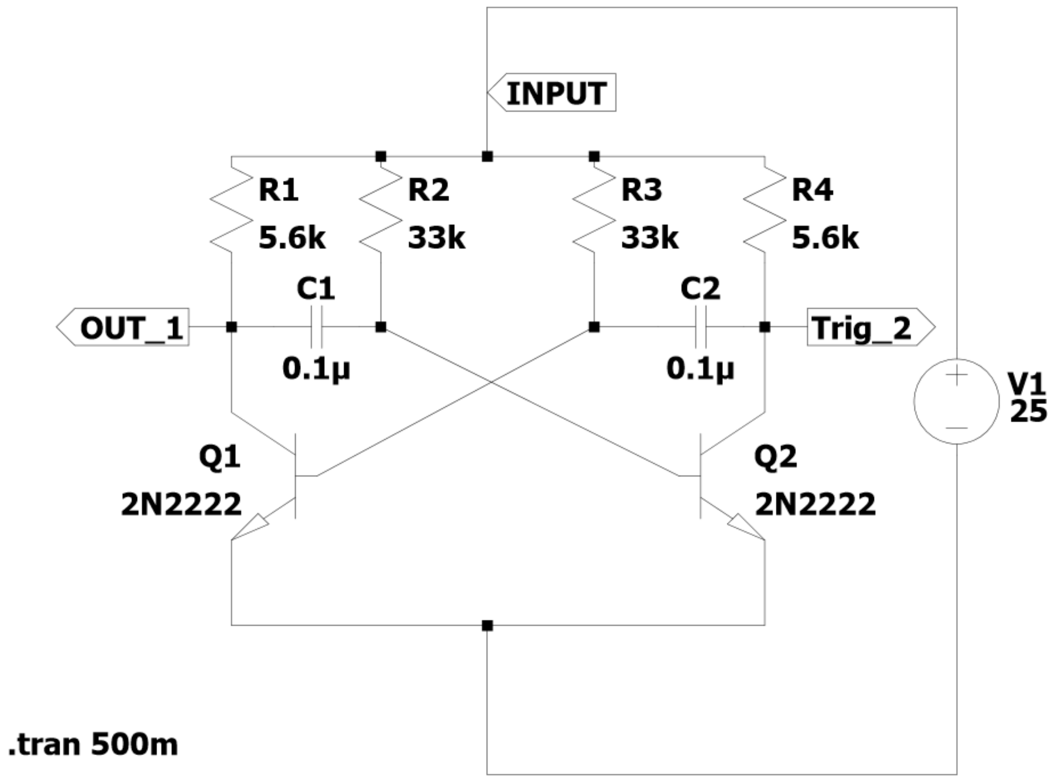



*Figure 20. LTSpice schematic of the 2N2222A astable multivibrator.*

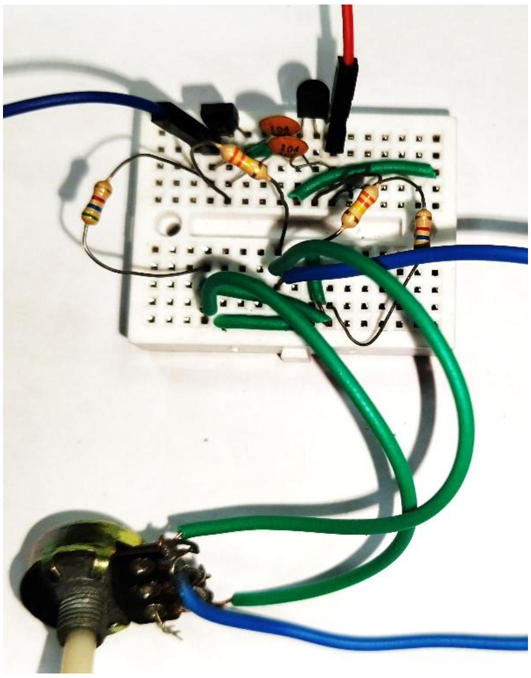


*Figure 21. Breadboard implementation of the 2N2222A astable multivibrator.*

### 5.6.6 Experimental Results

Oscilloscope measurements confirmed successful generation of complementary wave outputs from the collector terminals of the two transistors. The measured oscillation frequency was approximately (25-60) Hz by adjusting the 220 kΩ potentiometer connected to the charging resistors in series (common potentiometer was connected between both resistors to ensure equal changes at both the charging resistors), which remained stable during operation.

The output waveform exhibited a peak-to-peak amplitude of approximately 26 V, indicating that the selected resistor configuration enabled the output voltage to closely approach the applied supply voltage. In addition, the waveform demonstrated relatively sharp transitions between the ON and OFF states, indicating satisfactory switching performance of the transistors.

The experimental observations validated the suitability of the selected resistor values and confirmed that the circuit reliably generate complementary stimulation waveforms using a minimal number of components.

The Figure 21 shows the output waveforms measured at the collector terminals of the both the transistors, which is a differential voltage i.e. peak-to-peak voltage of around 24.30 V and frequency 25 Hz (Note: the frequency of the system could be varied using the potentiometer as per our requirements).

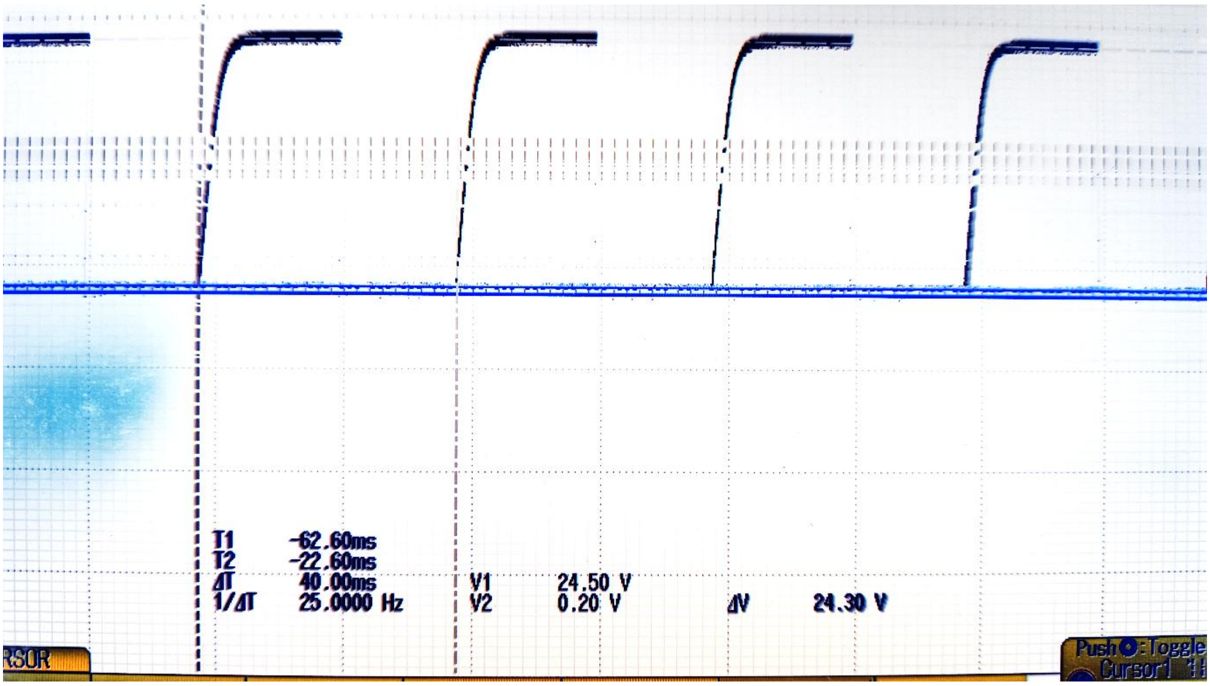



*Figure 22. Oscilloscope waveform of 2N2222A astable Multivibrator.*

### 5.6.7 Advantages

The 2N2222A based astable multivibrator offered several advantages:

- Simple and low-cost implementation.
- Minimal component count.
- Stable oscillation behaviour.
- Complementary output generation
- Output amplitudes approaching the supply voltage.
- Ease of implementation and debugging.
- Compatibility with future switching stages.

### 5.6.8 Limitations

Despite its successful operation, the circuit exhibited several limitations that restricted its direct use as a stimulation platform. Although complementary outputs were generated, the circuit was not specifically designed to provide controlled biphasic stimulation or active charge balancing.

These limitations motivated the investigation of alternative architectures involving MOSFET-based switching stages and H-bridge configurations.

### 5.6.9 Summary

The 2N2222A astable multivibrator represented the first fully validated hardware implementation developed during this work. Though systematic optimization of component values and experimental verification using oscilloscope measurements, astable oscillator capable of producing approximately 25 V complementary Square wave outputs was successfully realized. The knowledge gained from this stage formed the foundation for the subsequent development of MOSFET-based stimulation circuits and protection-oriented architectures.

## 5.7. LM358 based Astable Oscillator

### 5.7.1 Motivation

Following the successful implementation of the 2N2222A based astable multivibrator, an alternative architecture was investigated to explore the possibility of generating differential stimulation signals. The objective was to evaluate whether a bridge-tied load (BTL) configuration could provide a larger effective voltage across the stimulation electrodes without increasing the supply voltage.

In a BTL configuration, the load is connected between two actively driven outputs rather than between an output and ground. As a result, the voltage difference across the load can be increased making the approach attractive for stimulation applications where maximizing the electric field is desirable.

### 5.7.2 Circuit Description

The proposed circuit as shown in *Figure 23* and implemented on breadboard as shown in *Figure 24* employed comparator-based oscillators configured to generate complementary output signals. A resistor capacitor network was used to establish the oscillation frequency, while positive feedback provided regenerative switching between the output states.

The architecture generated two output signals, designated as OUT1 and OUT2, which were intended to operate in opposite phases. The stimulation load could then be connected between these outputs, forming a bridge-tied load configuration.

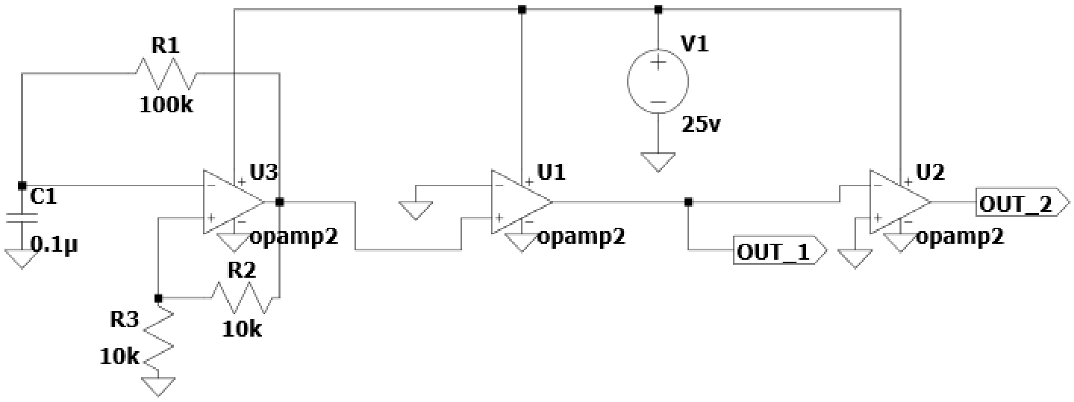



*Figure 23. LTSpice schematic of the comparator based BTL architecture.*

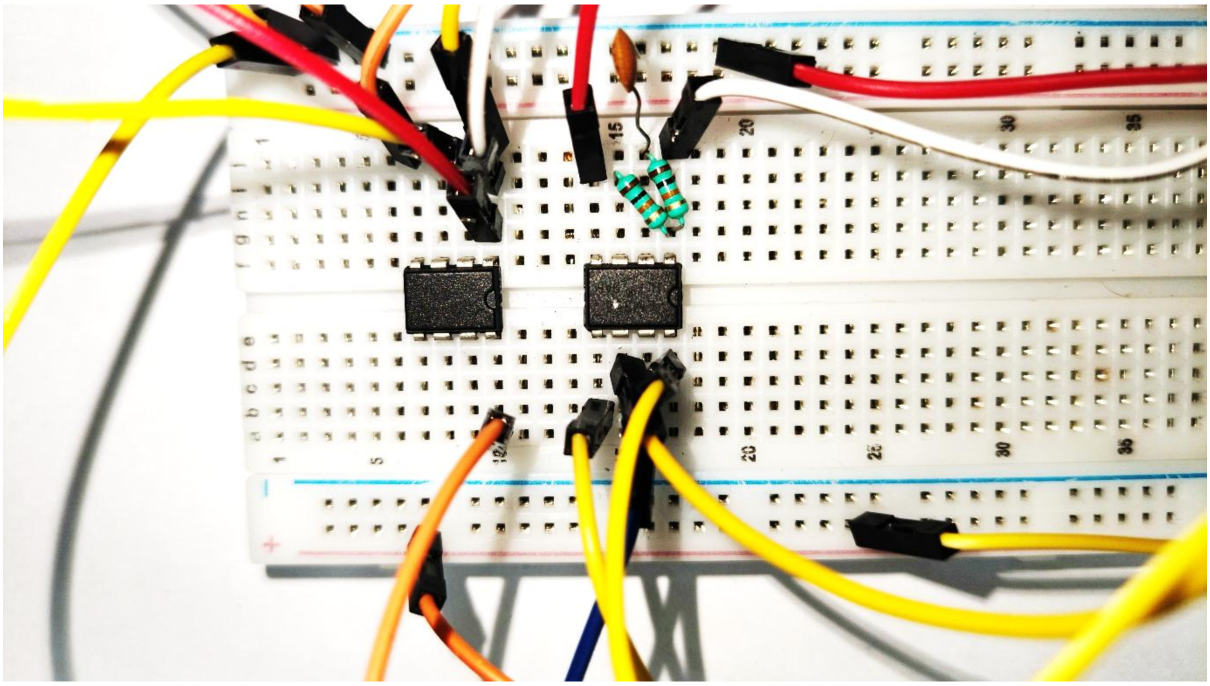


*Figure 24. Breadboard implementation of the BTL architecture.*

### 5.7.3 Operating Principle

The oscillator section generated periodic switching between two voltage states. Through the feedback network, the comparator output alternated between its high and low states, causing the timing capacitor to repeatedly charge and discharge.

The resulting output waveform was then used to produce complementary signals at the output terminals. When connected to a load, the voltage difference between OUT_1 and OUT_2 established a simulation voltage.

Unlike a conventional single ended configuration, the load experiences the differential voltage between the two outputs rather than the voltage between a single output and ground.

### 5.7.4 Experimental Investigation

The circuit was initially simulated and subsequently implemented on breadboard for practical evaluation. Oscilloscope measurements were performed to examine waveform quality, frequency stability and switching behaviour.

Although oscillations shown in *Figure 25* measured at the outputs OUT_1 and OUT_2 respectively (as shown in *Figure 23)* were successfully achieved, several practical challenges were observed during the implementation. The circuit exhibited increased sensitivity to breadboard parasitic capacitances and component tolerances. Small changes in wiring layout and component placement influenced waveform behaviour.

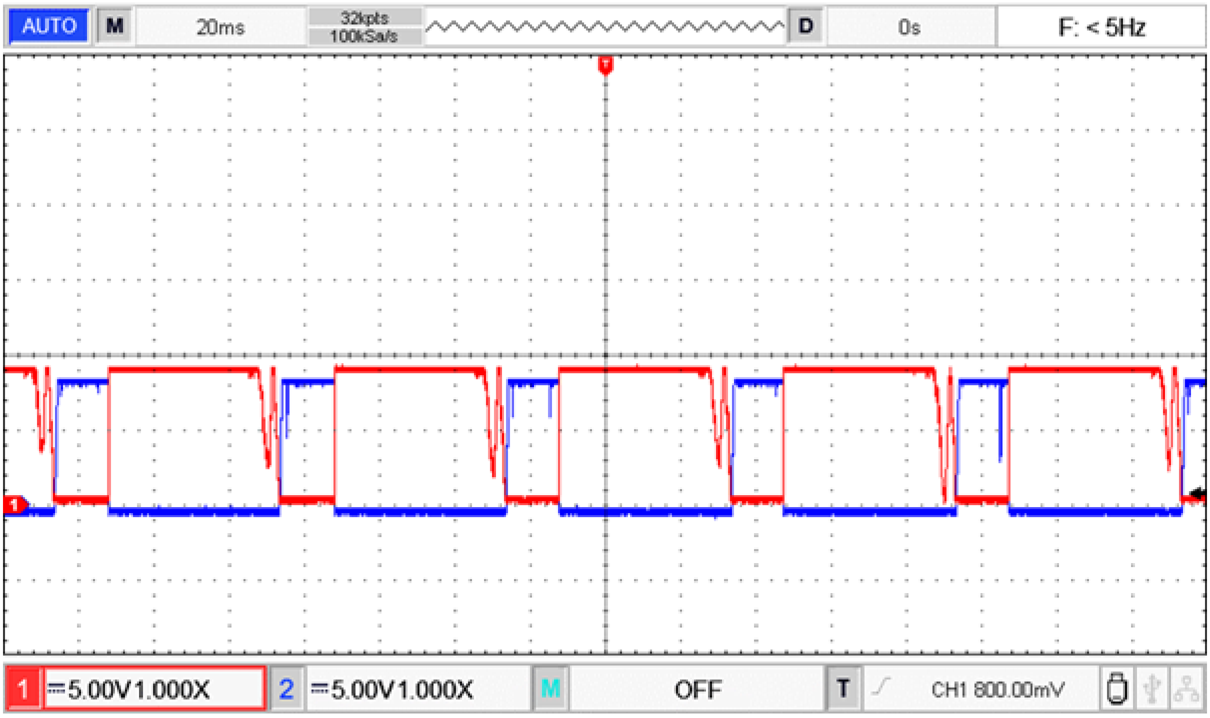



*Figure 25. Oscilloscope waveform of oscillator output obtained from the implemented BTL architecture.*

### 5.7.5 Advantages

The investigated architecture offered several potential advantages:

- Differential output operation.
- Increased effective voltage across the stimulation load.
- Potential pathway toward biphasic stimulation.
- Flexibility in waveform generation.
- Compatibility with future control-oriented architectures.

### 5.7.6 Limitations

Despite its theoretical advantages, several practical limitations were identified during experimental evaluation.

The circuit exhibited greater sensitivity to parasitic effects present in breadboard implementations, resulting in waveform asymmetry and increased susceptibility to noise.

For the intended application involving a lightweight and compact stimulation platform suitable for future animal mounted implementation, the additional complexity could not be justified based on the experimentally observed performance improvements.

### 5.7.7 Design Decision

Although the bridge-tied-load architecture successfully demonstrated differential waveform generation, it was not selected as the primary stimulation circuit. The increased component count, greater implementation complexity and sensitivity to practical nonidealities reduced its suitability for the intended proof-of-concept platform.

Since the principal objective of the present work was to develop a compact lightweight and experimentally verifiable stimulation system, subsequent efforts focused on power switching architecture based on MOSFET devices that could provide comparable output voltages with significantly reduced complexity.

### 5.7.8 Summary

The comparator-based bridge tied load architecture served as an exploratory investigation into differential stimulation techniques. While the concept demonstrated potential advantages in terms of differential voltage generation, practical implementation consideration favoured simpler transistor and MOSFET based solutions. The insights gained from this investigation informed the subsequent development of MOSFET switching architectures and the final proof of concept stimulation system.

## 5.8. Astable Multivibrator Driving a MOSFET H-Bridge

### 5.8.1 Motivation

Following successful implementation of the 2N2222A based astable multivibrator an investigation was carried out to determine whether the generated complementary waveforms could be used to drive a MOSFET-based H-Bridge. The objective was to explore switching architecture capable of providing alternating polarity across the stimulation load and serving as a foundation for future biphasic stimulation systems.

H-Bridge configurations are widely employed in motor control, power electronics and stimulation systems because they allow the direction of current through a load to be periodically reversed. Such architectures are particularly attractive in bioelectrical stimulation applications where alternating current paths can help reduce charge accumulation and electrode polarization effects.

The investigation was therefore undertaken to evaluate the feasibility of combining a simple transistor-based oscillator with a power switching stage implemented using power MOSFET devices.

### 5.8.2 Circuit Description

The architecture consisted of two major functional blocks:

- A 2N2222A based astable multivibrator responsible for generating complementary square wave control signals.
- A power MOSFET H-Bridge stage responsible for switching the stimulation load.

The H-Bridge was implemented using:

| Device Type | Component |
|---|---|
| N-Channel Power MOSFET | IRF 540N |
| P-Channel Power MOSFET | IRF 9540N |

The complementary outputs of the astable multivibrator were connected to the MOSFET gate network as shown in *Figure 26* and implemented on breadboard as shown in *Figure 27* with the intention of alternately switching the high side and low side devices.

Under ideal conditions one diagonal pair of MOSFETs conducts during one half cycle while the opposite diagonal pair conducts during the next half cycle. This produces an alternating voltage across the load and allows bidirectional current flow.

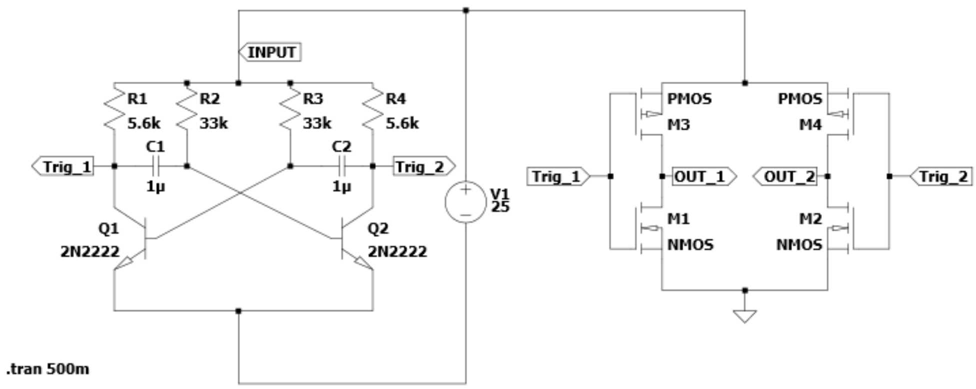



*Figure 26. LTSpice Schematic of the transistor-driven MOSFET H-Bridge.*

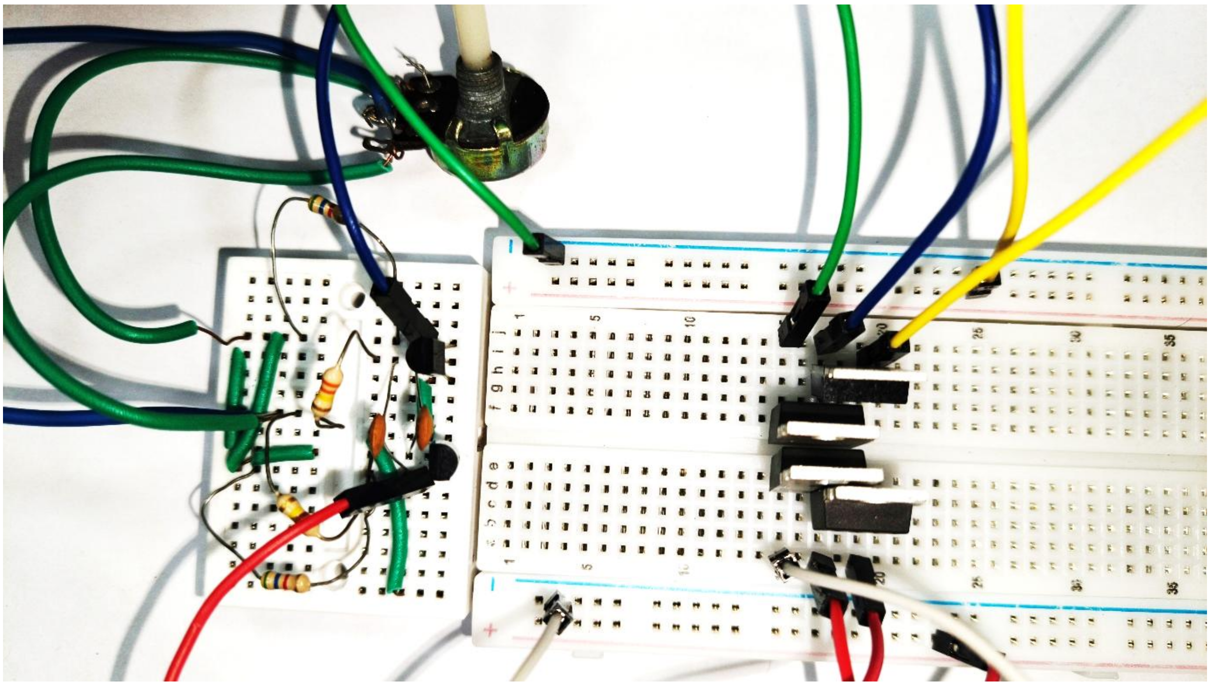


*Figure 27. Breadboard implementation of the power MOSFET H-Bridge architecture.*

### 5.8.3 Expected Operation

The complementary outputs generated by the astable multivibrator were expected to drive the MOSFET gates in an alternating fashion. During one half cycle, one pair of MOSFETs would conduct, applying the supply voltage across the load in one direction. During the subsequent half cycle, the opposite pair would conduct, reversing the polarity across the load.

Such an arrangement would provide differential excitation across the stimulation electrodes and could potentially serves as a foundation for future biphasic stimulation systems.

### 5.8.4 Experimental Observations

The H-bridge architecture was implemented and experimentally evaluated on a breadboard. While the astable multivibrator operated correctly in isolation, the behaviour of the overall system changed significantly after the H-Bridge stage was connected.

A substantial reduction in the effective supply was observed during operation. Although the oscillator generated the expected trigger signals, the addition of the H-Bridge stage resulted in a noticeable drop in the measured supply voltage, reducing the available output voltage to approximately 9 V under certain operating conditions.

As a consequence, the desired switching behaviour could not be reliably maintained and the expected load excitation was not achieved. The following modifications were tried to improve the performance.

### 5.8.5 Design Modifications and Troubleshooting

To identify the cause of the observed behaviour and improve switching performance, several modifications were investigated.

- Gate Delay Network:

  A resistor was introduced in series with each MOSFET gate to intentionally slow the turn on transition. The purpose of this modification was to reduce the possibility of simultaneous conduction between the P-Channel and N-Channel MOSFETs located on the same side of H-Bridge.

  A diode was connected in parallel with the gate resistor to provide a low-resistance discharge path during turn off. This arrangement allowed the MOSFET to turn OFF rapidly while maintaining a controlled turn ON transition.

  The objective of this modification was to introduce a degree of dead time into the switching process and reduce the possibility of shoot through current. Although the switching characteristics were altered the supply voltage reduction persisted.

- Pull-Up and Pull-down Resistors

  Additional pull-up and pull-down resistors were introduced into the gate drive network to ensure that the MOSFET gates remained at defined voltage levels during switching transitions and when the driving signal was inactive. The intention was to prevent floating gate conditions, improve switching reliability and ensure complete turn ON and turn OFF operation of the MOSFETs.

  Despite the inclusion of these biasing resistors, no significant improvement in the circuit performance was observed. The supply voltage reduction and unreliable switching behaviour remained unchanged.

- Midpoint Gate Biasing:

  A voltage divider network was subsequently introduced to generate an intermediate bias voltage of approximately 12.5 V from the 25 V supply.

  The MOSFET gate network was referenced to this mid-point voltage with the intention of improving the switching relationship between P-Channel and N-Channel devices. The objective was to ensure that only one device on each side of the H-Bridge remained active at any given instant.

  Despite the modification, reliable switching performance was not obtained and the supply voltage reduction remained present.

- Zener Diode Protection Network:

  Zener diodes were also incorporated into the gate-drive circuitry to limit the gate to source voltage of the MOSFETs and protect the devices from excessive gate voltage excursions. The intention was to stabilize the gate-drive conditions and prevent any abnormal switching behaviour caused by overvoltage conditions.

The addition of the Zener diode did not produce any noticeable improvement in circuit operation. The measured supply voltage continued to collapse during operation, indicating that excessive gate voltage was unlikely to be the primary cause of the observed behaviour.

- 2N2222A Driver Stage:

A dedicated transistor driver stage based on the 2N2222A was then inserted between the astable multivibrator and the MOSFET H-Bridge.

The purpose of this stage was to provide increase capability for charging and discharging the MOSFET gate capacitances. Since the MOSFET gates behave as capacitive loads, the additional driver stage was expected to improve switching speed and reduce gate drive limitations.

Although the driver stage improved gate drive capability, the overall architecture continued to exhibit unsatisfactory behaviour and the desired performance could not be achieved under the tested conditions.

The *Figure 28* shows the overall improved version of schematic that was considered to reach the desired outputs and the *Figure 29* shows the breadboard implementation of the improved version.

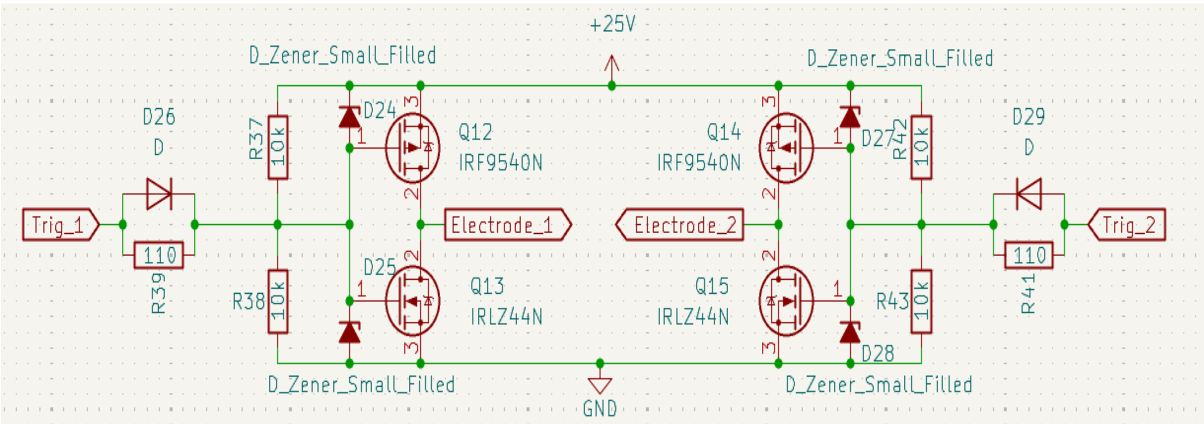



*Figure 28. Improved MOSFET H-Bridge architecture.*

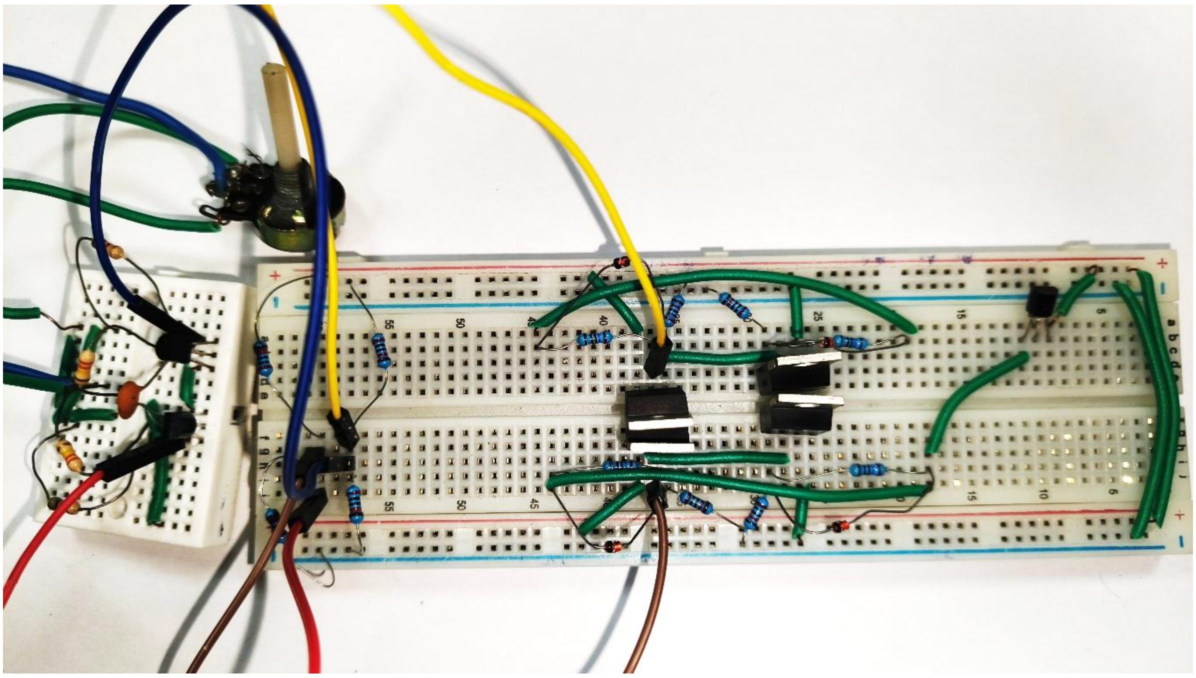


*Figure 29. Breadboard implementation of the improved MOSFET H-Bridge architecture.*

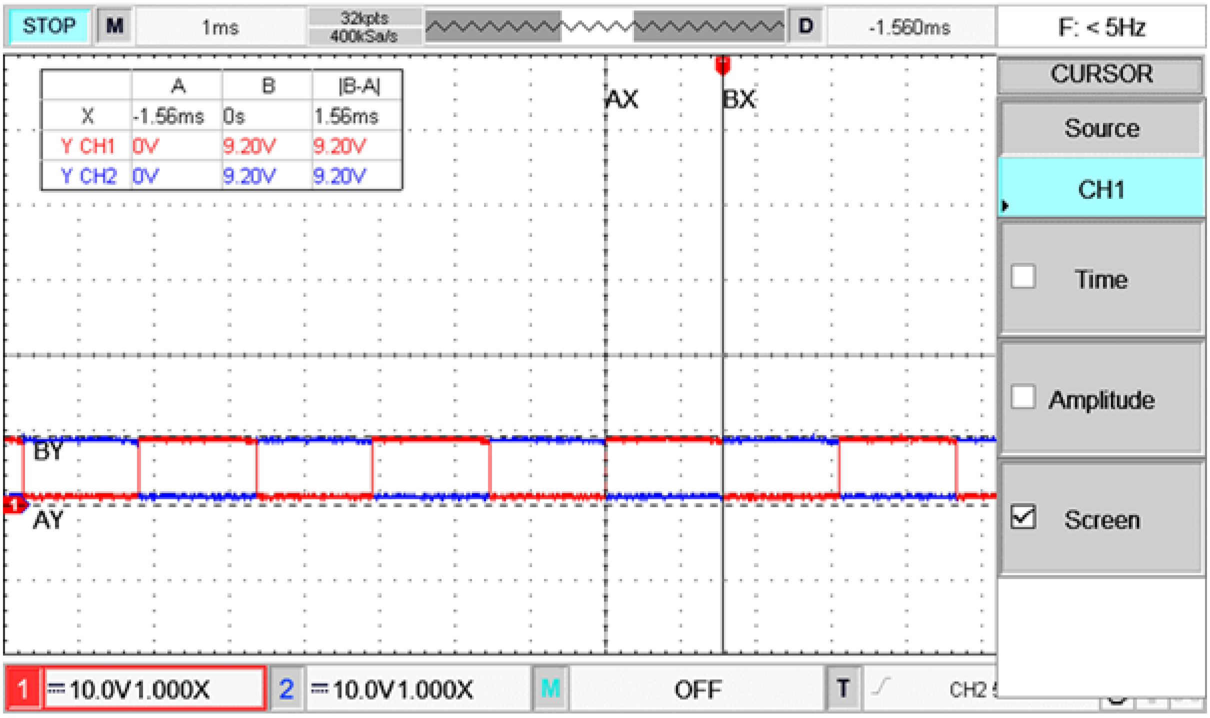



*Figure 30. Oscilloscope waveforms at the output stage of Astable Multivibrator driven H-Bridge in the improved MOSFET H-Bridge architecture.*

*Figure 30* shows the waveforms taken at the drain terminals from the left (Electrode_1) and right half (Electrode_2) of the circuit as shown in *Figure 28* respectively.

### 5.8.6 Discussion

The experimental observations suggest that the practical implementation of the H-Bridge was influenced by several factors.

The IRF540N and IRF9540 MOSFETs require appropriate gate drive conditions to operate efficiently. Inadequate gate drive voltage or current may cause the devices to operate within their linear region rather than as ideal switches, resulting in increased power dissipation and excessive current draw.

Although gate delay networks, pull-up and pull-down resistors, midpoint gate biasing, Zener diode protection and a dedicated transistor driver stage were implemented these modifications did not eliminate the observed voltage collapse. This suggests that the underlying limitations were likely associated with the gate drive requirements of the selected MOSFETs switching dynamics within the H-Bridge and the practical constraints of breadboard implementation.

Several possible causes may explain the persistent reduction in supply voltage:

- Shoot through current: even brief overlap between conduction periods of the P-channel and N-channel MOSFETs could create temporary low resistance path between the supply rails resulting in large current spikes and a corresponding drop in a supply voltage.
- Insufficient dead time: Although gate delay networks were introduced the delay may not have been sufficient to completely prevent simultaneous conduction during switching transitions.
- Inadequate gate drive voltage: The IRF540N is not a logic level MOSFET and requires adequate gate to source voltage for efficient switching. If the gate voltage was insufficient the device may have operated in its linear region increasing current consumption and power dissipation.
- High side drive limitations: Proper control of the IRF9540N P-channel MOSFET gate requires gate voltages referenced to its source terminal. The astable multivibrator may not have provided the required gate drive conditions for reliable high side switching.
- MOSFET gate capacitance: The astable multivibrator was originally designed as an oscillator rather than a dedicated gate driver. The gate capacitances of the MOSFETs may have imposed a significant load on the oscillator affecting the switching performance.
- Breadboard parasitic effects: Stray capacitance, contact resistance and wiring inductance associated with breadboard construction can significantly influence switching circuits and may have contributed to unstable operation.
- Supply current limitations: The large transient currents associated with MOSFET switching may have exceeded the capability of the power source or introduced voltage drops within the supply path.
- Device mismatch and switching asymmetry: Differences in the switching characteristics of the IRF540N and IRF9540N may have resulted in unequal switching times increasing likelihood of transient cross conduction.

The fact that the supply voltage continued to collapse even after multiple corrective measures strongly suggests that the issue was not caused by a single component or wiring error. Rather, it was likely the result of a combinations of gate drive limitations, transient shoot through currents, MOSFETs switching characteristics and breadboard related parasitic effects.

The results indicate that reliable operation of the H-bridge would likely require a dedicated gate driver integrated circuit, carefully controlled dead time generation, improved power supply decoupling and PCB based implementation optimized for high speed switching.

To further investigate whether the observed behaviour was primarily associated with the external oscillator stage or with the H-Bridge architecture itself a self-oscillating MOSFET H-Bridge configuration was subsequently explored.

### 5.8.7 Advantages of the architecture

The investigated architecture offered several potential advantages:

- Capability of differential load excitation.
- Potential pathway toward biphasic stimulation.
- Higher power handling capability.
- Compatibility with future protection and monitoring circuits.
- Commonly used architecture in stimulation systems.

### 5.8.8 Limitations

The experimental investigation revealed several practical limitations:

- Increased component count.
- Higher implementation circuitry.
- Requirement for dedicated gate drive circuitry.
- Sensitivity to switching delays.
- Greater susceptibility to breadboard parasitic effects.
- Increased power consumption.
- Reduced suitability for rapid proof of concept limitations.

### 5.8.9 Design Decision

Although the H-Bridge architecture offered several theoretical advantages, the additional complexity and implementation challenges outweighed its benefits at the present stage development.

The repeated investigations involving gate delay networks, pull-up and pull-down resistors, mid-point gate biasing, Zener diode protection and dedicated transistor driver stages demonstrated that further optimization would be required before reliable operation could be achieved.

Since the primary objective of this work was to develop a compact, lightweight and experimentally verifiable platform suitable for future animal mounted implementation, a simpler MOSFET based architecture was subsequently investigated.

Consequently, attention was directed toward the development of a self-oscillating MOSFET architecture that could provide the required stimulation waveform using a significantly lower component count and reduced implementation complexity.

The self-oscillating architecture was subsequently investigated and experimentally evaluated as a potential alternative before arriving at the final stimulation circuit selected for prototype implementation.

### 5.8.10 Summary

The astable multivibrator driven H-Bridge represented an important intermediate stage in the design process. The investigation provided valuable insight into MOSFET switching behaviour, gate drive requirements, biasing considerations, protection strategies and practical implementation challenges associated with differential excitation architectures.

Although the architecture was not selected for further development, the lessons learned during its investigation directly influenced the subsequent development of a self-oscillating MOSFET H-bridge architecture. The findings obtained from both H-Bridge investigations ultimately contributed to the selection of a simpler IRF540N based oscillator for final prototype.

## 5.9. Self-Oscillating MOSFET H-Bridge

### 5.9.1 Motivation

Following the investigation of the astable multivibrator driven MOSFET H-Bridge architecture a self-oscillating MOSFET H-Bridge was explored as an alternative approach for generating alternating stimulation waveforms. The motivation behind this architecture was to eliminate the requirement for an external oscillator stage and to allow the power switching network itself to generate the oscillatory behaviour required for stimulation. Such an approach had the potential to reduce overall component count, simplify the circuit topology and improve compactness for future wearable or animal mounted stimulation systems.

The architecture was initially designed and verified through LTSpice simulations before being implemented practically on breadboard platform. Both stimulation and experimental investigations were carried out to evaluate its suitability for non-invasive electrical stimulation applications.

### 5.9.2 Circuit Architecture

The self-oscillating H-Bridge consisted of two IRF9540 P-channel and two IRF540N-channel MOSFETs arranged in a complementary bridge configuration. Unlike conventional H-Bridge architectures that rely only on an external oscillator or gate driver, oscillation was generated through a regenerative feedback network formed using a cross coupled capacitors and biasing resistors.

The switching nodes of the H-Bridge were interconnected through capacitive feedback paths. When one side of the bridge transitioned from one state to another, the resulting voltage change was capacitively coupled to the opposite side, forcing it into the complementary state. This regenerative action continuously repeated, resulting in sustained oscillation without requiring a dedicated oscillator stage.

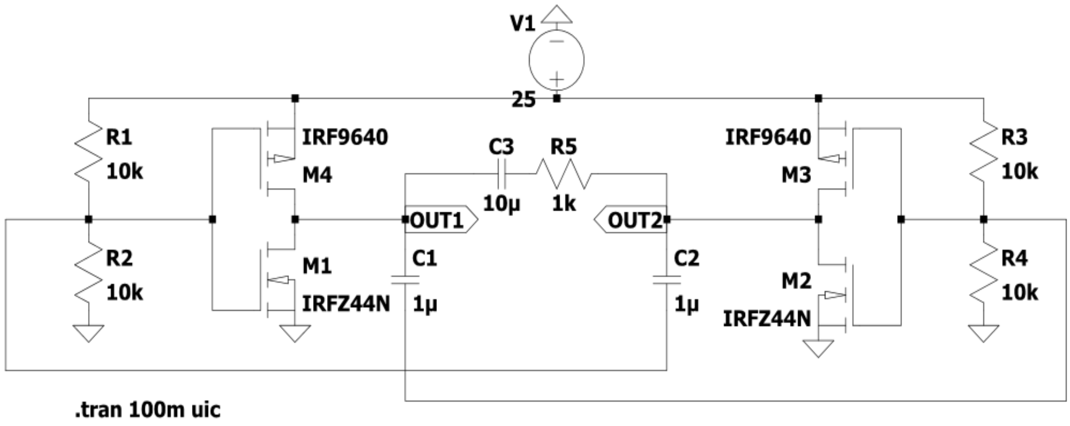



*Figure 31. LTSpice schematic of Self-Oscillating MOSFET H-Bridge.*

### 5.9.3 Principle of Operation

Initially, minor mismatches between device characteristics cause one side of the bridge to conduct slightly earlier than the other. The resulting voltage transition is coupled through the cross coupled capacitor network, forcing the opposite side into the complementary state. This transition creates a regenerative switching action that continuously alternates between two states of the bridge.

As the capacitors charge and discharge through the associated resistor network the gate voltages vary periodically producing alternating conduction of the MOSFET pairs. Consequently, complementary output waveforms are generated at the two bridge outputs.

Since the outputs switch in opposite directions, a differential voltage is developed across the load. Ideally this differential output approaches the full supply voltage swing, thereby providing a bipolar stimulation waveform without the need for an additional H-Bridge driver or inverter stage.

### 5.9.4 Simulation Results

The circuit was first analysed using LTSpice simulations to verify oscillation behaviour and evaluate the expected output waveforms. Simulation results as shown in *Figure 32* demonstrated stable oscillation and nearly complementary switching of the bridge outputs.

The individual output nodes alternated between approximately 0 V and 25 V, while the differential voltage between the two outputs approaches ±25 V. the generated waveform exhibited alternating polarity and therefore possessed characteristics suitable for electrical stimulation applications.

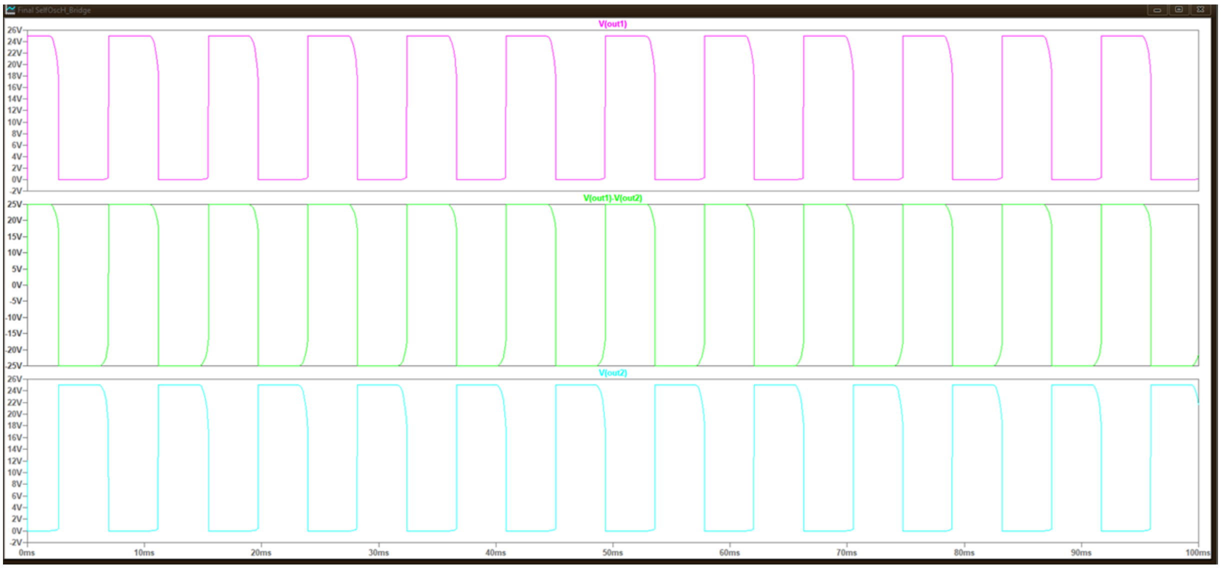



*Figure 32. LTSpice simulated waveforms of output of Self-Oscillating MOSFET H-Bridge.*

The simulation results suggested that the architecture could theoretically generate the required stimulation waveform while maintaining a relatively low component count.

### 5.9.5 Practical Implementation

Following successful simulation, the circuit was implemented on a solderless breadboard as shown in *Figure 33* using IRF540N and IRF9540N MOSFETs. Multiple resistors and capacitor combinations were investigated experimentally to evaluate their influence on oscillation frequency, waveform quality and output amplitude.

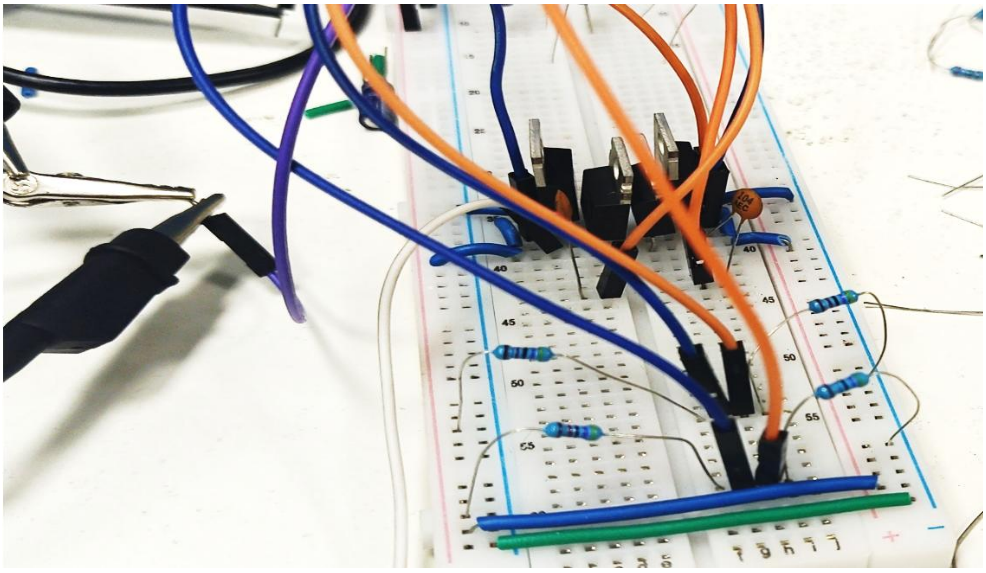


*Figure 33. Breadboard implementation of Self-Oscillating MOSFET H-Bridge.*

A series of experimental trials were performed using different timing resistor and capacitor values. The observed output voltages and oscillation characteristics are summarized below.

*Table 25. Experimental Evaluation of Self-Oscillating MOSFET H-Bridge:*

| Gate Resistors | Capacitor | Output Voltage |
|---|---|---|
| 100 kΩ | 0.1 µF | 8.4 V |
| 100 kΩ | 1 µF | 10.4 V |
| 10 kΩ | 0.1 µF | 9.2 V |
| 10 kΩ | 1 µF | 8.2 V |
| 10 kΩ | 2.2 µF | 8.2 V |
| 10 kΩ | 3.3 µF | ~10 V |
| 10 kΩ | 4.7 µF | ~11.2 V |
| 10 kΩ | 10 µF | 9.6 V |
| 20 kΩ | 100 µF | 13.6 V |
| 100 kΩ | 220 µF | ~18.8 V |

### 5.9.6 Experimental Observations

Although oscillation was successfully obtained, the practical implementation did not reproduce the performance predicted by simulation.

The most significant observation was the reduction in output voltage. For many resistor capacitor combinations, the measured output voltage remained considerably lower than the supply voltage. Typical output amplitudes ranged between approximately 8 V and 11 V despite the circuit being supplied from a 25 V source.

Furthermore, the oscillation characteristics were found to be highly sensitive to resistor and capacitor values. Small changes in component selection resulted in noticeable variations in both frequency and output amplitude.

Another important observation was the requirement for relatively large capacitors to achieve improved output voltages. For example, using a 220 µF capacitors with a 100 kΩ resistor increased the output voltage to approximately 18.8 V as shown in *Figure 34*. However, this improvement came at the expense of significantly slower oscillation frequencies and increased component size.

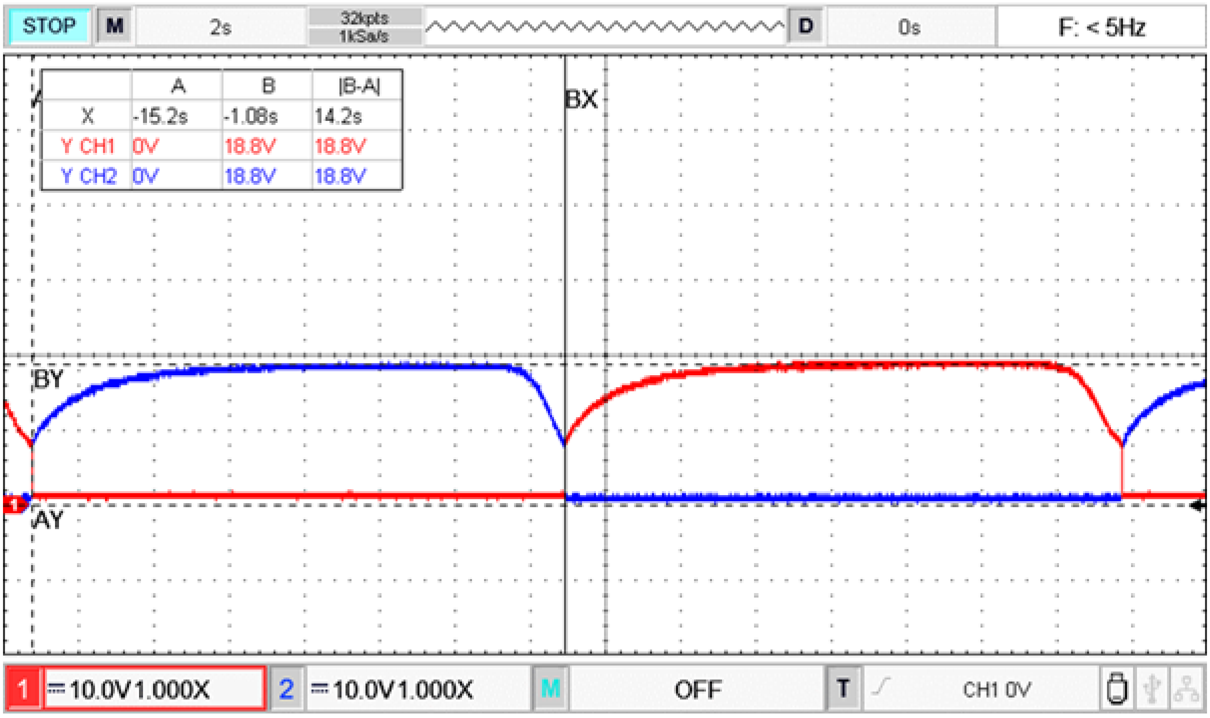



*Figure 34. Oscilloscope waveforms at output stage of implemented Self-Oscillating MOSFET H-Bridge.*

### 5.9.7 Troubleshooting and Optimization Attempts

To improve switching behaviour and investigate the cause of the reduced output voltage, several modifications were experimentally evaluated.

- Gate Delay Network:

  Series resistors were introduced at the MOSFET gates to intentionally slow the turn-on process. The objective was to reduce simultaneous conduction of complementary MOSFETs and thereby minimize shoot through current.

- Diode-Assisted Gate Discharge:

Diodes were connected in parallel with the gate resistors to provide a low impedance discharge path during turn-off. This arrangement was intended to accelerate gate discharge and improve switching transitions.

- Midpoint Biasing:

  A resistor divider network was introduced to establish an intermediate gate bias voltage of approximately half the supply voltage. This modification was intended to improve switching symmetry and prevent simultaneous conduction of MOSFET pairs.

- External Driver Stage:

  An intermediate transistor driver stage based on the 2N2222A transistor was also investigated. The purpose of this stage was increase gate drive capability and improve switching performance of the H-Bridge.

  Despite these modifications, the overall output characteristics did not improve sufficiently to justify further development of the architecture.

- Pull-Up and Pull-Down Resistors:

  Despite the inclusion of these biasing resistors, no significant improvement in the circuit performance was observed. The supply voltage reduction and unreliable switching behaviour remained unchanged.

- Zener Diode Protection Network:

  The addition of the Zener diode did not produce any noticeable improvement in circuit operation. The measured supply voltage continued to collapse during operation, indicating that excessive gate voltage was unlikely to be the primary cause of the observed behaviour.

The *Figure 35* shows the consolidated improved version of the circuit as discussed above and was implemented on breadboard as shown in *Figure 36*.

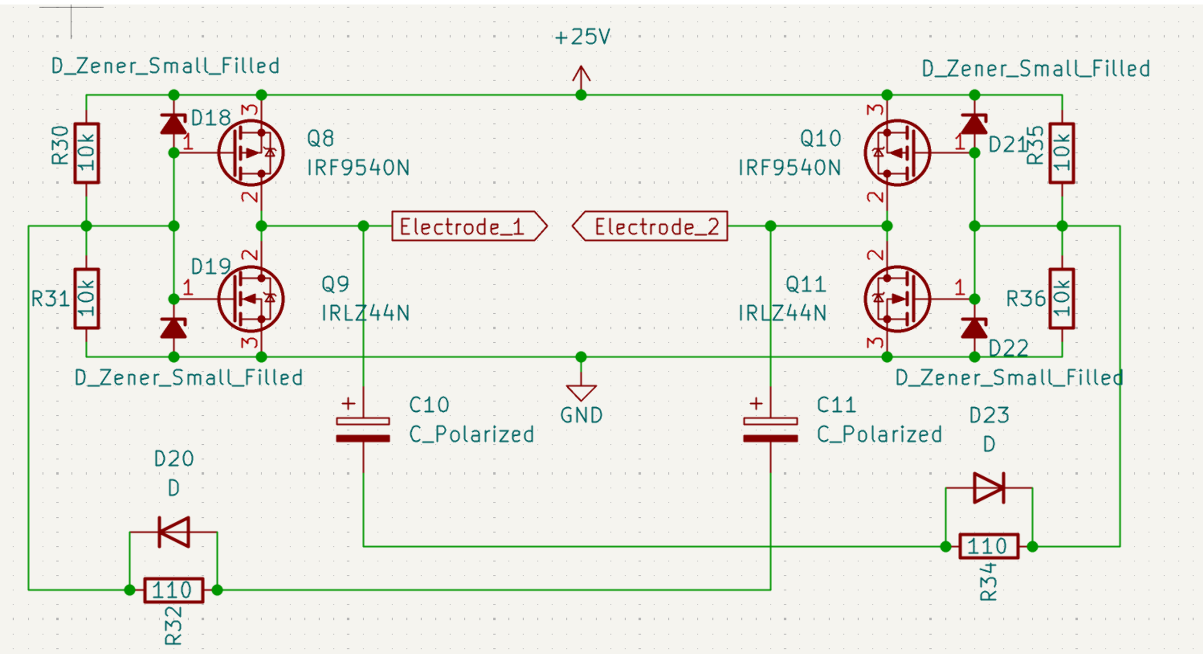



*Figure 35. Improved version of Self-Oscillating MOSFET H-Bridge architecture.*

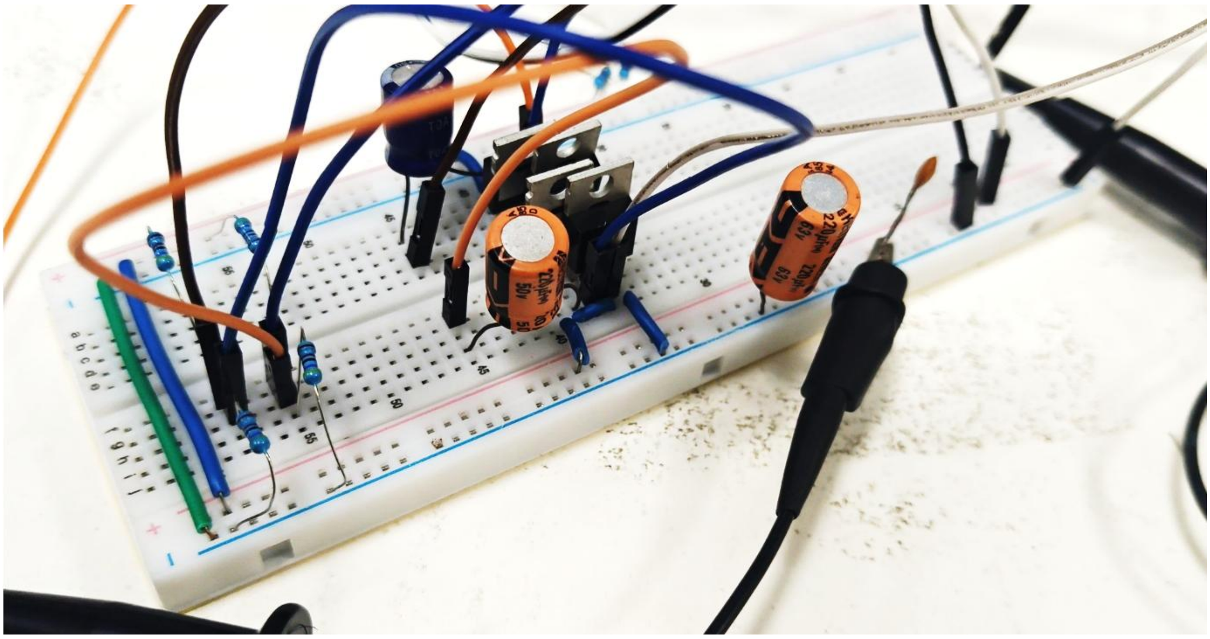


*Figure 36. Breadboard implementation of improved Self-Oscillating MOSFET H-Bridge architecture.*

### 5.9.8 Discussion

The practical investigations revealed that the self-oscillating H-Bridge was significantly more sensitive to component tolerances and parasitic effect than initially expected. The oscillation mechanism depended directly on MOSFET threshold voltages, gate capacitances, resistor tolerances and capacitor charging characteristics.

Consequently, small variations in component values produced substantial changes in circuit behaviour. Breadboard parasitic capacitances and wiring inductances further influenced switching performance and reduced predictability.

Although acceptable oscillation could be achieved under certain operating conditions, the architecture exhibited lower robustness and repeatability than desired for a stimulation system intended for future biological experimentation.

### 5.9.9 Advantages

The self-oscillating MOSFET H-Bridge offered several advantages:

- Elimination of an external oscillator stage.
- Reduced component count.
- Direct generation of alternating stimulation waveforms.
- Potential for compact implementation.
- Ability to generate bipolar output waveforms.

### 5.9.10 Limitations

Several limitations were identified during practical implementation:

- Reduced output voltage compared to the supply voltage.
- High sensitivity to compared tolerances.
- Dependence on large timing capacitors for improved performance.
- Significant influence of breadboard parasitic effects.
- Difficult optimization process.
- Reduced predictability under varying operating conditions.

### 5.9.11 Design Decision

Although the self-oscillating MOSFET H-Bridge successfully demonstrated oscillatory behaviour in both simulation and practical implementation, it did not consistently achieve the output performance required for the proposed stimulation system.

The architecture required extensive tuning, exhibited strong dependence on component values and produced lower output amplitudes than expected. Furthermore, the improvements obtained through troubleshooting and optimization did not sufficiently outweigh the additional complexity associated with the design.

Consequently, the self-oscillating H-Bridge architecture was not selected for final prototype implementation. Instead a simpler IRF540N based astable mutlivibrator architecture was adopted as the final stimulation due to its reduced component count improved predictability, ease of implementation and reliable experimental performance.

The investigation of the self-oscillating H-Bridge nevertheless provided valuable insights into MOSFET switching behaviour, feedback-based oscillation mechanisms and practical limitations associated with self-oscillating power electronic architectures. These findings contributed significantly to the final design decisions adopted in the developed stimulation platform.

## 5.10. Current Monitoring and Protection Architecture

### 5.10.1 Motivation

Although the final IRF540N based astable multivibrator successfully generated required alternating stimulation waveform, long term deployment of electrical stimulation systems required additional safety mechanisms to prevent excessive current delivery and potential hardware faults. Consequently, a current monitoring and protection architecture was investigated as an extension of the stimulation system.

The primary objective of this architecture was to provide a means of monitoring the stimulation current and automatically disabling the oscillator whenever the current exceeded a predefined threshold. Such protection mechanisms become increasingly important when stimulation system are intended for biological experimentation, wearable applications or future animal studies.

### 5.10.2 Design Objectives

The current monitoring architecture was designed with the following objectives:

- Continuous monitoring of stimulation current.
- Adjustable current limit setting.
- Automatic shutdown during overcurrent conditions.
- Prevention of excessive power dissipation.
- Protection of stimulation circuitry.
- Compatibility with future MOSFET based implementations.
- Expandability towards additional safety features.

The architecture was therefore developed as a proof-of-concept safety subsystem that could be incorporated into future versions of the stimulation platform.

### 5.10.3 Principle of Operation

The proposed protection architecture operates by measuring the voltage developed across a low value current sensing placed in the stimulation current path.

According to the Ohm's law:

The voltage developed across the sensing resistor is directly proportional to the stimulation current.

This sensed voltage is applied to one input of an LM358 operational amplifier configured as comparator. A reference voltage is applied to the second input using an adjustable potentiometer.

When the sensed voltage remains below the reference voltage, the stimulation circuit continues to operate normally.

However, if the stimulation current increases beyond the pre-set limit, the voltage across the sensing resistor exceeds the reference threshold. The comparator output then changes state and activates a switching transistor which disconnects power from the oscillator stage.

As a result stimulation output is automatically terminated whenever the current exceeds the selected limit.

### 5.10.4 Circuit Architecture

The experimental protection circuit as shown in *Figure 37* consisted of:

- Current sensing resistor.
- LM358 operational amplifier.
- Adjustable threshold potentiometer.
- BC557 PNP transistor.
- Astable multivibrator stage.

The current sensing resistor was placed in the low side return path of the circuit. This configuration simplified measurement because the sensed voltage remained reference to ground.

The LM358 continuously compared the sensed voltage against the adjustable threshold voltage generated by potentiometer. When an overcurrent condition was detected, the comparator output drove a transistor-based shutdown stage.

The shutdown stage utilized a BC557 transistor operating as a high side switch. Under normal operating conditions the BC557 supplied power to the oscillator circuit. During an overcurrent event the comparator output forced the BC557 into the OFF state disconnecting power from the oscillator and preventing further current flow.

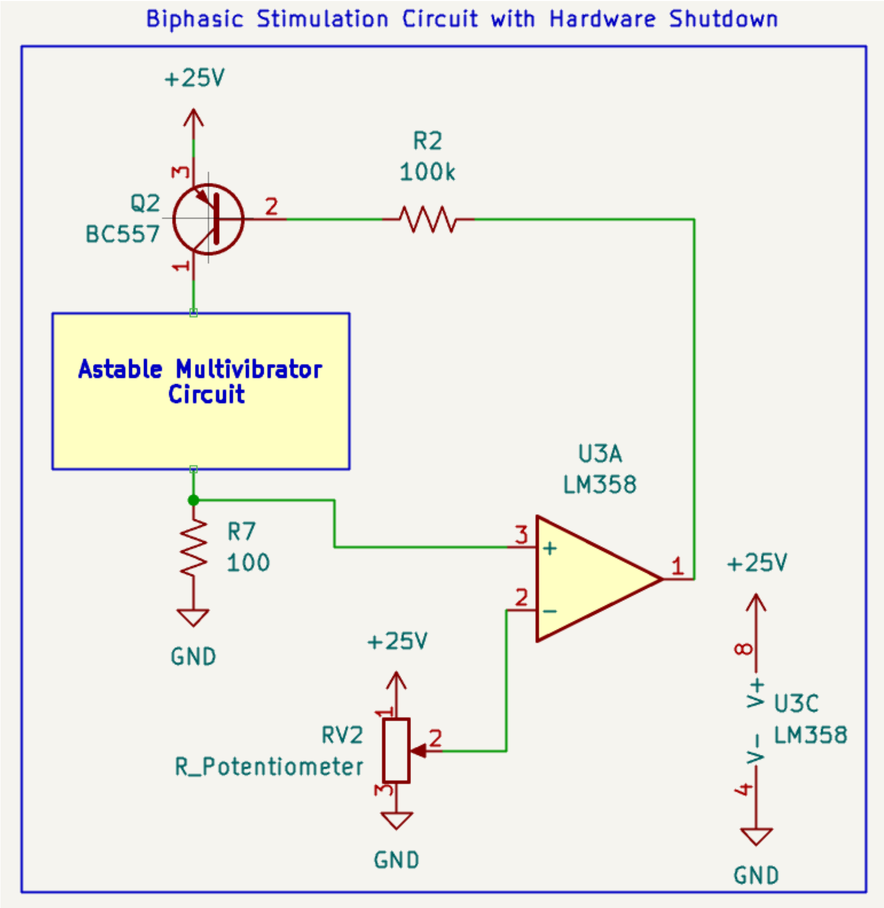



Figure 37. Schematic of the current monitoring.

### 5.10.5 Practical implementation

Due to limited availability of IRF540N devices during the experimental phase of this investigation the shutdown mechanism was implemented using bipolar junction transistors.

The experimental prototype employed.

- BC557 as the high side disconnect switch.
- 2N2222A as the comparator interface transistor.
- LM358 comparator stage.
- Adjustable threshold network.

This implementation successfully demonstrated the fundamental operating principle of automatic overcurrent protection.

Although bipolar transistors were used for proof-of-concept verification, the same architecture can be directly translated to a MOSFET based implementation in future versions of the system. Replacing the bipolar switching stage with the power MOSFETs would reduce the power dissipation, improve switching efficiency and increase current handling capability.

### 5.10.6 Experimental Verification

The protection architecture was assembled and evaluated as a standalone subsystem as shown in *Figure 38*.

The following functions were experimentally verified.

- Comparator operation.
- Adjustable current threshold.
- Transistor switching behaviour.
- Automatic oscillator shutdown.
- Restoration of operation after removal of fault conditions.

The circuit successfully demonstrated the ability to interrupt power delivery whenever the sensed current exceeded the user defined threshold.

These results confirmed feasibility of implementing active current protection within the stimulation platform.

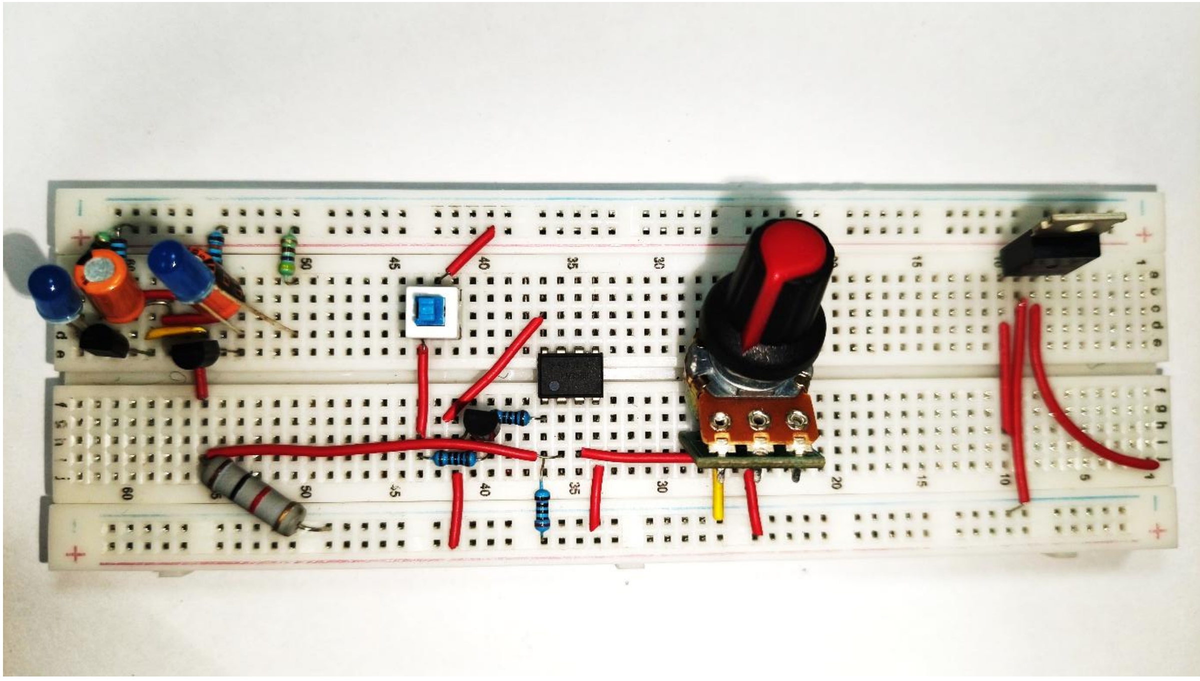


*Figure 38. Breadboard implementation of current monitoring circuit.*

### 5.10.7 Significance for Electrical Stimulation Systems

Safety is a critical consideration in electrical stimulation systems intended for biological applications.

Excessive stimulation currents may lead to:

- Tissue heating.
- Electrode degradation.

- Device damage.
- Unpredictable stimulation behaviour.
- Reduced experimental reliability.

Although the present work focused primarily on proof-of-concept validation incorporation of current monitoring demonstrates a pathway toward developing a more robust and clinically relevant stimulation platform.

The proposed architecture therefore serves as a foundation upon which additional protection mechanisms can be integrated.

### 5.10.8 Future Expansion of Protection Architecture

The developed current monitoring circuit represents only first stage of comprehensive safety system.

Future protection features may include:

- Open Electrode Detection: Detection of disconnected electrodes or broken interconnects.
- Short Circuit Detection: Identification of abnormally low load impedance conditions.
- DC Leakage Monitoring: Verification that the stimulation waveform remains alternating and does not introduce sustained DC current.
- Charge Balancing Circuitry: Amount of charge delivered in Positive half cycle and negative half cycle to be equal to avoid polarization.
- Fault Latch Circuit: Permanent shutdown following a fault condition until manual reset.
- Battery Undervoltage protection: Disabling stimulation when battery voltage falls below a safe operating limit.
- Thermal monitoring: Protection against exercise device temperature.
- Watchdog Supervision: Monitoring oscillator operation and detecting abnormal switching behaviour.
- Electrode Impedance Monitoring: To monitor if electrodes are connected or not to the surface.

Integration of these features would significantly improve operational safety and move the system closer to requirements typically associated with experimental animal studies and advanced stimulation devices.

### 5.10.9 Advantages

The proposed protection architecture offers several advantages:

- Simple implementation.
- Low component count.
- Adjustable protection threshold.

- Real time fault monitoring.
- Automatic shutdown capability.
- Compatibility with existing oscillator circuits.
- Expandability toward future safety features.

These characteristics make the architecture suitable for incorporation into future versions of the stimulation platform.

### 5.10.10 Limitations

The current implementation possesses several limitations.

The architecture does not presently incorporate:

- Fault latching mechanisms.

Furthermore, quantitative calibration of the current threshold was not performed during the present investigation and remains an area of future work.

Consequently, the circuit should be considered a proof-of-concept protection architecture intended to demonstrate feasibility rather than a fully validated safety subsystem.

## 5.11. Chapter Summary and Discussion

This chapter presented the systematic investigation of multiple stimulation circuit architectures for the development of a proof-of-concept transcutaneous electrical stimulation platform. Several oscillator and output stage configurations were analysed through simulation and practical implementation to evaluate their suitability for generating controlled stimulation waveforms.

The initial investigations focused on transistor based astable multivibrators implemented using BC547 and 2N2222A transistors. Experimental observations demonstrated that the 2N2222A implementation provided improved switching characteristics and higher current handling capability compared to the BC547 based design. Multiple resistor and capacitor combinations were evaluated to study their influence on output voltage, frequency, duty cycle and waveform quality.

Alternative approaches based on LM358 operational amplifiers and bridge-tied-load architectures were also investigated. Although functional waveform generation was achieved, practical limitations including waveform asymmetry, increase sensitivity to breadboard parasitic effects and additional circuit complexity reduced their ability for the intended application.

To further increase the output drive capability, both astable multivibrator driven H-bridge and self-oscillating MOSFET H-bridge architectures were explored. Several gate drive optimization techniques including gate resistors, discharge diodes, voltage divider biasing networks and transistor driver stages were experimentally evaluated. While these architectures successfully generated alternating output waveforms practical testing revealed challenges associated with supply voltage collapse, increase component count and reduced implementation simplicity.

Based on the experimental observation, it was concluded that increasing architectural complexity did not necessarily result in improved practical performance for the intended proof-of-concept application. Consequently, emphasis was placed on identifying a simpler and more reliable solution that could be readily integrated into a portable stimulation platform.

In parallel with waveform generation studies, a current monitoring and hardware protection architecture was developed to establish a foundation for future safety-oriented system designs. The proposed protection scheme incorporated current sensing, threshold comparison and automatic stimulation shutdown mechanisms intended to improve operational safety during future biological experiments.

The findings obtained from the various circuit investigations provided valuable insight into the trade-offs between performance, complexity, reliability and implementation feasibility. These observations ultimately guided the selection of the final stimulation architecture presented in the following chapter.

# Chapter 6
# Final Implementation: IRF540N-based Astable Multivibrator

Compared to the previously investigated comparator-based oscillators, H-Bridge configurations and self-oscillating architectures the IRF540N based astable multivibrator provided a significantly simpler solution while still generating the required alternating output waveforms. The architecture was therefore selected as the basis of the final proof-of-concept stimulation prototype.

## 6.1. Design Objectives

The primary objective of the final circuit architecture were:

- Generation of alternating stimulation waveforms.
- Operation from a portable battery source.
- Minimal component count.
- Implementation on a compact PCB.
- Reduced overall system weight.
- Compatibility with future safety and monitoring circuitry.
- Suitability for future animal mounted implementation.

The circuit was therefore designed with simplicity, reliability and practical manufacturability as the primary considerations.

## 6.2. Circuit Architecture

The final stimulation circuit as shown in *Figure 39* consisted of two IRF540 N-channel MOSFETs configured as a cross coupled astable multivibrator. Oscillation was achieved through regenerative feedback provided by cross coupled capacitors and gate biasing resistors.

Unlike the previously investigated H-Bridge architectures, the circuit utilized only N-channel MOSFETs and did not require complementary high side switching devices. This significantly reduced circuit complexity and simplified practical implementation.

The complete circuit consisted of:

- Two IRF540N MOSFETs.
- Gate biasing resistors.
- Cross coupled timing capacitors.
- Power supply interface.

The resulting architecture provided complementary output waveforms while maintaining a very low component count.

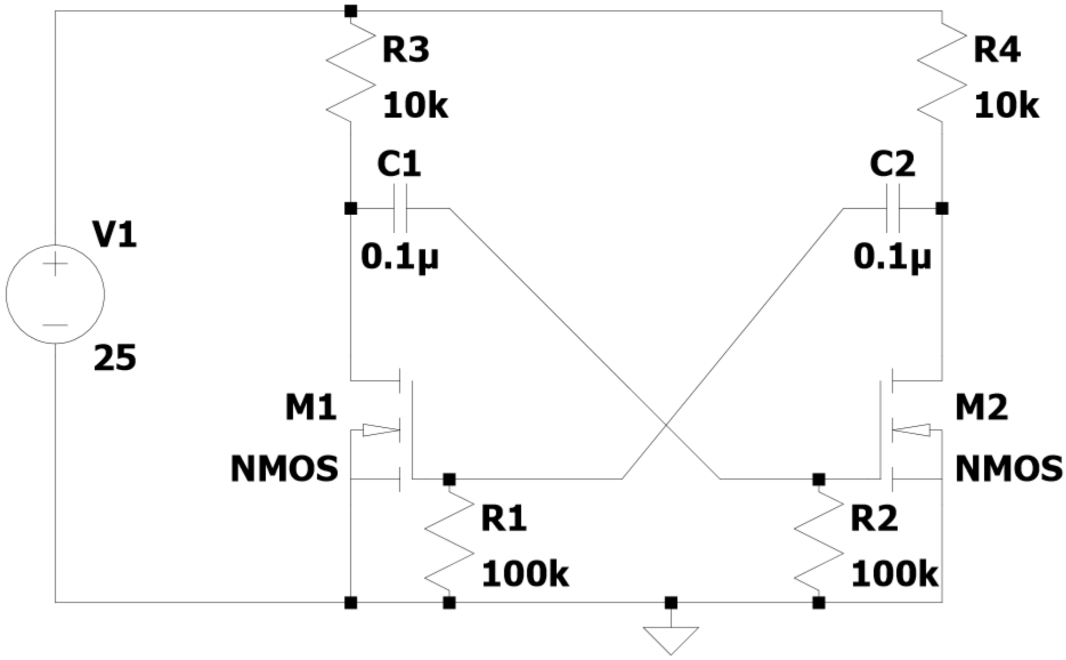



*Figure 39. Self-Oscillating MOSFET H-Bridge troubleshooting breadboard implementation.*

## 6.3. Principle of Operation

The circuit operates on the same fundamental principle as a conventional astable multivibrator.

When one MOSFET enters conduction, the voltage at its drain decreases. Through the cross coupled capacitor network this voltage transition is transferred to the gate of the opposite MOSFET, forcing it into the OFF state.

As the capacitor charges through the associated gate resistor, the gate voltage gradually changes until the opposite MOSFET begins conducting. The resulting voltage transition then switches the first MOSFET OFF.

This process continuously repeats, producing alternating conduction of the two MOSFETs and generating complementary square wave outputs.

The oscillation frequency is primarily determined by the resistor capacitor time constant of the feedback network.

## 6.4. Component Selection

Several resistor capacitor combinations were investigated experimentally before arriving at the final implementation.

Initial investigation using large resistor values and electrolytic capacitors produced slower oscillation frequencies but did not consistently provide output voltages approaching the supply voltage. Conversely, very small resistor values resulted in undesirable switching behaviour and reduced waveform quality.

Following experimental optimization, the final implementation utilized:

| Component | Value |
|---|---|
| Gate Resistor | 100kΩ |
| Timing Capacitor | 0.1μF |
| MOSFET | IRF540N |
| Supply Voltage | 25 V |

This configuration provided stable oscillation, sharp switching transitions and output amplitudes approaching the supply voltage.

## 6.5. Experimental Implementation

To facilitate hardware miniaturization and practical deployment a dedicated printed circuit board (PCB) schematic was also designed for the final IRF540N based astable multivibrator architecture. The PCB design incorporated the oscillator circuitry along with the necessary power connections and component footprints required for integration into the complete stimulation system.

Due to the materials available during the project, the PCB was designed as a single layer layout. This design choice was motivated by the availability of single sided copper clad substrate and the desire to maintain the fabrication process that could be readily reproduced using standard laboratory resources. The relatively low component count of the selected architecture also made single layer routing feasible without requiring complex interconnections.

The PCB layout was developed with consideration for compactness, ease of assembly and straightforward routing of the oscillator feedback network. Through-hole component footprints were selected to accommodate the components used throughout the experimental investigations.

Although the PCB design was completed, fabrication of the custom board was not undertaken during the present work. Nevertheless, the completed schematic and layout demonstrate the feasibility of transitioning the prototype from a Zero PCB implementation to a dedicated printed circuit board in future iterations.

## 6.6. Experimental Implementation

The final circuit was initially validated on breadboard platform and subsequently implemented on a Zero PCB for proof-of-concept prototype evaluation.

The completed hardware as shown in *Figure 40* incorporated:

- 3.7V 1500mAh battery.
- TP4056 lithium ion battery charging module.
- XL6009 DC-DC boost converter.
- IRF540N astable Multivibrator.
- External stimulation Electrode.

The implementation demonstrated the feasibility of realizing the complete stimulation system using commercially available components while maintaining a compact footprint.

Although the circuit was suitable for implementation on a custom PCB, this approach was not pursued during the present work, as PCB fabrication facilities were temporarily unavailable.

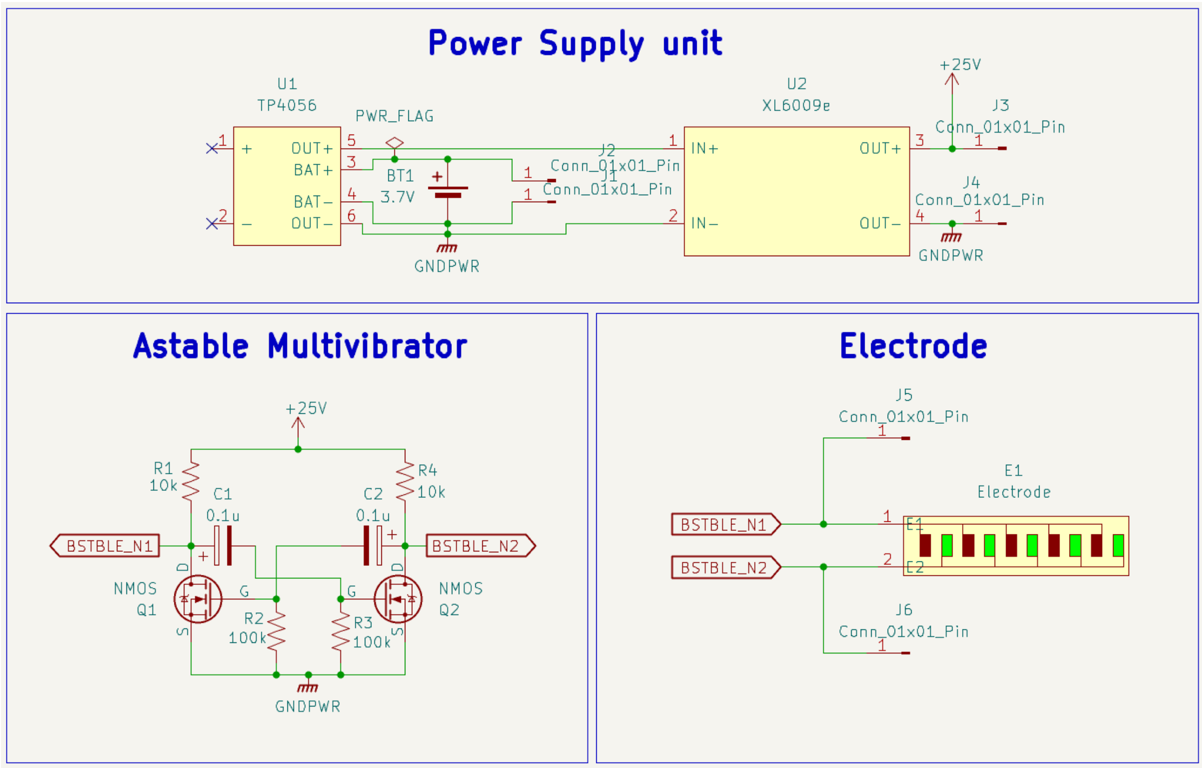



*Figure 40. PCB schematic of Final IRF540N Astable Multivibator.*

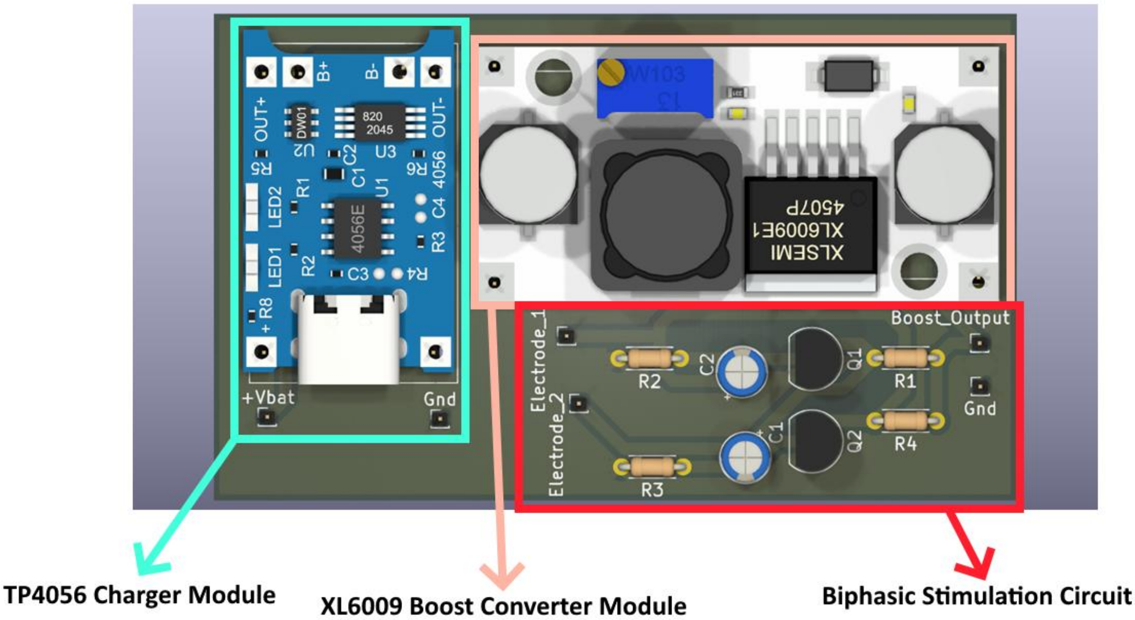



*Figure 41. Single-Layer PCB Layout designed for Fabrication of Final IRF540N Astable Multivibator.*

## 6.7. Power Management and Experimental Implementation

The final circuit was initially validated on a breadboard platform as shown in *Figure 42* and subsequently implemented on a Zero PCB for proof-of-Concept evaluation as shown in *Figure 43*.

The completed hardware incorporated:

- 3.7V, 1500mAh Li-ion Battery.
- TP4056 Lithium-ion Battery charging module.
- XL6009 DC-DC boost converter.
- IRF540N astable multivibrator.
- External Stimulation Electrode.

The power subsystem was designed around a single cell Li-ion battery to improve portability and facilitate future wearable implementation. Battery charging and protection were provided using a TP4056 charging module. The TP4056 was selected because it offers a compact and widely available solution for constant-current/ constant-voltage (CC/CV) charging of single cell lithium ion batteries. In addition to battery charging functionality, the module used in this work incorporates a DW01A battery protection integrated circuit together with a pair of back-to-back MOSFETs.

The DW01A continuously monitors battery operating condition and provides protection against overcharge, over discharge and over current conditions. The associated back-to-back MOSFET arrangement enables bidirectional current control, allowing the protection circuit to disconnect the battery during fault conditions regardless of current direction. This arrangement improves battery safety and helps prevent damage resulting from abnormal operating conditions.

To generate the voltage required by the stimulation circuitry, a DC-DC boost converter was employed. During the initial design phase, the MT3608 boost converter was identified as a suitable candidate due to its compact factor, low footprint and suitability for portable electronic systems. However, the MT3608 module was not readily available during the implementation phase of the project. Consequently, an XL6009 boost converter was used for experimental validation and prototype construction.

The XL6009 successfully provided the required boost supply voltage and enables complete functional verification of the stimulation circuit. Nevertheless, for future miniaturized implementations, the MT3608 remains an attractive alternative because of its smaller physical dimensions and reduced board area requirements. Adoption of an MT3608 based design could contribute to further reductions in overall system size and weight which are important considerations for wearables and anima mounted stimulation platforms.

Commercially available breakout modules were integrated directly for XL6009 and TP4056 into the prototype to accelerate development and focus on experimental efforts on the stimulation circuitry itself. The astable multivibrator was implemented separately using discrete components, while the charging and voltage conversion functions were provided by the respective modules.

Although this modular approach simplified construction and testing, it does not represent the most compact implementation. In a future design iteration, the charging circuit, battery protection circuitry, boost converter and stimulation oscillator could all be integrated onto a single custom PCB. Such integration would eliminate the need for separate breakout boards, reduce overall footprint and improve mechanical robustness. Furthermore, integrating all stages onto a common copper clad PCB would provide a clear pathway toward a fully miniaturized stimulation device suitable for practical deployment.

Although a dedicated PCB design was prepared, fabrication facilities available for the project were temporarily unavailable during the experimental phase. An alternative approach involving chemical etching of the single sided copper clad substrate was considered as shown in *Figure 41*. However, the majority of components used in the prototype were through hole devices which would have required drilling numerous mounting holes following the etching process. The necessary precision drilling equipment and fabrication tools were not readily available at the time of development. Consequently, a Zero PCB was selected as the most practical solution for constructing the proof-of-concept prototype. This approach enabled rapid assembly, straightforward modification during testing and reliable integration of all circuit modules without introducing significant delays to the project timeline.

Following circuit validation on the breadboard platform, the final stimulation architecture was assembled on a Zero PCB to improve mechanical robustness and facilitate practical testing. Through hole components including the IRF540N MOSFETs, TP4056 charging module, XL6009 boost converter, passive components and interconnecting wires were manually soldered onto the board according to the finalized circuit layout. The assembled prototype enabled stable electrical connections and provided a compact platform for subsequent waveform characterization and electrode integration.

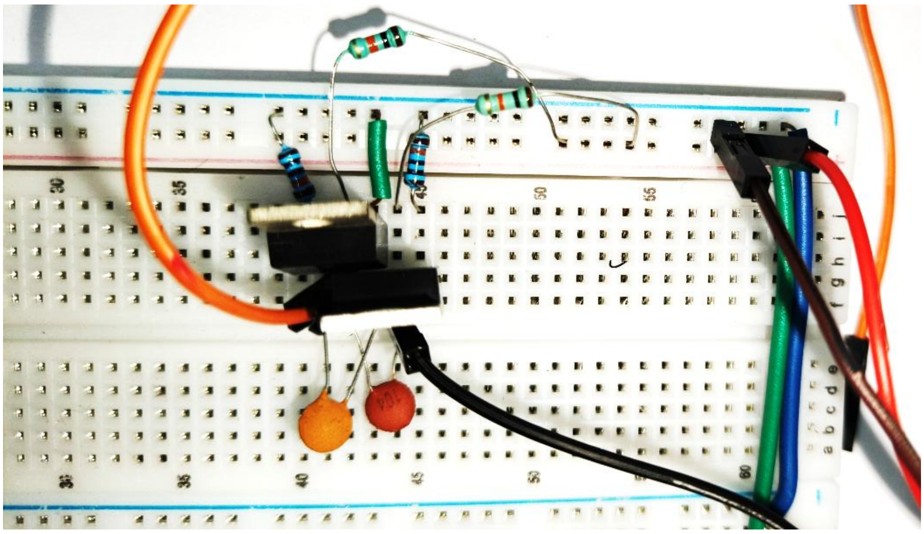


*Figure 42. Breadboard implementation of Final IRF540N Astable Multivibator.*

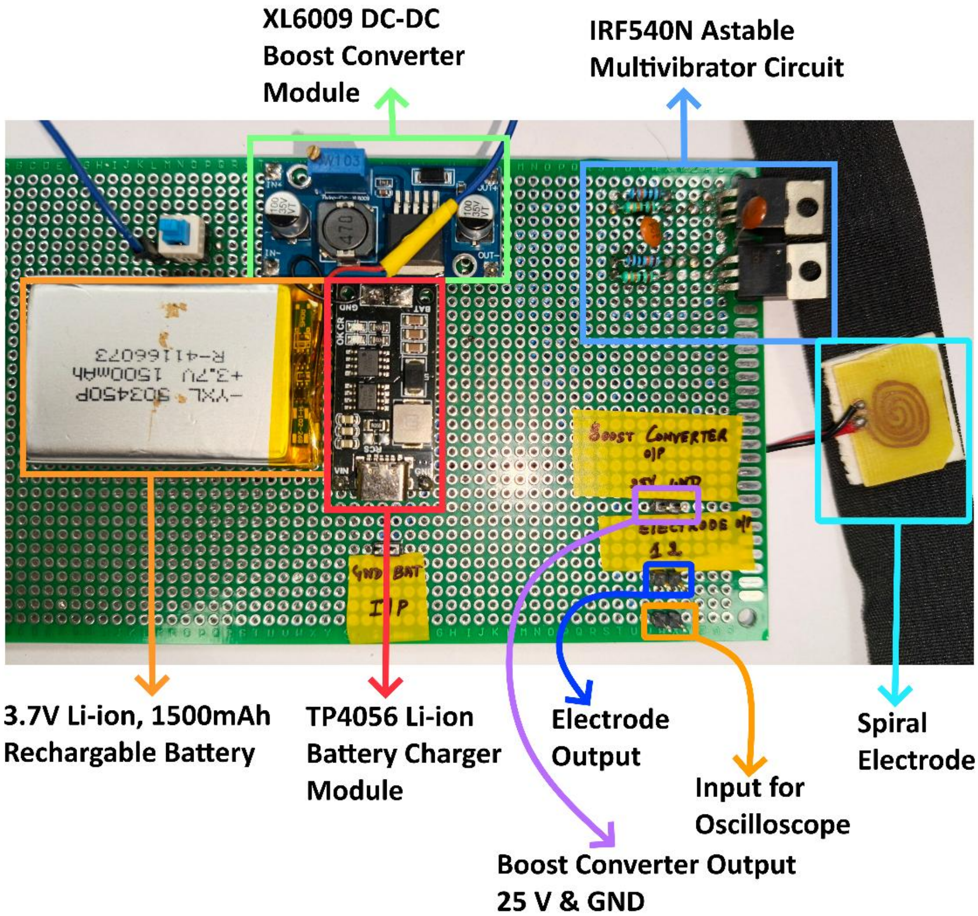



*Figure 43. Implementation of Final Prototype soldered on Zero PCB.*

## 6.8. Oscilloscope Characterization

The output waveforms generated by the implemented circuit were analysed using a digital oscilloscope.

Measurements confirmed the successful generation of alternating square wave outputs with amplitude approaching the intended operating voltage as shown in *Figure 44*. Under the unloaded conditions, corresponding to the case where the stimulation electrode was not connected, the circuit produced a stable square wave output with peak to peak voltage of approximately 52 V and a period of approximately 14.3 ms as shown in *Figure 45*.

The observed waveforms exhibited:

- Stable oscillation.
- Sharp switching transitions.
- Repeatable operation.
- Minimal distortion under unloaded conditions.

The measured oscillation period remained consistent with the selected resistor capacitor network and demonstrated satisfactory stability during operation.

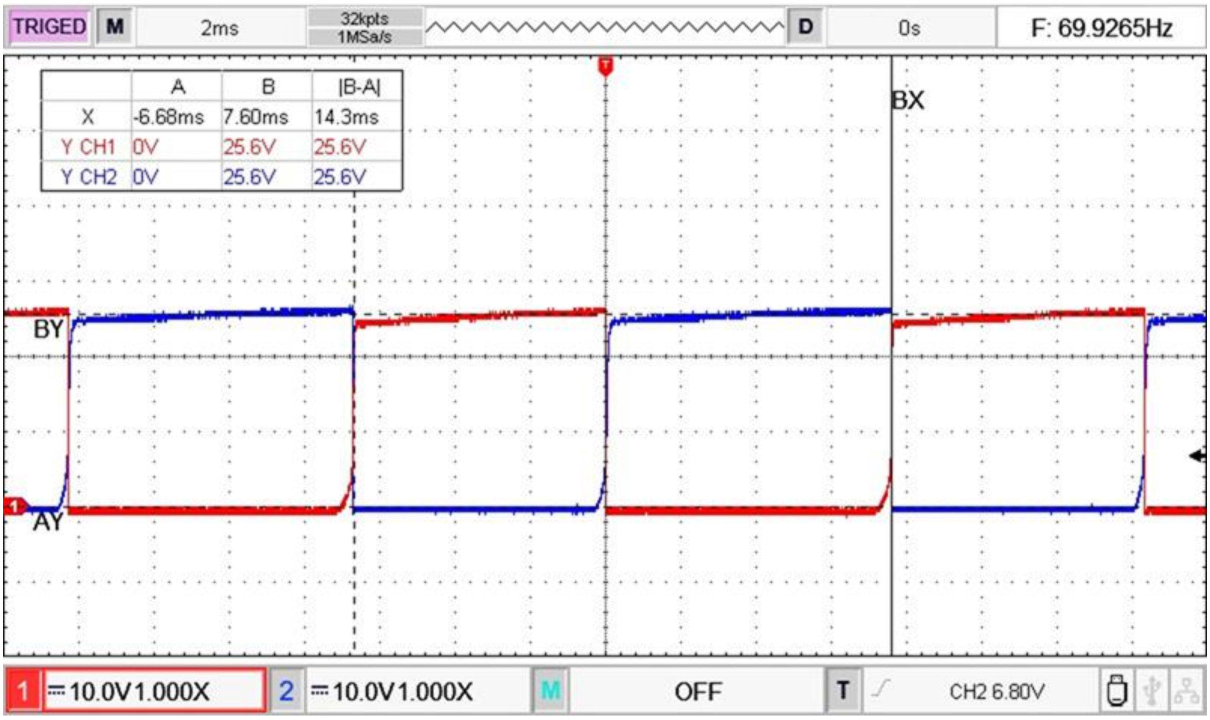



*Figure 44. Oscilloscope waveforms at the electrode terminals of Final IRF540N Astable Multivibrator, showing amplitude of 25.6 V on each electrode terminal and frequency of 66 Hz.*

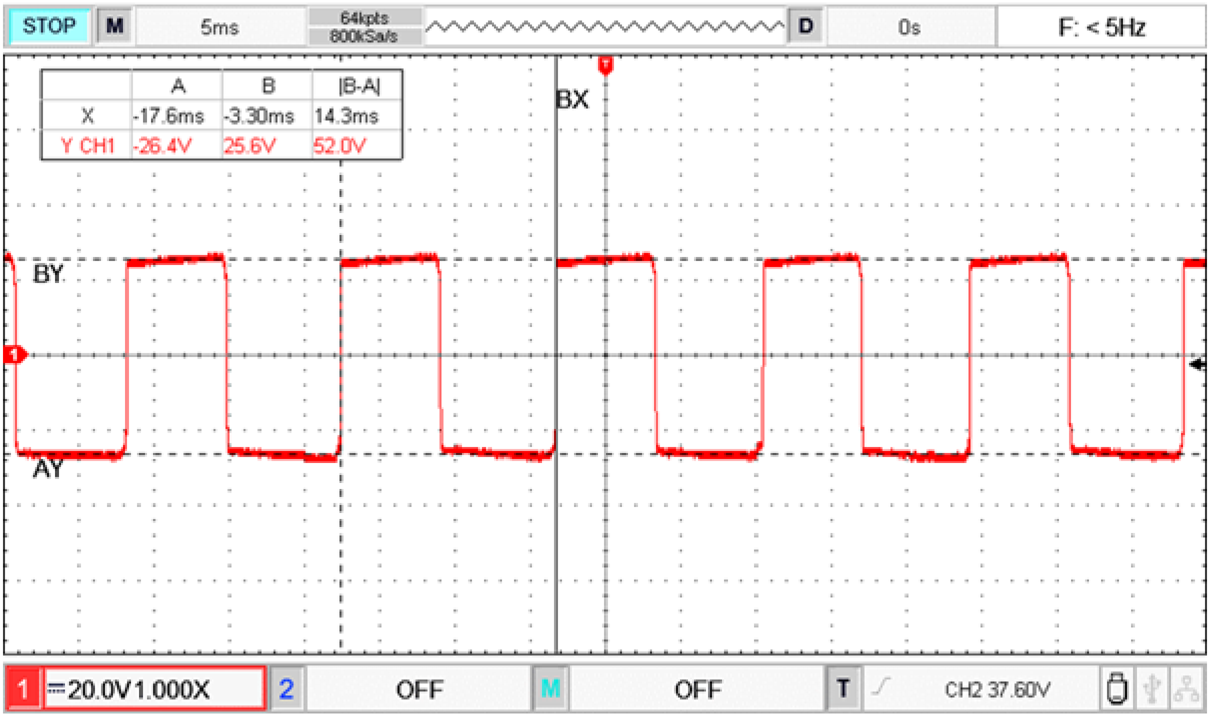



*Figure 45. Oscilloscope waveform of differential voltage appearing across the load from the implementation of Final IRF540N Astable Multivibrator. The obtained output is ±26 V, i.e. 52 V peak-to-peak at 70 Hz, fulfilling the objectives of the project.*

## 6.9. Electrode Loading Evaluation

The fabricated stimulation electrode was connected to the circuit and preliminary observations were performed to evaluate the influence of electrode loading on the generated waveform.

Measurements were obtained under three operating conditions:

- Electrode left Ideal (unloaded condition).
- Electrode placed in contact with skin.

*Table 26. Output waveform characteristics under different loading conditions:*

| Condition | Peak-to-Peak Voltage (V) | Period (ms) |
|---|---|---|
| Electrode left Ideal | 52.8 | 11.5 |
| Electrode placed on Skin | 42.4 | 13.5 |

When no electrode was connected, the circuit exhibited the highest output amplitude and shortest oscillation period as shown in *Figure 46*, the measured peak-to-peak voltage was approximately 52.8 V with an oscillation period of approximately 11.5 ms.

When the fabricated electrode was placed on the skin as shown in Figure 47, the measured peak-to-peak voltage was approximately 42.4 V with an oscillation period of approximately 13.5 ms as shown in *Figure 48*. The voltage observed in this condition may be attributed to contact area, skin impedance, pressure distribution, moisture content and electrode skin coupling characteristics.

Despite these variations, the oscillator continued to operate under all loading conditions and maintained a recognizable square wave output. The observed changes in the oscillation period and output amplitude further indicate that the electrode tissue interface introduces a load dependent impedance into the oscillator network. Such a behaviour is expected because biological tissues exhibit both resistive and capacitive characteristics. Although the oscillator remains functional under varying biological loading conditions.

At this stage the investigation was limited to electrical waveform characterization and proof-of-concept evaluation. Detailed impedance characterization, current measurements and biological efficacy studies remain subject to future work.

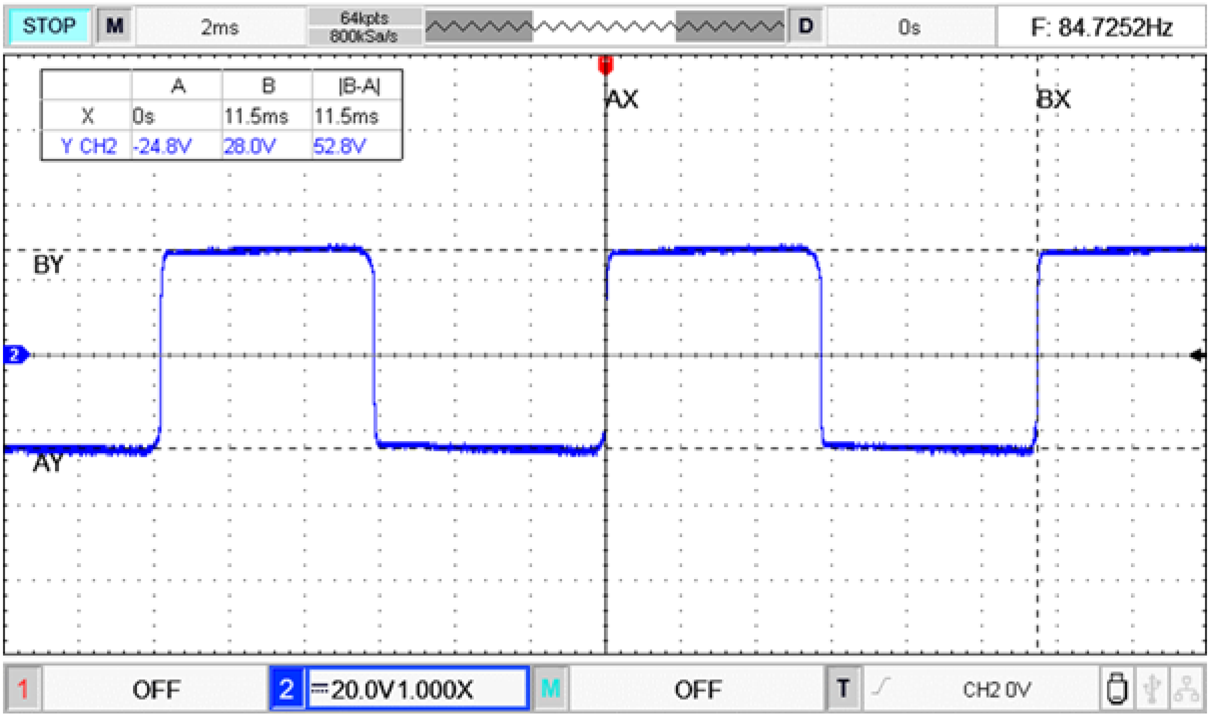



*Figure 46. Unloaded waveform of Final IRF540N Astable Multivibrator (Electrodes without skin contact).*

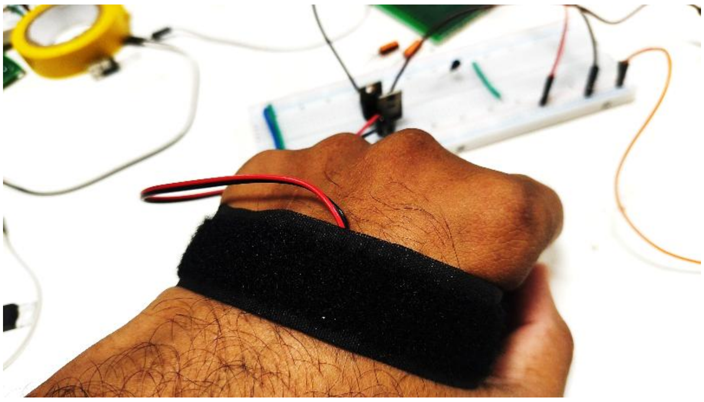


*Figure 47. Electrode in contact with the skin.*

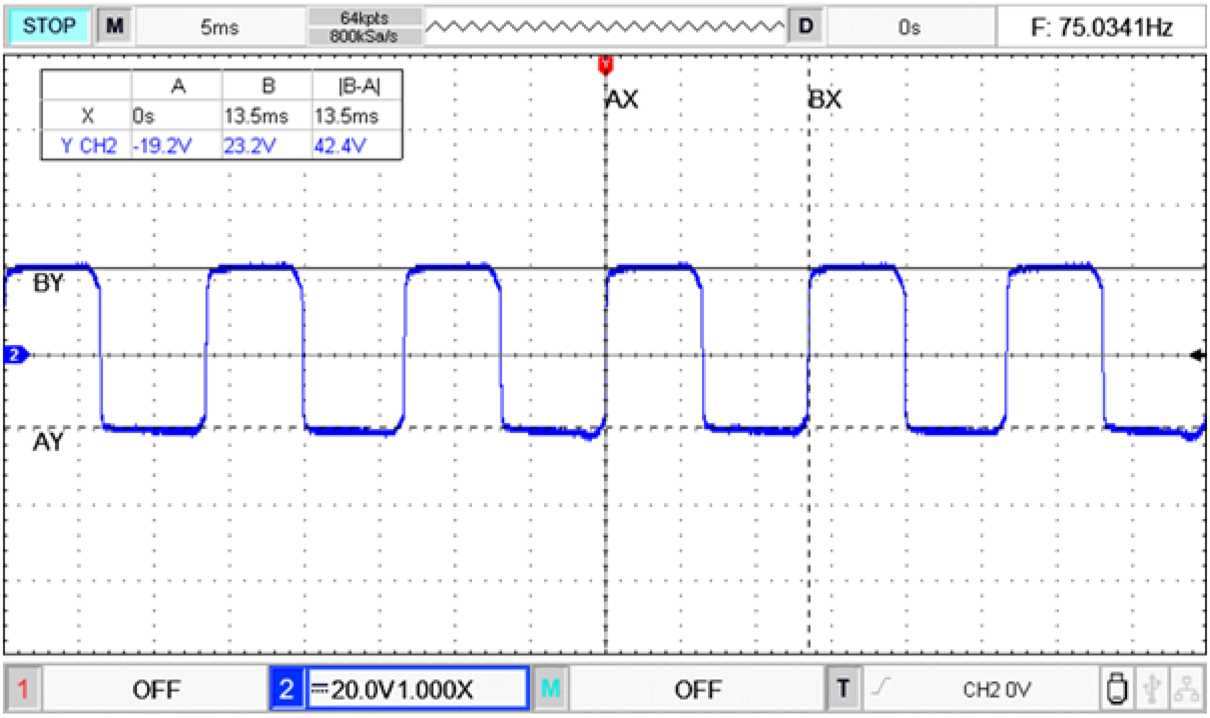



*Figure 48. Loaded waveform of Final IRF540N Astable Multivibrator with skin contact.*

## 6.10. Fabricated Electrode Considerations

The stimulation electrodes used during experimental evaluation were fabricated using a manual free hand drawing and chemical etching process on copper clad substrate. As a result, the fabricated electrodes did not possess the dimensional precision typically achievable through industrial PCB manufacturing processes.

Variations were present in several geometric parameters including:

- Electrode width.
- Inter electrode gap spacing
- Edge uniformity.
- Symmetry between electrode segments.
- Surface finish quality.

These geometric variations can influence the electrical characteristics of the electrode including current distribution, electric field concentration, contact impedance and electrode skin coupling behaviour. Consequently, the measured waveform characteristics obtained during testing should be interpreted as representative proof-of-concept results rather than definitive performance metrics of the final electrode design.

For future development, fabrication using machine-based PCB manufacturing techniques or precision photolithographic processes is expected to improve dimensional accuracy, repeatability and electrode symmetry. Such improvements may result in different electrical characteristics and potentially more consistent stimulation performance.

The manually fabricated electrode was used solely for proof-of-concept validation of the stimulation architecture and should not be interpreted as the final electrode implementation intended for experimental studies.

## 6.11. Advantage of the Selected Architecture

The final IRF540N based astable multivibrator offered several practical advantages compared to the previously investigated architectures.

- Lowest component count among all investigated circuits.
- Elimination of comparator circuits and gate driver stages.
- Reduced PCB area requirements.
- Simplified fabrication and assembly.
- Lower overall system weight.
- Stable and predictable operation.
- Ease of troubleshooting.
- Compatibility with future safety enhancements.
- Suitability for compact stimulation platforms.

These characteristics made the architecture particularly attractive for proof-of-concept implementation and future miniaturization.

## 6.12. Limitations

Although the circuit successfully achieved the objective of present work, several limitations remain.

The implemented prototype does not incorporate:

- Electrode impedance was not quantitively measured.
- Stimulation current was not directly characterized.
- Current monitoring.
- Open-electrode detection.
- Short circuit protection.
- DC leakage monitoring.
- Watchdog supervision.
- Active charge balancing.

Furthermore, biological efficacy studies were outside the scope present investigation.

Additional limitations include the manually fabricated electrode geometry and the use of discrete breakout modules for battery charging and voltage conversion. While these modules simplified development and testing a fully integrated PCB implementation would be preferable for future miniaturized designs.

Consequently, the developed system should be considered a proof-of-concept stimulation platform intended for preliminary and future development.

## 6.13. Design Decision

Among all investigated architectures the IRF540N based astable multivibrator provided the most favourable balance between simplicity, performance, manufacturability and experimental reliability.

The architecture successfully generated the intended alternating stimulation waveform required for proof-of-concept evaluation using a minimal number of components while avoiding the implementation challenges associated with comparator-based oscillators, H-bridge configurations and self-oscillating power stages.

For these reasons, the architecture was selected as the final stimulation circuit and served as the foundation for the developed proof-of-concept prototype.

## 6.14. Summary

The IRF540N based astable multivibrator represents the final stimulation architecture developed during this work. Through a systematic process of circuit exploration, optimization, troubleshooting, PCB design and experimental evaluation, a compact stimulation platform was successfully realized. The resulting prototype demonstrated stable waveform generation and successful integration with the fabricated stimulation electrode under loaded and biological loading conditions. Battery charging and protection were provided through a TP4056 module incorporating DW01A based protection circuitry, while voltage boosting was achieved using a XL6009 was used for experimental implementations due to component availability, future miniaturized versions may benefit from the smaller footprint offered by the MT3608. The use of commercially available breakout modules enabled rapid prototyping and functional verification, while also highlighting opportunities for future integration onto a single custom PCB. The completed single layer PCB design therefore establishes a clear pathway toward hardware miniaturization, precision electrode fabrication, integrated power management and the development of a fully featured stimulation system incorporating advanced safety and monitoring capabilities.

# Chapter 7
# Conclusion and Future Work

## 7.1. Introduction

This thesis presented the design, simulation, implementation and experimental validation of a proof-of-concept transcutaneous electrical stimulation platform intended to support future preclinical stimulation studies. The work was motivated by the need for compact, customizable and experimentally accessible stimulation system capable of supporting future investigations involving non-invasive electrical stimulation techniques.

The development process began with the design and evaluation of multiple electrode geometries using COMSOL Multiphysics simulation part of Pooja's (2024) work. Several interdigitated and spiral electrode configurations were investigated to study the influence of electrode geometry on electric field distribution. Based on simulation results and practical fabrication considerations, a spiral electrode geometry was selected for prototype implementation. PCB compatible electrode layouts were developed using KiCAD and a proof-of-concept electrode was fabricated using a ferric chloride chemical etching process due to the temporary unavailability of PCB fabrication facilities.

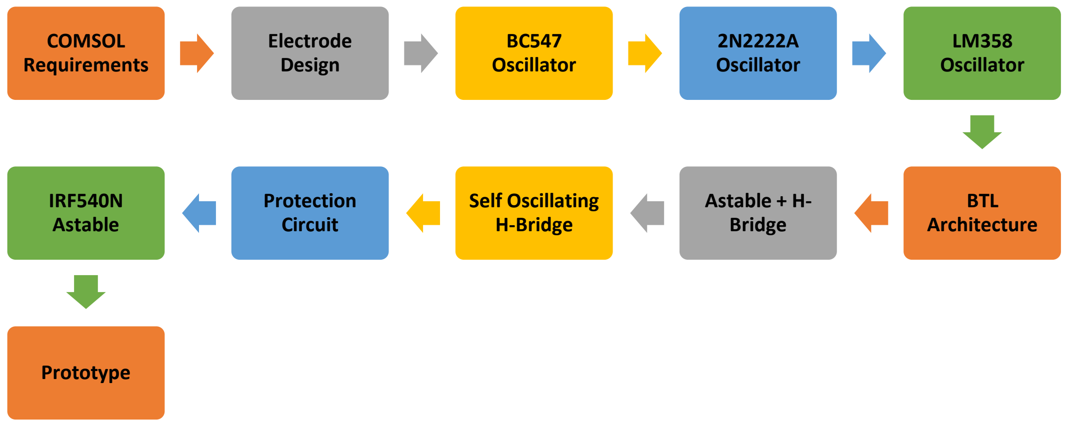
Multiple stimulation architectures were subsequently investigated through simulations and hardware implementation. BC547 based oscillators, 2N2222A astable multivibrator, LM358 based oscillators, bridge-tied-load configurations, astable multivibrator driven H-bridge and self-oscillating MOSFET H-bridges were designed, analysed and experimentally evaluated. Oscilloscope measurements and practical testing were used to compare waveform quality, voltage capability, implementation complexity and reliability.

The experimental investigations demonstrated that simpler architectures often provided superior practical performance compared to more complex implementations. Through successive design iterations and hardware validation an IRF540N based astable multivibrator architecture was identified as the most suitable solution for proof-of-concept stimulation generation. The selected architecture was integrated with a rechargeable lithium-ion battery, charging circuitry, voltage boosting stage and stimulation electrode to realize a portable stimulation prototype.

Experimental validation confirmed successful waveform generation and stable operation under electrode-loaded conditions. The developed prototype demonstrated the feasibility of delivering controlled stimulation waveforms through a custom fabricated electrode interface while maintaining a relatively simple and low component count implementation.

In addition to waveform generation a current monitoring and protection architecture was developed to establish a foundation for future safety integration. Although not fully incorporated into the final prototype the proposed protection circuitry demonstrated the feasibility of incorporating current limiting and automatic shutdown mechanisms into future system revisions.

Overall, this work established a complete engineering workflow encompassing simulation driven electrode design, circuit exploration, hardware implementation, prototype development and preliminary safety architecture development. The resulting platform provides a functional proof-of-concept system and a foundation for future wearable stimulation devices intended for small animal studies.




## 7.2.Major Contribution of the Thesis

The major contributions of this work are summarized below:

- Development of PCB compatible electrode layouts using KiCAD.
- Fabrication of a proof-of-concept stimulation electrode using ferric chloride chemical etching.
- Investigation and experimental validation of multiple stimulation circuit architectures.
- Design and implementation of an IRF540N based stimulation generator.
- Development of a portable stimulation prototype incorporating battery powered operation.
- Development of a current monitoring and protection architecture for future safety integration.
- Establishment of a design pathway toward wearable transcutaneous electrical stimulation systems for animal studies.

## 7.3.Future Work

Although the developed prototype successfully demonstrated proof-of-concept stimulation generation, several opportunities remain for future improvement and expansion.

The first area of future work involves fabrication of the electrode and stimulation circuitry using precision PCB manufacturing process. Machine fabricated electrodes would provide improved dimensional accuracy, consistent conductor width-to-gap rations and greater repeatability compared to manually fabricated prototypes.

Future versions of the stimulation circuitry may be implemented on custom designed PCB’s to significantly reduce system size, improve reliability and simplify integration. Multi-layer PCB designs may further enable compact routing and improved power distribution while reducing the overall footprint of the wearable system.

The current monitoring and protection architecture developed during this work may be fully integrated into future hardware revisions. Additional safety features including open electrode detection, DC leakage monitoring, fault latching mechanisms, over current protection and battery fault detection may also be incorporated to improve operational safety.

Comprehensive electrical characterization of the electrode tissue interface under different loading conditions should also be performed to better understand stimulation behaviour and optimize electrode geometry. Comparative evaluation of different electrode materials and flexible substrate technologies may further improve system performance and comfort.

The ultimate objective of this work is the development of a fully miniaturized wearable stimulation platform that can be mounted directly on a small laboratory animal as shown in *Figure 48*. Integration of custom fabricated flexible electrodes, compact stimulation electronics, rechargeable power systems and comprehensive safety monitoring circuitry would enable future preclinical investigations following appropriate ethical approvals and regulatory requirements.

The outcomes of the present work to provide the technical foundation necessary for pursuing these future developments and advancing toward a compact, safe and experimentally deployable transcutaneous electrical stimulation platform.

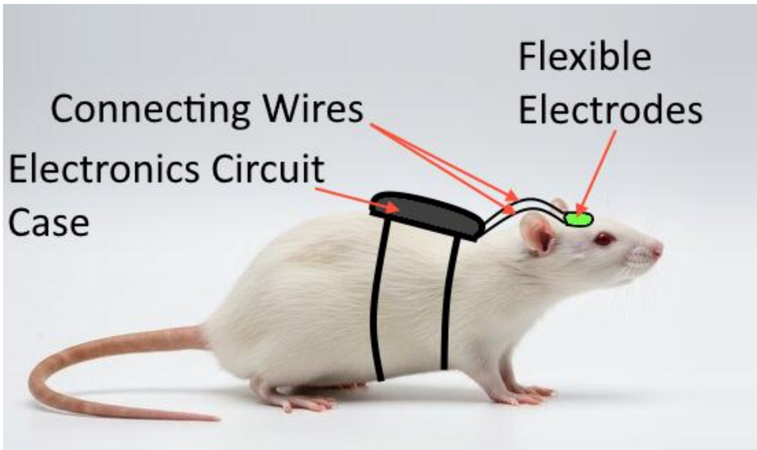



*Figure 49. Proposed wearable system for small animal studies*

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